SEVEN OPEN UNIT EQUILATERAL TRIANGLES DO NOT COVER A REGULAR UNIT HEXAGON

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Abstract. We prove that the regular hexagon of side length one cannot be covered by seven open equilateral triangles of side length one. The center and the six vertices canonically label the seven covering triangles. A finite classification routes every hypothetical cover to one of four mechanisms: trace-length deficit, propagation of boundary-reach lower bounds, area loss, or an equilateral-enclosure obstruction for nine forced points. The paper is organized as a single proof: the structural reduction is followed by one section for each mechanism, and repeated reader/verification layers have been removed. Long branchwise algebra remains in the propagation section, while the two mixed support-arc intersections are verified in a self-contained exact certificate appendix over $\mathbb{Q}(\sqrt{3})$.

Contents

2020 *Mathematics Subject Classification.* 52C15, 52A40, 51M04.

Key words and phrases. equilateral triangles, geometric covering, regular hexagon, boundary propagation, area loss, exact computer-assisted certificate.

1. Introduction and Proof Outline

1.1. The theorem and the canonical labeling. Let

$$O = (0,0), \qquad V_i = \left(\cos\frac{i\pi}{3}, \sin\frac{i\pi}{3}\right), \qquad H = \operatorname{conv}\{V_0, \dots, V_5\},$$

where indices lie in $\mathbb{Z}/6\mathbb{Z}$. Write

$$e_{i,i+1} = [V_i, V_{i+1}], \qquad r_i = [O, V_i], \qquad M_i = \frac{1}{2}V_i,$$

and put

$$S = \partial H \cup \bigcup_{i=0}^{5} r_i.$$

Then

$$\mathcal{H}^1(\partial H) = 6, \qquad \mathcal{H}^1\left(\bigcup_{i=0}^{5} r_i\right) = 6, \qquad \mathcal{H}^1(S) = 12.$$

Theorem 1.1 (Main theorem)**.** *The hexagon H cannot be covered by seven open equilateral triangles of side length one.*

Corollary 1.2 (Soifer's Hexagon Problem)**.** *For every $L > 1$, the regular hexagon $H_L = LH$ cannot be covered by seven closed unit equilateral triangles.*

The seven points $O, V_0, \dots, V_5$ are pairwise at distance at least one, whereas any two points in an open unit equilateral triangle are at distance strictly less than one. Hence the seven covering triangles admit a unique labeling

$$U_C, U_0, \dots, U_5$$

such that

$$O \in U_C, \qquad V_i \in U_i.$$

We call U_C the *center triangle* and U_i the *vertex triangle at V_i*. Set

$$T_C = \overline{U_C}, \qquad T_i = \overline{U_i}.$$

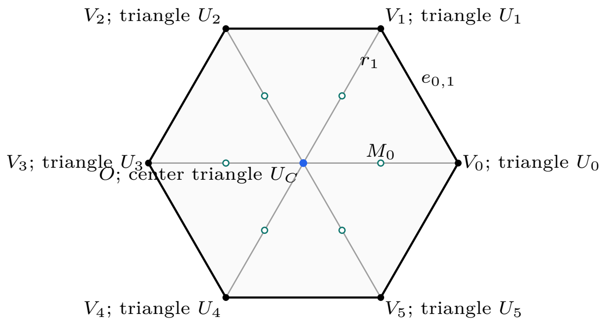


FIGURE 1. The center, vertices, radial segments, and radial midpoints of H.

1.2. **Finite classifications.** The center type is the number of boundary edges on which T_C has a positive-length trace. The only possibilities are $0, 1, 2$, denoted by CE0, CE1, CE2.

For a vertex triangle, let o be the number of its triangle vertices outside H and let n be the number of adjacent radial segments met in positive length. The exhaustive types are

type	(o, n)
Vd0	$(1,0), (2,0), (3,0)$
Vd1	$(1,1)$
Vd2	$(1,2)$
T3-like	$(2,1)$.

Let A_i, B_i, C_i be the maximal reaches of T_i toward V_{i-1}, V_{i+1}, O, respectively. The vertex triangle T_i is *supercritical* when $A_i + B_i > 1$, and

$$N_+ = |\{i : A_i + B_i > 1\}| .$$

Lowercase (a_i, b_i, c_i) always denotes selected lower bounds for these reaches and never the maxima defining N_+.

On $e_{i,i+1}$, parametrized from V_i to V_{i+1}, the two incident open traces have the form $[0, B_i)$ and $(1 - A_{i+1}, 1]$. Their nonempty complement

$$[B_i, 1 - A_{i+1}]$$

is a *boundary gap*; equality gives a singleton gap, which remains uncovered. Let N_{gap} be the number of positive center traces that contain a boundary gap.

Let N_{sp} be the number of vertex triangles of type Vd1, Vd2, or T3-like. The common trace calculation proves that the number of vertex triangles that are either supercritical or meet an adjacent radial segment in positive length is

$$N_+ + N_{\text{sp}}.$$

In particular, every CE1/CE2 configuration with $N_+ + N_{\text{sp}} \geq 3$ is impossible by a trace-length bound on S.

1.3. **Exhaustive routing.** Table 1 is read from top to bottom in the CE1/CE2 part. The first applicable row is the unique method for a hypothetical cover. The full classification proof is Proposition 2.13.

center type and N_+	exhaustive refinement	method
CE0, $N_+ = 0$	all vertex types	1
CE0, $N_+ = 1$	all six triangles Vd0	4
CE0, $N_+ = 1$	a Vd1 or Vd2 triangle	1
CE0, $N_+ = 1$	no Vd1/Vd2 triangle and a T3-like triangle	3
CE0, $N_+ \geq 2$	all vertex types	3
CE1/CE2	$N_+ + N_{\mathrm{sp}} \geq 3$	1
CE1/CE2, $N_+ = 0$	all six triangles Vd0	2
CE1/CE2, $N_+ = 0$	a Vd1 or Vd2 triangle	1
CE1/CE2, $N_+ = 0$	no Vd1/Vd2 triangle and a T3-like triangle	2
CE1/CE2, $N_+ = 1$	all Vd0, no boundary gap	4
CE1/CE2, $N_+ = 1$	all Vd0, a boundary gap	2
CE1/CE2, $N_+ = 1$	exactly one T3-like, no Vd1/Vd2	2
CE1/CE2, $N_+ = 1$	exactly one Vd1/Vd2, no T3-like	1 / 1+2

Table 1. The exhaustive routing. Method 1 is a trace-length argument, Method 2 propagates reach bounds, Method 3 sums area loss, and Method 4 uses a nine-point enclosure obstruction. In the last row, the first alternative is for CE1 and the second for CE2.

1.4. **Organization of the proof.** Section 2 proves the structural classification, strict handoff selection, midpoint facts, and the signed center normal form. Section 3 gives the complete trace-length argument. Section 4 develops the local admissible set and all boundary-reach propagation cases. Section 5 proves the local and cyclic area-loss estimates. Section 6 constructs the nine forced points and proves the equilateral-enclosure obstruction. Section 7 then applies Table 1 once and closes the proof.

2. Structural Reduction and Common Geometry

This section supplies the complete finite reduction used by every later method. It proves the open–closed scaling equivalence, the center and vertex classifications, strict boundary handoffs, midpoint facts, and exhaustive routing. It then records the signed center normal form used in both the trace and propagation arguments. Each result is stated and proved once.

2.1. **Structural Reduction to Finite Cases.** Throughout, indices are taken in $\mathbb{Z}/6\mathbb{Z}$. Put

$$O = (0,0), \qquad V_i = \left(\cos\frac{i\pi}{3}, \sin\frac{i\pi}{3}\right), \qquad H = \operatorname{conv}\{V_0, \dots, V_5\}.$$

Thus H is the regular hexagon of side length one. We use the notation

$$e_{i,i+1} = [V_i, V_{i+1}], \qquad r_i = [O, V_i], \qquad M_i = \frac{1}{2}V_i,$$

and, when needed,

$$D = \bigcup_{i=0}^{5} r_i, \qquad S = \partial H \cup D, \qquad S_{1/2} = \partial H \cup \{O, M_0, \dots, M_5\}.$$

The arms r_i meet only at O, up to sets of one-dimensional measure zero, and consequently

$$\mathcal{H}^1(\partial H) = 6, \qquad \mathcal{H}^1(D) = 6, \qquad \mathcal{H}^1(S) = 12. \tag{1}$$

For $L > 0$, $H_L := LH$ denotes the concentric regular hexagon of side length L. All symmetries used below belong to the dihedral group D_6; applying a symmetry always transports the indices, orientations, and incoming and outgoing labels together.

2.1.1. *Open, closed, and scaled formulations.*

Proposition 2.1 (Compactness and scaling equivalence)**.** *The following assertions are equivalent.*

(1) *The set H is covered by seven open equilateral triangles of side length* 1*.*
(2) *For some $\varepsilon > 0$, the set H is covered by seven closed equilateral triangles of side length $1-\varepsilon$.*
(3) *For some $L > 1$, the set H_L is covered by seven closed equilateral triangles of side length* 1*.*

Proof. Suppose that open unit triangles $U_1, \dots, U_7$ cover H. Define

$$m_j(p) = \operatorname{dist}(p, \mathbb{R}^2 \setminus U_j), \qquad m(p) = \max_{1\le j\le 7} m_j(p).$$

The continuous function m is positive on the compact set H, so $\eta := \min_{p\in H} m(p) > 0$. Choose

$$0 < \rho < \min\left\{\eta, \frac{\sqrt{3}}{6}\right\}.$$

Moving each side of U_j inward through distance ρ gives a closed equilateral triangle K_j of side $1-2\sqrt{3}\rho$. Every $p \in H$ has $m_j(p) \ge \eta > \rho$ for some j, and hence belongs to K_j. This proves (1)$\Rightarrow$(2), with $\varepsilon = 2\sqrt{3}\rho$.

Conversely, enlarging a closed triangle of side $1-\varepsilon$ about its incenter to an open triangle of side 1 preserves containment, proving (2)$\Rightarrow$(1). Finally, dilation by $(1-\varepsilon)^{-1}$ identifies (2) with (3), with

$$L = (1-\varepsilon)^{-1} > 1.$$

□

We conduct the geometric argument in the open side-one model. Denote the original open triangles by U_C and U_i for $0 \le i \le 5$, and put

$$T_C = \overline{U_C}, \qquad T_i = \overline{U_i}.$$

The distinction between U_i and T_i will be maintained: closure may add endpoints or tangencies, although it does not change area or one-dimensional trace measure.

Lemma 2.2 (Distinct triangles)**.** *The seven triangles can be labelled so that*

$$O \in U_C, \qquad V_i \in U_i \quad (0 \le i \le 5),$$

and these seven triangles are distinct. Consequently

$$O \in \operatorname{int}(T_C), \qquad V_i \in \operatorname{int}(T_i).$$

Proof. The seven distinguished points $O, V_0, \dots, V_5$ are pairwise at distance at least 1. Any two points of an open unit equilateral triangle are at distance strictly less than its diameter 1. Thus one open triangle contains at most one distinguished point. Selecting a covering triangle for each of the seven points gives an injection from seven points to seven triangles, hence a bijection and the asserted labelling. □

2.1.2. *The center classification.* For a closed center triangle define

$$N_E(T_C) = |\{i : T_C \cap e_{i,i+1} \text{ contains a positive-length interval}\}|.$$

Point contacts, endpoint contacts, and tangencies do not contribute. We say that T_C has type CE0, CE1, or CE2 according as $N_E(T_C)$ is 0, 1, or 2.

Proposition 2.3 (Exhaustive center types)**.** *Every center triangle has exactly one of the types CE0, CE1, and CE2. Equivalently,*

$$N_E(T_C) \leq 2.$$

Proof. Among any three edges of the six-cycle ∂H, two have no common endpoint. It is therefore enough to show that relative-interior points of two nonadjacent edges are more than unit distance apart. Up to symmetry there are two cases. For $0 < t, s < 1$, put

$$x = (1-t)V_0 + tV_1.$$

If $y = (1-s)V_2 + sV_3$, direct calculation gives

$$\|x-y\|^2 - 1 = (1-t)(2-t) + s(s+t) > 0.$$

If instead $y = (1-s)V_3 + sV_4$ and $u = t+s$, then

$$\|x-y\|^2 - 1 = (u-1)^2 + 2 > 0.$$

A unit equilateral triangle has diameter 1. It therefore cannot contain relative-interior points on two nonadjacent hexagon edges. Three positive-length edge traces would supply such a pair, a contradiction. ☐

2.1.3. *Vertex types and the T3-like normalization.* For an original vertex triangle T_i, let

$$o(T_i) = |\{\text{vertices of } T_i \text{ outside } H\}|$$

and let $n(T_i)$ be the number of adjacent arms r_{i-1}, r_{i+1} having positive-length intersection with T_i. We use the names

type	(o, n)
Vd0	$(1,0), (2,0), (3,0)$
Vd1	$(1,1)$
Vd2	$(1,2)$
T3-like	$(2,1)$.

Lemma 2.4 (Local wedge)**.** *Every original vertex triangle has $1 \leq o \leq 3$. If it has positive-length intersection with one adjacent arm, at least one of its vertices belongs to H; if it has positive-length intersection with both adjacent arms, at least two of its vertices belong to H.*

Proof. Normalize to $i = 0$ and write

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0).$$

Then $V_0 = (0,0)$,

$$\|(x,y)\|^2 = x^2 + y^2 - xy,$$

and, within the closed unit ball about V_0, membership in H is equivalent to membership in the wedge

$$W = \{(x,y) : x \geq 0, y \geq 0\}.$$

Indeed,

$$x^2+y^2-xy=\Big(y-\frac{x}{2}\Big)^2+\frac{3x^2}{4}=(x-y)^2+xy$$

shows that $x,y\ge 0$ and norm at most one imply the remaining hexagon inequalities. The adjacent arms are

$$r_1=\{(1,y):0\le y\le 1\},\qquad r_5=\{(x,1):0\le x\le 1\}.$$

Suppose $T_0\cap r_1$ has positive length, and let m be the largest x-coordinate of a vertex of T_0. If $m>1$, a maximizing vertex (m,y) lies in the unit ball and hence satisfies $0<y<1$; it belongs to H. If $m=1$, positive-length contact with the supporting line $x=1$ forces a side of T_0 to lie there, and both endpoints satisfy $1+y^2-y\le 1$, hence belong to H. The same argument applies to r_5. If both arms are met but only one triangle vertex belonged to H, that vertex would have to have both $x>1$ and $y>1$. This is impossible because $(x-y)^2+xy>1$.

Finally, if all three triangle vertices belonged to the convex set H, then $T_0\subset H$. This is incompatible with the extreme point V_0 lying in the interior of T_0. Hence $o\ge 1$. □

Proposition 2.5 (Exhaustive vertex types)**.** *Every original vertex triangle has exactly one of the four types Vd0, Vd1, Vd2, and T3-like.*

Proof. We have $n\in\{0,1,2\}$. Lemma 2.4 gives $o\le 2$ when $n\ge 1$, and $o\le 1$ when $n=2$. Together with $o\ge 1$, the possibilities are

n	possible o
0	$1,2,3$
1	$1,2$
2	$1,$

which are precisely the four named classes. □

The next normalization is useful for trace estimates, but its final warning is essential.

Proposition 2.6 (T3-like closed-trace domination)**.** *Let T be the closure of an original T3-like V_i-triangle. There is a translate $\widehat{T}$ of T, with the same orientation and side length, such that*

$$V_i\in \operatorname{relint}(\text{a side of }\widehat{T}),\qquad T\cap H\subsetneq \widehat{T}\cap H,$$

and $\widehat{T}$ remains of type $(o,n)=(2,1)$. Every old open interior point in $H\setminus\{V_i\}$ remains an interior point of $\widehat{T}$. The point V_i itself becomes a boundary point, so $\widehat{T}$ is a closed-trace majorant and is not a replacement open triangle.

Proof. Use the wedge coordinates of Lemma 2.4. Set

$$D_t=1-t+t^2,\qquad z=\sqrt{D_t},\qquad 0\le t\le 1.$$

After relabelling triangle vertices, every orientation belongs to one of the following two families:

	first side vector	second side vector
I	$z^{-1}(1,t)$	$z^{-1}(1-t,1)$
II	$z^{-1}(t,-(1-t))$	$z^{-1}(1,t).$

For Type I write

$$U=\alpha+y-tx,\qquad V=\beta+x-(1-t)y,\qquad T=\{U\ge 0,V\ge 0,U+V\le z\}.$$

Here $\alpha, \beta > 0$ and $\alpha + \beta < z$. Direct inversion shows that if exactly two triangle vertices are outside W, the sole wedge vertex has both coordinates strictly below 1: in the two possible cases its coordinates are bounded respectively by $(1-t)/z$ and t/z, with the endpoint strictness supplied by $\alpha > 0$ or $\beta > 0$. The support argument in Lemma 2.4 then excludes positive-length contact with either $x = 1$ or $y = 1$. Thus Type I cannot be T3-like.

For Type II write

$$U = \alpha + (1-t)x + ty, \qquad V = \beta + tx - y, \qquad T = \{U \ge 0, V \ge 0, U + V \le z\}.$$

Reflecting in $x = y$ if necessary, assume the supported adjacent arm is r_1. The two outside vertices have respectively a negative x- and a negative y-coordinate, while the unique wedge vertex is

$$Q = \left(\frac{z - \alpha - t\beta}{D_t}, \frac{t(z-\alpha) + (1-t)\beta}{D_t} \right).$$

The endpoint cases $t = 0, 1$ give no positive adjacent interval, so $0 < t < 1$. The exact axis reaches are

$$A = \beta, \qquad B = z - \alpha - \beta,$$

and T3-like support is equivalent to

$$Q_x > 1, \qquad Q_y \le 1.$$

Define $\widehat{T}$ by replacing α with 0 and keeping t, β fixed. Since the linear map $(x, y) \mapsto (U, V)$ is nonsingular, this changes only the translation. At the origin, $\widehat{U} = 0$ and $0 < \widehat{V} = \beta < z$, so V_0 is in the relative interior of a side. For $(x, y) \in W$,

$$\widehat{U} = (1-t)x + ty \ge 0, \qquad \widehat{V} = V, \qquad \widehat{U} + \widehat{V} = U + V - \alpha.$$

Thus $T \cap W \subset \widehat{T} \cap W$. The old x-axis reach is B and the new one is $B + \alpha$; because $B < 1$ and $\alpha > 0$, the inclusion is strict inside H.

The two outside vertices remain outside, whereas V_0 and the new wedge vertex lie in H, so $o = 2$. Moreover $\widehat{Q}_x > Q_x > 1$. From $Q_x > 1$ one gets $t\beta < z - \alpha - D_t$, and hence

$$tD_t(1 - \widehat{Q}_y) > D_t(1 - z) + (1-t)\alpha > 0.$$

Thus $\widehat{Q}_y < 1$ and $n = 1$. Finally, for every nonzero point of W, $\widehat{U} > 0$; the displayed slack relations show that every old interior point other than the origin remains interior. □

2.1.4. *Reach coordinates, supercritical vertex triangles, and gaps.* For the original vertex triangle U_i, define its maximal incoming, outgoing, and radial reaches by

$$A(U_i) = \sup_{\substack{t \in [0,1] \\ V_i + t(V_{i-1} - V_i) \in U_i}} t,$$

$$B(U_i) = \sup_{\substack{t \in [0,1] \\ V_i + t(V_{i+1} - V_i) \in U_i}} t,$$

$$C(U_i) = \sup_{\substack{t \in [0,1] \\ V_i + t(O - V_i) \in U_i}} t.$$

Closure does not change the suprema, so

$$A(U_i) = A(T_i), \qquad B(U_i) = B(T_i), \qquad C(U_i) = C(T_i).$$

For compact indexed formulas, put

$$A_i := A(U_i) = A(T_i), \qquad B_i := B(U_i) = B(T_i), \qquad C_i := C(U_i) = C(T_i).$$

The closure T_i contains the corresponding endpoints. We reserve lowercase (a_i, b_i, c_i) for selected or prescribed reach lower bounds whenever actual and selected values occur together. A realizing triangle then satisfies

$$A_i \geq a_i, \qquad B_i \geq b_i, \qquad C_i \geq c_i.$$

An actual vertex triangle is supercritical if $A_i + B_i > 1$, and

$$N_+ = |\{i : A_i + B_i > 1\}| \,. \tag{2}$$

This definition never counts a smaller reach pair in place of the actual vertex triangle.

Parametrize $e_{i,i+1}$ by

$$X_i(x) = V_i + x(V_{i+1} - V_i), \qquad 0 \leq x \leq 1.$$

The two incident open traces are

$$U_i \cap e_{i,i+1} = [0, B_i), \qquad U_{i+1} \cap e_{i,i+1} = (1 - A_{i+1}, 1]$$

in this coordinate. A *boundary gap* on this edge is the nonempty closed complement between these two traces,

$$[B_i, 1 - A_{i+1}] \qquad \text{when } B_i \leq 1 - A_{i+1}.$$

In particular, equality gives a singleton gap. This convention retains the actual open traces; a point missing from both open triangles is not erased merely because both closures contain it.

Lemma 2.7 (Exhaustiveness of the gap split)**.** *On each edge the complement of the two incident vertex-triangle traces is either empty or one closed interval; a singleton interval counts as a gap. Every edge having a gap has positive-length intersection with the open center triangle. Consequently a* CE0 *center permits no gaps, a* CE1 *center permits at most one, and a* CE2 *center permits at most two.*

Proof. The two incident traces are intervals issuing from the two endpoints, so their complement is exactly the interval displayed above when nonempty. A nonincident vertex triangle cannot contain a relative-interior point of the edge: that point is more than unit distance from the triangle's distinguished vertex. Thus every point of a gap lies in U_C. Because U_C is open, its intersection with that edge contains a relative neighborhood and hence has positive length. Different gap edges therefore give different edges counted by $N_E(T_C)$. Proposition 2.3 gives the three stated bounds, including singleton gaps. □

Lemma 2.8 (T3-like vertex triangles are nonsupercritical)**.** *Every original T3-like vertex triangle satisfies*

$$A_i + B_i \leq 1. \tag{3}$$

In particular, it is not counted by N_+.

Proof. The proof of Proposition 2.6 showed that a T3-like triangle has Type II orientation with

$$A_i = \beta, \qquad B_i = z - \alpha - \beta, \qquad z = \sqrt{1 - t + t^2} \leq 1.$$

Thus $A_i + B_i = z - \alpha \leq 1$. The same conclusion holds after reflection. □

We state the precise passage from maximal reaches to strict handoff lower bounds. Its hypothesis that the six vertex triangles themselves cover the boundary is part of the statement.

Proposition 2.9 (Strict boundary handoffs)**.** *Assume that $U_0, \dots, U_5$ cover ∂H. On each edge choose*

$$\xi_i \in (1 - A_{i+1}, B_i)$$

and define

$$a_i = 1 - \xi_{i-1}, \qquad b_i = \xi_i.$$

Then $0 < a_i < A_i$, $0 < b_i < B_i$, the two selected anchors lie in U_i, and

$$a_i^2 + a_i b_i + b_i^2 < 1.$$

Moreover:

(1) *if exactly one actual vertex triangle, indexed by p, is supercritical, every strict selection has exactly one selected supercritical vertex triangle, also indexed by p;*
(2) *if at least two actual vertex triangles are supercritical, some simultaneous strict selection has at least two selected supercritical vertex triangles.*

Proof. Because V_i is interior to a diameter-one closed triangle,

$$0 < A_i < 1, \qquad 0 < B_i < 1.$$

No nonincident vertex triangle contains a relative-interior point of $e_{i,i+1}$, since such a point is more than unit distance from its distinguished vertex. The two displayed incident traces therefore cover the edge and, being relatively open and nonempty, overlap. Hence each interval $(1 - A_{i+1}, B_i)$ is nonempty. The anchor and distance assertions follow from the choice and from the fact that two points of U_i are at distance strictly less than one.

Put $\ell_i = 1 - A_{i+1}$ and $u_i = B_i$. Then

$$A_i + B_i > 1 \iff \ell_{i-1} < u_i, \qquad a_i + b_i > 1 \iff \xi_{i-1} < \xi_i. \tag{4}$$

If actual vertex triangle i is nonsupercritical, then

$$\xi_i < u_i \leq \ell_{i-1} < \xi_{i-1},$$

so every selected version of it is strictly subcritical. Also

$$\sum_{i=0}^{5} (a_i + b_i) = \sum_{i=0}^{5} (1 - \xi_{i-1} + \xi_i) = 6. \tag{5}$$

If p is the sole actual supercritical index, the other five selected sums are strictly below one; equation (5) forces the pth sum above one. This proves (1).

For (2), first take two nonadjacent supercritical indices p, q. For each $r \in \{p, q\}$ choose

$$\ell_{r-1} < z_r < u_r,$$

then choose

$$\xi_{r-1} \in (\ell_{r-1}, \min\{u_{r-1}, z_r\}), \quad \xi_r \in (\max\{\ell_r, z_r\}, u_r).$$

The two pairs of variables are disjoint and give ascents at p, q. Any set of at least three indices on a six-cycle contains two nonadjacent indices. It remains to consider exactly two adjacent supercritical indices $p, p{+}1$. For the other four nonsupercritical vertex triangles, successive inequalities $\ell_i < u_i \le \ell_{i-1}$ give

$$\ell_{p-1} < \ell_{p+1}.$$

Together with supercriticality at $p, p+1$, this implies

$$\max\{\ell_{p-1}, \ell_p\} < \min\{u_p, u_{p+1}\}.$$

Choose ξ_p between these bounds and then choose $\xi_{p-1} < \xi_p < \xi_{p+1}$ in their respective overlap intervals. By (4), both adjacent vertex triangles are selected supercritical. □

2.1.5. *Midpoint forcing.* We first need a local fact for vertex triangles. Take unit vectors

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \qquad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right).$$

Lemma 2.10 (Self-midpoint obstruction)**.** *Let a closed unit triangle contain*

$$0, \qquad au, \qquad bv, \qquad \frac{u+v}{2}.$$

Then $a + b \le 1$. Consequently, for every vertex triangle,

$$M_i \in T_i \quad \Longrightarrow \quad A_i + B_i \le 1.$$

If the two demanded boundary points and the midpoint are contained with a positive margin, the conclusion is strict.

Proof. Suppose $a + b > 1$ and set

$$K = \operatorname{conv}\left\{0, au, \frac{u+v}{2}, bv\right\}.$$

For outward unit normals n_0, n_1, n_2 separated by 120°, the least equilateral triangle with those normals and containing K has side length

$$\frac{2}{\sqrt{3}} \sum_{j=0}^{2} h_K(n_j).$$

The finite-caliper theorem, Theorem 6.1, reduces its minimum to an outward normal of one of the four edges of K.

For the edge from 0 to au, direct projection gives

$$S_{0A} = \max\left\{\frac{\sqrt{3}}{4}, \frac{\sqrt{3}a}{2}\right\} + \frac{\sqrt{3}b}{2} > \frac{\sqrt{3}}{2};$$

if $a \ge 1/2$ this is $(\sqrt{3}/2)(a{+}b)$, and if $a < 1/2$ then $b > 1/2$. The edge through bv is symmetric. For the edge from au to $(u{+}v)/2$, put $D_a = \sqrt{4a^2 - 2a + 1}$. Projection gives

$$S_{AM} = \begin{cases} \dfrac{\sqrt{3}(a+b)}{2D_a}, & a \le \frac{1}{2}, \\ \dfrac{\sqrt{3}(2a^2+b)}{2D_a}, & a > \frac{1}{2}. \end{cases}$$

The first value is strictly above $\sqrt{3}/2$ because $D_a \le 1$. In the second case, $b > 1 - a$ and

$$(2a^2 - a + 1)^2 - D_a^2 = a^2(2a-1)^2 \ge 0,$$

so again $S_{AM} > \sqrt{3}/2$. The fourth edge is symmetric. Every enclosing equilateral triangle therefore has side greater than one, a contradiction.

If equality $a+b=1$ occurred while all relevant points had positive margin, both boundary reach lower bounds could be increased slightly; the closed conclusion would then be violated. □

The corresponding center fact is exact and uses the interior condition at O.

Proposition 2.11 (Unique center midpoint)**.** *Let T_C be a closed unit equilateral triangle with $O \in \operatorname{int}(T_C)$. If T_C has positive-length intersection with a boundary edge of H, then it contains exactly one of $M_0, \dots, M_5$.*

Proof. Normalize the positive trace by a dihedral symmetry. Proposition 2.14 gives a single signed side-slack model for both CE1 and CE2 and proves directly that its midpoint set is $\{M_0\}$. Undoing the normalization proves the assertion. □

Remark 2.12. The hypothesis $O \in \operatorname{int}(T_C)$ cannot be weakened to $O \in T_C$: the triangle $\operatorname{conv}\{O, V_0, V_1\}$ has a boundary-edge overlap and contains both M_0 and M_1.

Proposition 2.13 (Exhaustive structural reduction)**.** *Every hypothetical cover by seven open unit equilateral triangles admits the distinct center and vertex triangles used in Table 1, and its maximal reaches determine exactly one vertex triangle of that table.*

Proof. Lemma 2.2 gives the seven triangles. Propositions 2.3 and 2.5 give the exhaustive and mutually exclusive center and vertex types. The maximal reaches determine exactly one of $N_+ = 0$, $N_+ = 1$, and $N_+ \geq 2$. For CE0 these data and the vertex classification give the five displayed vertex triangles.

Now suppose the center is CE1 or CE2. It contains exactly one radial midpoint by Proposition 2.11. Lemma 3.16 gives $q = N_+ + m$, and Lemma 3.5, together with the disjoint count, shows that $N_+ \geq 2$ forces a third short triangle. Hence every such branch either enters the first CE1/CE2 vertex triangle or has $q \leq 2$ and $N_+ \leq 1$. If $N_+ = 0$, the vertex classification gives, in order, all Vd0, at least one Vd1/Vd2, or no such triangle and at least one T3-like. If $N_+ = 1$, then $q \leq 2$ implies $m \leq 1$: the triangles are all Vd0, exactly one T3-like, or exactly one Vd1/Vd2. In the all-Vd0 case, Lemma 2.7 supplies the exhaustive zero-gap/nonempty-gap split. These alternatives are disjoint, proving the claim. □

2.2. **Signed CE1/CE2 Reductions.** This section gives one signed center model and the common one-gap vertex triangle interface. The transfer notation has already been fixed in Section 4.1: the exact nonsupercritical backward-reach map is $g_c^\vee$ whenever the radial lower bound exceeds $1/2$; for a lower radial lower bound the proof uses $B \leq 1 - A$ directly.

2.2.1. *One signed center normal form.*

Proposition 2.14 (Signed CE1/CE2 center normal form)**.** *Let T_C be a closed unit equilateral triangle with $O \in \operatorname{int}(T_C)$ and a positive-length boundary trace. Normalize such a trace to $e_{0,1}$ and use affine coordinates*

$$X = V_0 + b(V_1 - V_0) + a(V_5 - V_0).$$

There are parameters

$$0 < R < 1, \qquad W = 1 - R,$$

$$E = \sqrt{1 - RW}, \qquad \eta = 1 - E, \qquad P = E(1 - E),$$

and nonnegative center slacks α, δ such that, with

$$k = \eta + \alpha + \delta,$$

the center triangle is

$$\boxed{\begin{aligned} F_0 &= R + \alpha - a + Wb, \\ F_1 &= Rb + Wa - k, \\ F_2 &= W + \delta - b + Ra, \end{aligned}} \qquad T_C = \{F_0, F_1, F_2 \geq 0\}. \tag{6}$$

Define

$$\Delta_R = P - \alpha - W\delta, \qquad \Delta_L = P - R\alpha - \delta. \tag{7}$$

Then

$$T_C \cap e_{0,1} = \left[\frac{k}{R}, W + \delta\right], \qquad \mathcal{H}^1(T_C \cap e_{0,1}) = \frac{\Delta_R}{R}, \tag{8}$$

and the possible companion trace is

$$T_C \cap e_{5,0} = \left[\frac{k}{W}, R + \alpha\right] \quad \textit{when } \Delta_L > 0, \qquad \mathcal{H}^1(T_C \cap e_{5,0}) = \frac{[\Delta_L]_+}{W}. \tag{9}$$

The normalization requires $\Delta_R > 0$, and

$$T_C \textit{ is CE1} \iff \Delta_L \leq 0, \qquad T_C \textit{ is CE2} \iff \Delta_L > 0. \tag{10}$$

The six center exits, measured from O, are

$$\boxed{\begin{aligned} d_0^C &= E - \alpha - \delta, \qquad & d_1^C &= \frac{\delta}{R}, \quad & d_2^C &= \delta, \\ d_3^C &= \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\}, \quad & d_4^C &= \alpha, \quad & d_5^C &= \frac{\alpha}{W}. \end{aligned}} \tag{11}$$

For closures of original open center triangles one has $\alpha, \delta > 0$.

Proof. Write

$$T_C \cap e_{0,1} = [s, t], \qquad 0 \leq s < t \leq 1.$$

The two sides through these endpoints have a positive slope ratio. Reflect across the perpendicular bisector of $e_{0,1}$ if necessary and call the ratio $R \leq 1$; put $W = 1 - R$. Direct equations for the three side lines give

$$F_1 = Rb + Wa - Rs, \qquad F_2 = -b + Ra + t, \qquad F_0 = Wb - a + E + Rs - t,$$

where $E = \sqrt{1 - RW}$ is forced by the unit side-length relation. If $R = 1$, then $F_0(O) = s - t < 0$, contrary to $O \in \operatorname{int}(T_C)$; hence $0 < R < 1$.

Set

$$\alpha = F_0(O), \qquad \delta = F_2(O).$$

Then

$$t = W + \delta, \qquad Rs = \eta + \alpha + \delta = k,$$

and substitution gives (6). The gradients are the three unit-equilateral side-normal directions and

$$F_0 + F_1 + F_2 = E,$$

so the three inequalities define the same unit equilateral triangle.

On $e_{0,1}$ one has $a = 0$. The active inequalities are

$$Rb \geq k, \qquad b \leq W + \delta,$$

which gives (8); its length is

$$W + \delta - \frac{k}{R} = \frac{P - \alpha - W\delta}{R}.$$

On $e_{5,0}$ one has $b = 0$, and the active inequalities are

$$Wa \geq k, \qquad a \leq R + \alpha.$$

Their signed interval length is

$$R + \alpha - \frac{k}{W} = \frac{P - R\alpha - \delta}{W},$$

which proves (9).

The positive right trace gives

$$\delta < \frac{P}{W} = \frac{ER}{1+E} < \frac{R}{2}.$$

On $e_{1,2}$, parameterized by $b = 1 + q$, $a = q$, the slack F_2 is

$$\delta - R - Wq < 0.$$

Thus a second positive trace cannot occur on $e_{1,2}$. The CE classification says that a center triangle has at most two positive boundary traces and that two such traces are adjacent. Hence $e_{5,0}$ is the only possible companion edge, and its signed length proves (10). Equality $\Delta_L = 0$ is only a point contact and is therefore CE1.

For the radial exits, substitute the six ray parametrizations in (6). On r_1 the decreasing slack is $\delta - Rq$; on r_2 it is $\delta - q$; on r_3 the two decreasing slacks are $\alpha - Rq$ and $\delta - Wq$. Reflection gives r_4, r_5, and on r_0 the decreasing slack is $E - \alpha - \delta - q$. This proves (11).

Finally,

$$W(\alpha + \delta) \leq \alpha + W\delta < P,$$

so

$$E - \alpha - \delta > E - \frac{P}{W} = \frac{E(E-R)}{W} > \frac{1}{2},$$

where the last inequality is equivalent to $2E(E-R) - W = (E-R)^2 > 0$. Together with

$$\alpha < P < \min\left\{\frac{R}{2}, \frac{W}{2}\right\}, \qquad \delta < \frac{R}{2},$$

the midpoint tests give the exact midpoint set $\{M_0\}$. For the closure of an original open center triangle, all three center slacks are positive; hence $\alpha, \delta > 0$. □

Remark 2.15 (Legacy variables). For CE1, take

$$\lambda = R, \qquad s = \frac{k}{R}, \qquad t = W + \delta, \qquad C_0 = \alpha, \qquad C_2 = \delta.$$

For CE2, take

$$x = \frac{k}{W}, \qquad \nu = R + \alpha, \qquad y = \frac{k}{R}, \qquad v = W + \delta.$$

If $S = x + y$ and $D = \sqrt{x^2 + xy + y^2}$, then

$$S = \frac{k}{RW}, \qquad D = \frac{Ek}{RW},$$

and the old coupling equation is

$$(\nu + v)S - xy = D.$$

Thus the CE1 and CE2 formula lists are two coordinate readings of Proposition 2.14.

2.2.2. *Boundary contribution and diameter transfer.*

Corollary 2.16 (Common center-boundary formula)**.** *The full center-boundary contribution is*

$$L_{\partial H}(T_C) = \frac{\Delta_R}{R} + \frac{[\Delta_L]_+}{W}. \tag{12}$$

In CE1,

$$L_{\partial H}(T_C) \le P \le \frac{\sqrt{3}}{2} - \frac{3}{4}.$$

In CE2,

$$L_{\partial H}(T_C) = \frac{E}{1+E} - \frac{E^2(\alpha+\delta)}{RW} < \frac{1}{2}.$$

Proof. Equation (12) is the sum of (8) and (9). If $\Delta_L \le 0$, then $R\alpha + \delta \ge P$. Multiplying by W and using $\alpha \ge RW\alpha$ gives

$$\alpha + W\delta \ge WP.$$

Thus $\Delta_R \le RP$, so the contribution is at most P. Since $E \ge \sqrt{3}/2$ and $E(1-E)$ decreases on this interval, the CE1 cap follows.

If $\Delta_L > 0$, direct addition gives

$$\begin{aligned} L_{\partial H}(T_C) &= \frac{P - \alpha - W\delta}{R} + \frac{P - R\alpha - \delta}{W} \\ &= \frac{P - E^2(\alpha+\delta)}{RW}. \end{aligned}$$

Here $W + R^2 = W^2 + R = E^2$ and $P/(RW) = E/(1+E)$, so the last expression is strictly below $1/2$. ☐

Lemma 2.17 (Adjacent-edge diameter transfer)**.** *For* $0 \le q \le 1$*, put*

$$\beta(q) = \frac{-q + \sqrt{4 - 3q^2}}{2}.$$

If a unit equilateral triangle reaches parameters q, b *on two adjacent boundary edges, then*

$$b \le \beta(q).$$

For $q > 0$*,*

$$\beta(q) < 1 - \frac{q}{2}, \qquad 1 - \beta(q) > \frac{q}{2}.$$

Proof. The squared distance between the two boundary points is $q^2 + qb + b^2$, so it is at most one. Solving the quadratic in b gives the first assertion. For $q > 0$, $\sqrt{4 - 3q^2} < 2$, which gives both strict linear comparisons. ☐

2.2.3. *The local propagation interface.* For the high-radial parameters used below, the exact nonsupercritical reach transfer is the raw map $g_c^\vee$. It is nondecreasing and extensive:

$$g_c^\vee(z) \geq z. \tag{13}$$

The exact reflection duality is

$$g_c^\vee(a) \leq z \iff g_c^\vee(1-z) \leq 1-a. \tag{14}$$

Let a nonsupercritical vertex triangle have maximal reaches (A_i, B_i, C_i) satisfying

$$A_i \geq z, \qquad C_i \geq c_i > \frac{1}{2}.$$

Then its forward reach satisfies

$$B_i \leq g_{c_i}(1-z).$$

If the next boundary handoff gives $A_{i+1} \geq 1 - B_i$, then

$$\begin{aligned} A_{i+1} &\geq 1 - B_i \\ &\geq 1 - g_{c_i}(1-z) \\ &= g_{c_i}^\vee(z). \end{aligned} \tag{15}$$

Every composition below is shorthand for repeated applications of this inequality to specified actual vertex triangles.

2.2.4. *The common one-gap interface.*

Proposition 2.18 (Common five-vertex-triangle one-gap reduction)**.** *Assume every vertex triangle is Vd0, $N_+ = 1$, and the normalized right center trace contains a V-gap, possibly a singleton. In the CE2 case assume the companion trace contains no V-gap. Put*

$$X = R - \delta, \qquad H = \frac{k}{2R}, \qquad m_3 = \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\},$$

and

$$\begin{aligned} c_1 &= 1 - \frac{\delta}{R}, \quad c_2 = 1 - \delta, \qquad c_3 = 1 - m_3, \\ c_4 &= 1 - \alpha, \quad\; c_5 = 1 - \frac{\alpha}{W}, \quad c_0 = k. \end{aligned}$$

Let

$$z_0 = X, \qquad z_i = g_{c_i}^\vee(z_{i-1}) \quad (1 \leq i \leq 5),$$

and $Z = z_5$. Then the maximal reaches satisfy

$$A_0 \geq Z, \qquad A_0 \leq g_k\left(1 - \frac{k}{R}\right) < 1 - H. \tag{16}$$

Consequently it is enough to prove

$$[g_{1-\alpha}^\vee \mid g_{1-m_3}^\vee \mid g_{1-\delta}^\vee](H) > W + \delta. \tag{17}$$

Proof. The exact-midpoint argument makes vertex triangles $1, \ldots, 5$ nonsupercritical and vertex triangle 0 uniquely supercritical. Gap containment gives

$$B_0 \geq \frac{k}{R}, \qquad A_1 \geq X.$$

Suppose inductively that $A_i \ge z_{i-1}$. Radial coverage gives actual radial reach at least $c_i > 1/2$, and the raw high-radial bound gives

$$B_i \le g_{c_i}(1 - z_{i-1}).$$

The next boundary handoff therefore gives

$$A_{i+1} \ge 1 - B_i \ge g_{c_i}^{\vee}(z_{i-1}) = z_i.$$

For the final handoff, the CE1 companion trace is empty and the CE2 companion trace is gap-free, so $A_0 \ge z_5 = Z$.

The same vertex triangle has boundary reach at least k/R and radial reach at least k. Reflection and down-closedness of $\mathcal{A}$ give

$$\left(\frac{k}{R}, A_0, k\right) \in \mathcal{A},$$

so

$$A_0 \le g_k\left(1 - \frac{k}{R}\right).$$

Lemma 2.17 gives

$$g_k\left(1 - \frac{k}{R}\right) \le \beta\left(\frac{k}{R}\right) < 1 - \frac{k}{2R} = 1 - H,$$

which proves (16).

The first and fifth maps are extensive and the middle maps are nondecreasing, so it is enough to prove

$$[g_{c_2}^{\vee} \mid g_{c_3}^{\vee} \mid g_{c_4}^{\vee}](X) > 1 - H.$$

Three applications of (14) show that its negation is equivalent to the negation of (17). This proves the reduction. ☐

The CE1 sign $\Delta_L \le 0$ is completed by the selected-Q_+ relaxation in Proposition 4.27. We now give the shorter CE2 scalar clause.

Lemma 2.19 (CE2 total slack and two thresholds)**.** *Assume $\Delta_R > 0$ and $\Delta_L > 0$, and put*

$$T = \alpha + \delta, \qquad Q = \frac{\eta + T}{2R}.$$

Then

$$T < \frac{\eta}{E}, \tag{18}$$

and

$$\min\{e(\alpha), e(\delta)\} < Q. \tag{19}$$

Proof. The CE2 inequalities are

$$\alpha + W\delta < P, \qquad R\alpha + \delta < P.$$

Multiply the first by W, the second by R, and add. Since $W + R^2 = W^2 + R = E^2$, this gives

$$E^2 T < P = E\eta,$$

which is (18).

Let $d = \min\{\alpha, \delta\}$. Since $d \le T/2$ and the rational low-root bound is increasing,

$$\min\{e(\alpha), e(\delta)\} < \frac{2d}{1 - 2d} \le \frac{T}{1 - T}.$$

It remains to prove
$$\frac{T}{1-T} < \frac{\eta+T}{2R}.$$
Define
$$\Phi(t) = (\eta+t)(1-t) - 2Rt.$$
This function is concave, so its minimum on $[0, \eta/E]$ is at an endpoint. Clearly $\Phi(0) = \eta > 0$. Direct simplification using $E^2 = 1 - R + R^2$ gives
$$\Phi\left(\frac{\eta}{E}\right) = \frac{R(E-R)^2}{E^2} > 0.$$
Thus $\Phi(T) > 0$, proving (19). □

Proposition 2.20 (Short CE2 one-gap completion)**.** *Under the CE2 hypotheses of Proposition 2.18, the one-gap state is impossible in either orientation.*

Proof. The right-gap initial input is $z_0 = R - \delta$. The strict CE2 domain gives
$$Wz_0 = RW - W\delta = \eta + P - W\delta > \eta + \alpha.$$
For
$$\pi(q) = (\eta+q)(1-2q) - 2qW,$$
one has
$$\pi(0) = \eta > 0, \qquad \pi(P) = \eta(\eta + 2ER^2) > 0.$$
Concavity therefore gives $\pi(\alpha) > 0$, and the low-root bound yields
$$z_0 > \frac{\eta+\alpha}{W} > \frac{2\alpha}{1-2\alpha} > e(\alpha). \tag{20}$$

If $e(\alpha) < Q$, extensivity carries (20) to the vertex triangle-4 map $g^{\vee}_{1-\alpha}$, which gives an output at least $1 - e(\alpha) > 1 - Q$. All later maps preserve it. If $e(\alpha) \geq Q$, Lemma 2.19 gives $e(\delta) < Q$, and
$$z_0 > e(\alpha) \geq Q > e(\delta).$$
The vertex triangle-2 map $g^{\vee}_{1-\delta}$ then gives an output above $1 - Q$, again preserved by all later maps. Thus $Z > 1 - Q = 1 - H$, contradicting (16).

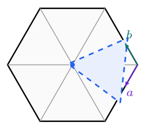
Reflection exchanges
$$R \leftrightarrow W, \qquad \alpha \leftrightarrow \delta$$
and reverses the vertex-triangle order. The signed CE2 domain and the proof above are invariant under this exchange, so the left-gap orientation is impossible as well. □

Remark 2.21 (Why the CE1 scalar clause remains separate)*.* The geometric one-gap interface is common, but (19) is not valid on the full CE1 sign domain. Thus the merger stops at Proposition 2.18; the CE1 selected-Q_+ clause and the CE2 threshold clause are genuinely different scalar proofs of the same target (17).

(e) signed parameters and affine center chart

the chart formulas are exact; parameter callouts are non-metric

$\chi : \mathbb{R}^2 \to \mathbb{R}^2$ as displayed, $\quad 0 < R < 1, \quad W := 1 - R$
$E := \sqrt{1 - RW}, \quad \eta := 1 - E, \quad P := E(1 - E), \quad k := \eta + \alpha + \delta$



$F_0, F_1, F_2 : \mathbb{R}^2 \to \mathbb{R},$
$F_0(a, b) := R + \alpha - a + Wb,$
$F_1(a, b) := Rb + Wa - k,$
$F_2(a, b) := W + \delta - b + Ra.$

$T_C(R, \alpha, \delta) := \{\chi(a, b) : F_0(a, b) \geq 0,\ F_1(a, b) \geq 0,\ F_2(a, b) \geq 0\}$
$O = \chi(1, 1)$, hence $F_0(1, 1) = \alpha,\ F_1(1, 1) = 1 - k,\ F_2(1, 1) = \delta.$

$\chi(a, b) = V_0 + b(V_1 - V_0) + a(V_5 - V_0)$

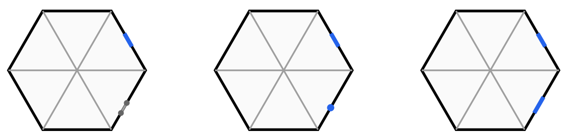
$\Delta_R := P - \alpha - W\delta, \qquad \Delta_L := P - R\alpha - \delta$

$\mathcal{D}_{\mathrm{sgn}} := \{(R, \alpha, \delta) \in \mathbb{R}^3 : 0 < R < 1,\ \alpha > 0,\ \delta > 0,\ \Delta_R > 0\}$

R is an affine slope parameter; only $\ell_R, \ell_L, d_i^C, c_i^C$ in the trace-and-exit panels are displayed as skeleton lengths.

FIGURE 2. The affine center chart and the exact dependency structure of the signed parameters.

(f) companion-surplus sign on $e_{5,0}$



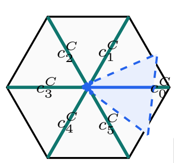
$\Delta_L < 0:$ CE1 $\qquad \Delta_L = 0:$ CE1 $\qquad \Delta_L > 0:$ CE2

candidate endpoint coordinates on $e_{5,0}$ from V_0: $k/W,\ R + \alpha$

$$\ell_L = \frac{[\Delta_L]_+}{W}, \qquad \ell_R = \frac{\Delta_R}{R} > 0 \text{ on the normalized active edge } e_{0,1}.$$

FIGURE 3. The exact sign test for the candidate companion trace, including the CE1 point-contact case.

(h) center exits and complementary radial reach lower bounds



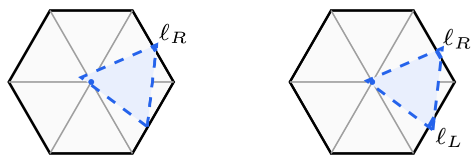
$d_0^C = E - \alpha - \delta$
$d_1^C = \delta/R$
$d_2^C = \delta$
$d_3^C = \min\{\alpha/R, \delta/W\}$
$d_4^C = \alpha$
$d_5^C = \alpha/W$

$c_i^C := 1 - d_i^C$
d_i^C: measured from O
c_i^C: measured from V_i

CE2 sample: $(d_0^C, \ldots, d_5^C) \approx (.8018, .0667, .04, .05, .03, .075)$.

FIGURE 4. The six exact center exits d_i^C and complementary radial lower bounds $c_i^C = 1 - d_i^C$.

(i) boundary-contribution bounds on the skeleton



$$\text{CE1}: \quad L_{\partial H}(T_C) = \ell_R \leq P \leq \frac{\sqrt{3}}{2} - \frac{3}{4}$$

$$\text{CE2}: \quad L_{\partial H}(T_C) = \ell_R + \ell_L < \frac{1}{2}$$

FIGURE 5. The exact CE1 and CE2 center-boundary contributions and their general bounds.

3. Trace-Length Bounds

The first mechanism compares the total one-dimensional trace available from the seven triangles with the length of either the perimeter ∂H or the full skeleton S. The local trace estimates, the master perimeter deficit, the conditional skeleton theorem, and every routed application are proved in this section.

3.1. **Detailed Trace-Length Estimates.** This section develops the boundary and conditional-skeleton trace bounds used by the length mechanism.

For a closed triangle T and a rectifiable target E, write

$$L_E(T) = \mathcal{H}^1(T \cap E).$$

The passage from an original open triangle U to its closure $T = \overline{U}$ can only enlarge the trace. Hence a cover of E by the open triangles implies

$$\mathcal{H}^1(E) \leq L_E(T_C) + \sum_{i=0}^{5} L_E(T_i). \tag{21}$$

All estimates in this section concern full traces. They therefore remain valid for any assigned measurable subsets of those traces.

The strategy uses only the perimeter ∂H and the full skeleton S, whose measures were recorded in (1). Every skeleton estimate below is conditional on its stated center and vertex triangle hypotheses; no unconditional noncoverage assertion for S is made.

∂H: **length** 6 $\qquad$ S: **length** 12

the full-skeleton estimate is used only under its stated branch hypotheses

FIGURE 6. The two one-dimensional targets. The full skeleton is used only under the hypotheses of the corresponding trace budget.

3.1.1. *Conditional skeleton trace caps.* We now develop the skeleton estimate used only when the center is CE1 or CE2. The first lemma is the center contribution.

Lemma 3.1 (Center skeleton cap)**.** *If T_C is a CE1 or CE2 center triangle and contains exactly one radial midpoint, then*

$$L_S(T_C) < \frac{3}{2}.$$

Proof. Normalize the unique midpoint as M_0 and the positive boundary trace to $e_{0,1}$. Use the signed variables of Proposition 2.14; in particular

$$W = 1 - R, \qquad E = \sqrt{1 - RW}, \qquad \eta = 1 - E, \qquad P = E(1 - E).$$

The positive trace condition is

$$\alpha + W\delta < P. \tag{22}$$

The two possible boundary contributions are

$$\ell_{01} = W + \delta - \frac{\eta + \alpha + \delta}{R}, \qquad \ell_{50} = \frac{[P - (R\alpha + \delta)]_+}{W}.$$

The six radial contributions are bounded by

i	0	1	2	3	4	5
$\mathcal{H}^1(T_C \cap r_i)$	$E - \alpha - \delta$	δ/R	δ	$\min\{\alpha/R, \delta/W\}$	α	$\alpha/W.$

Adding gives

$$L_S(T_C) \le K_R + \mathcal{E}(\alpha, \delta),$$

where

$$K_R = W + E - \frac{\eta}{R}$$

and

$$\mathcal{E}(\alpha, \delta) = -\frac{\alpha}{R} + \frac{\alpha}{W} + \delta + \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\} + \frac{[P - (R\alpha + \delta)]_+}{W}.$$

Splitting only according to the minimum and the positive part gives

$$\mathcal{E}(\alpha, \delta) \le \frac{P}{W}. \tag{23}$$

For example, if $\alpha/R \le \delta/W$, the expression without the positive part is $\alpha/W + \delta$; when the positive part is active, its excess over P/W is $\alpha - R\delta/W \le 0$. The reflected order gives excess $\delta - W\alpha/R \le 0$. If the positive part vanishes, (22) gives the same bound strictly.

Put $A = 1 + R - R^2$. Then

$$\frac{3}{2} - K_R - \frac{P}{W} = \frac{1 + A - 2EA}{2RW}.$$

Both sides of $1 + A > 2EA$ are positive, and

$$(1 + A)^2 - 4A^2E^2 = R^2W^2(5 + 4R - 4R^2) > 0.$$

Thus $K_R + P/W < 3/2$. □

Lemma 3.2 (Positive-support skeleton cap)**.** *Let T_i be the closure of an original open vertex triangle, so $V_i \in \operatorname{int}(T_i)$. If T_i has positive-length intersection with an adjacent radial arm, then*

$$L_S(T_i) < \frac{3}{2}.$$

Proof. Normalize to $i = 0$ and translate the hexagon by V_0:

$$\widetilde{H} = V_0 + H, \qquad \widetilde{O} = V_0, \qquad W_j = V_0 + V_j.$$

Then $W_3 = O$, $W_2 = V_1$, and $W_4 = V_5$. The five original skeleton components on which a unit triangle containing V_0 can have positive-length trace are

$$[V_0, V_1],\ [V_0, V_5],\ [O, V_1],\ [O, V_0],\ [O, V_5].$$

All five belong to the skeleton $\widetilde{S}$ of $\widetilde{H}$. Since $V_0 = \widetilde{O}$ is interior to T_0, the triangle is a legitimate center triangle for $\widetilde{H}$. Its positive adjacent-arm trace is a positive boundary-edge trace of $\widetilde{H}$, so it is CE1 or CE2 by Proposition 2.3. Proposition 2.11 and Lemma 3.1 therefore give

$$L_{\widetilde{S}}(T_0) < \frac{3}{2}.$$

A nonlocal component of S is more than unit distance from the interior point V_0, apart from zero-measure endpoints. Hence $L_S(T_0) \le L_{\widetilde{S}}(T_0)$. □

Lemma 3.3 (No-support skeleton cap)**.** *If a vertex triangle has $A_i + B_i \le 1$ and no positive-length trace on either adjacent radial arm, then*

$$L_S(T_i) \le 2.$$

Proof. Its boundary contribution is $A_i + B_i \le 1$ by (26). For $P = sV_j \in r_j \setminus \{O\}$,

$$\|V_i - P\|^2 = 1 + s^2 - 2s\cos\frac{(j-i)\pi}{3} > 1$$

when $j \notin \{i{-}1, i, i{+}1\}$. The two adjacent arms have zero-length trace by hypothesis, so only the own unit arm can contribute positive length. This adds at most one. □

For the remaining cap we use one specialized part of the exact local equilateral enclosure criterion. In local directions u, v meeting at $120°$, suppose a closed unit triangle contains

$$0, \quad au, \quad bv, \quad c(u + v),$$

with $0 \le a \le b \le 1$ and $a + b > 1$. The support-line calculation gives the selected supercritical cell

$$\begin{aligned} &a^2 + ab + b^2 \le 1, \qquad c \le \frac{1}{2}, \\ &(a^2 - 1)c^2 + (2ab^2 + b)c + (b^4 - b^2) \le 0, \end{aligned} \tag{24}$$

where c lies on the connected component containing $c = 0$. To see the selector directly, the quadratic is negative at $c = 0$, positive at $c = 1/2$, and opens downward. For the middle assertion, four times its value at $1/2$ is increasing in a and, at $a = 1 - b$, equals $b^2(2b - 1)^2$; it is therefore positive when $a + b > 1$. Hence $c \le 1/2$

selects its smaller-root component. Formula (24) follows by fixing in the support sum the three active contacts $\{B\}$, $\{B, c(u+v)\}$, and $\{A\}$, and squaring the positive unit-side equation.

Lemma 3.4 (Supercritical skeleton cap)**.** *If $A_i + B_i > 1$, then*

$$L_S(T_i) < \frac{3}{2}.$$

Proof. By Proposition 2.5, a vertex triangle is Vd0, Vd1, Vd2, or T3-like. Proposition 3.6 makes every Vd1/Vd2 vertex triangle nonsupercritical, and Lemma 2.8 does the same for T3-like vertex triangles. Hence a supercritical triangle is Vd0 and, by definition, has no positive-length trace on either adjacent arm. Thus, after normalizing and writing $a = A_i$, $b = B_i$, the skeleton trace has length $a + b + c$, where c is the own-arm reach. Reflect so that $a \le b$, and put

$$s = a + b, \qquad \delta = b - a, \qquad c_0 = \frac{3}{2} - s.$$

The diameter constraint gives

$$1 < s \le \frac{2}{\sqrt{3}}, \qquad 0 \le \delta \le \sqrt{4 - 3s^2}.$$

Let $P(c)$ denote the quadratic in (24). Substitution and multiplication by 16 give

$$\begin{aligned} 16P(c_0) = Q(s,\delta) = {} & \delta^4 + 8\delta^3 s - 6\delta^3 + 14\delta^2 s^2 - 18\delta^2 s + 5\delta^2 \\ & - 8\delta s^3 + 30\delta s^2 - 34\delta s + 12\delta \\ & + s^4 - 6s^3 - 19s^2 + 60s - 36. \end{aligned}$$

Its derivative factors as

$$\frac{\partial Q}{\partial \delta} = 2(2\delta + 4s - 3)\big(\delta^2 + \delta(4s - 3) + (s - 1)(2 - s)\big) \ge 0.$$

Therefore

$$Q(s,\delta) \ge Q(s,0) = (s - 1)(s^3 - 5s^2 - 24s + 36) > 0.$$

Indeed, the cubic is decreasing on $[1, 2/\sqrt{3}]$ and at its right endpoint equals $88/3 - 136\sqrt{3}/9 > 0$. Hence $P(c_0) > 0$. Since (24) selects the smaller-root component, $P(c) \le 0$ forces $c < c_0$. Thus $a + b + c < 3/2$. □

Lemma 3.5 (A positive-support adjacent exceptional triangle is forced)**.** *Suppose T_C is CE1 or CE2 and, after symmetry, contains exactly M_0. If at least two vertex triangles are supercritical, then a vertex triangle distinct from a chosen supercritical triangle has positive-length adjacent-arm support.*

Proof. Choose a supercritical index $s \ne 0$. Lemma 2.10 gives $M_s \notin T_s$, and $M_s \notin T_C$ by the unique-midpoint property. In the original open cover, M_s is therefore contained by some other open vertex triangle. Diameter one restricts this triangle to U_{s-1} or U_{s+1}. Because it contains an interior point of the adjacent arm, its intersection with that arm has positive length. □

3.1.2. *Vd1/Vd2 corner estimates.* We next isolate the much stronger trace loss caused by Vd1 and Vd2 geometry. For the following two results use the temporary coordinates

$$X = V_0 + x(V_5 - V_0) + y(V_1 - V_0).$$

Thus $V_0 = (0,0)$, $V_5 = (1,0)$, $V_1 = (0,1)$, $O = (1,1)$, and $\|(x,y)\|^2 = x^2 + y^2 - xy$.

Proposition 3.6 (Vd1/Vd2 corner normal form)**.** *Let T be the closure of an original Vd1 or Vd2 triangle at V_0, and let a, b be its exact reaches toward V_5, V_1. There is a unique $t > 0$ such that, with $d = \sqrt{t^2+t+1}$,*

$$T = \left\{ (x,y) : \begin{array}{r} x - (t+1)y \le a, \\ ty - (t+1)x \le tb, \\ tx + y \le d - a - tb \end{array} \right\}.$$

The raw traces, in coordinates measured from the indicated outer vertex, are

$$\begin{array}{c|c} r_0 & 0 \le s \le \dfrac{d-a-tb}{t+1} \\ r_1 & \dfrac{t(1-b)}{t+1} \le s \le \dfrac{d-a-tb-1}{t} \\ r_5 & \dfrac{1-a}{t+1} \le s \le d-a-tb-t. \end{array}$$

Intersecting with $[0,1]$ *gives the actual traces. Moreover,*

$$a + b < \frac{1}{2}. \tag{25}$$

Proof. A Vd1/Vd2 triangle has one vertex W outside H and two vertices U, Z in H. Since V_0 is an interior convex combination of them, both coordinates of W are negative. The two axis endpoints $(a,0), (0,b)$ lie on distinct sides incident with W. Put $p = U - W$, $q = Z - W$. Both vectors have positive coordinates and satisfy

$$\|p\| = \|q\| = \|p - q\| = 1.$$

The order in which the two sides cross the positive axes selects the $60°$ rotation

$$q = (p_x - p_y, p_x), \qquad p_x > p_y > 0.$$

Writing $t = (p_x - p_y)/p_y$ gives uniquely

$$p = \frac{(t+1,1)}{d}, \qquad q = \frac{(t,t+1)}{d}.$$

The two side lines through the axis endpoints and the parallel third side are exactly those in (3.6). Substitution of $(s,s), (s,1), (1,s)$ gives (3.6).

If the r_1 trace has positive length, its lower endpoint is smaller than its upper endpoint. Clearing denominators gives

$$(t+1)a + tb < (t+1)(d-1) - t^2.$$

Since the left side is at least $t(a+b)$,

$$a + b < \frac{(t+1)(d-1) - t^2}{t} < \frac{1}{2}.$$

For the last inequality, both sides are positive and

$$\left(t^2 + \frac{3t}{2} + 1\right)^2 - (t+1)^2(t^2+t+1) = \frac{t^2}{4} > 0.$$

Reflection proves the same conclusion when the supported arm is r_5. $\square$

Lemma 3.7 (Vd2 neighboring-midpoint cap)**.** *If an original Vd2 triangle contains either neighboring midpoint, then its exact boundary reaches satisfy*

$$a+b<\frac{1}{3}.$$

Proof. In Proposition 3.6, reflection sends (t,a,b,M_1,M_5) to $(1/t,b,a,M_5,M_1)$, so assume $t\ge 1$ and put $U=a+tb$.

If $M_5=(1,1/2)$ is contained, the third half-plane in (3.6) gives

$$a+b\le U\le d-t-\frac{1}{2}\le\sqrt{3}-\frac{3}{2}<\frac{1}{3};$$

the middle expression decreases for $t\ge 1$.

If $M_1=(1/2,1)$ is contained, the second and third half-planes give

$$b\ge\frac{t-1}{2t},\qquad a+b\le S(t):=d-t-\frac{1}{2t}.$$

Positive-length contact with the other adjacent arm r_5 gives, by (3.6),

$$a+b<R(t):=\frac{2(t+1)d-3t^2-t-2}{2t}.$$

Direct differentiation shows that S is increasing and R decreasing on $[1,\infty)$. For example, the sign calculation for S' reduces to

$$(2t+1)^2t^4-(t^2+t+1)(2t^2-1)^2=(t+1)(t^3+3t^2-1)>0,$$

while $R'<0$ follows from $d>t+1/2$. Splitting at $t=4/3$ yields

$$\begin{aligned} 1\le t\le\frac{4}{3}:&\quad a+b\le S(4/3)=\frac{8\sqrt{37}-41}{24}<\frac{1}{3},\\ t\ge\frac{4}{3}:&\quad a+b<R(4/3)=\frac{7\sqrt{37}-39}{12}<\frac{1}{3}. \end{aligned}$$

The final inequalities follow respectively from $8\sqrt{37}<49$ and $7\sqrt{37}<43$. $\square$

3.1.3. *Boundary trace caps.* The signed center normal form gives the CE1 and CE2 center contributions from one formula; no separate maximal-interval calculation is needed here.

Theorem 3.8 (Boundary trace table)**.** *For closures of original open triangles, the following bounds hold:*

triangle	*hypothesis*	$L_{\partial H}$
T_C	CE0	0
T_C	CE1	$\le\frac{\sqrt{3}}{2}-\frac{3}{4}$
T_C	CE2	$<\frac{1}{2}$
T_i	Vd0, $A_i+B_i\le 1$	≤ 1
T_i	Vd0, $A_i+B_i>1$	$\le\frac{2}{\sqrt{3}}$
T_i	Vd1/Vd2	$<\frac{1}{2}$
T_i	*T3-like*	$<1.$

Proof. For every vertex triangle,

$$L_{\partial H}(T_i) = A_i + B_i. \tag{26}$$

Indeed, the two incident traces are initial intervals of these lengths. A relative-interior point of any nonincident edge is more than unit distance from V_i, whereas V_i is interior to the diameter-one triangle T_i. This excludes any other positive-length boundary trace.

The CE0 vertex triangle is the definition. Both remaining center vertex triangles are the two class specializations of the common signed formula in Corollary 2.16.

For a nonsupercritical Vd0 vertex triangle, (26) is enough. For a supercritical vertex triangle, its incident endpoints are at squared distance $A_i^2 + A_iB_i + B_i^2 \le 1$, and hence

$$\frac{3}{4}(A_i + B_i)^2 \le 1.$$

This proves the $2/\sqrt{3}$ cap. The Vd1/Vd2 vertex triangle is (25). Finally, the Type II calculation in Proposition 2.6 gives for an original T3-like triangle

$$A_i + B_i = z - \alpha < z < 1$$

because $0 < t < 1$ and $\alpha > 0$. □

3.1.4. *Final length contradictions.*

Proposition 3.9 (Boundary-length branches)**.** *The following branches are impossible:*

(1) *CE0 with $N_+ = 0$;*
(2) *CE0 with $N_+ = 1$ and at least one Vd1/Vd2 triangle;*
(3) *CE1 or CE2 with $N_+ = 0$ and at least one Vd1/Vd2 triangle;*
(4) *CE1 with $N_+ = 1$ and at least one Vd1/Vd2 triangle;*
(5) *CE2 with $N_+ = 1$ and at least two Vd1/Vd2 triangles.*

Proof. In CE0, the center open triangle contains no relative-interior boundary point. No nonincident vertex triangle can contain such a point either. Hence on every edge the two incident open vertex traces form two nonempty relatively open sets covering a connected interval; they overlap in positive length. Thus

$$1 < B_i + A_{i+1}$$

on every edge. Summing gives $6 < \sum_i (A_i + B_i)$, contradicting $N_+ = 0$. This proves (1).

Items (2)–(5) are respectively items (1)–(4) of the common budget Corollary 3.17. That corollary applies the one master perimeter deficit to the center cap and the indicated strict Vd1/Vd2 caps. □

Proposition 3.10 (Vd2 neighboring-midpoint branch)**.** *Assume a CE2, $N_+ = 1$ configuration has exactly one Vd1/Vd2 triangle, that triangle is Vd2 and contains a neighboring midpoint, and all other vertex triangles are Vd0. Then the boundary cannot be covered.*

Proof. This is item (5) of Corollary 3.17, using Lemma 3.7. □

Proposition 3.11 (Branches closed by Method 1)**.** *Direct length-sum obstructions close the following routing entries and hybrid subcase:*

(1) *CE0, $N_+ = 0$;*

(2) *CE0, $N_+ = 1$, with a Vd1/Vd2 triangle;*
(3) *CE1/CE2 with $q = N_+ + m \geq 3$;*
(4) *CE1/CE2, $N_+ = 0$, with a Vd1/Vd2 triangle;*
(5) *CE1, $N_+ = 1$, with exactly one Vd1/Vd2 triangle and no T3-like triangle;*
(6) *the Vd2 neighboring-midpoint subcase of the CE2 hybrid routing entry.*

Proof. Items (1), (2), (4), and (5) are Proposition 3.9. Item (3) is the short-triangle count and the three-short-triangle theorem in Corollary 3.17. Item (6) is Proposition 3.10. This is only the indicated subcase of the exceptional CE2 vertex triangle; its remaining placements are handled by the boundary-reach argument. □

3.2. **Common perimeter and skeleton budgets.** The signed center formula also removes repeated length arithmetic. We first state a generic perimeter deficit, then count all short triangles before entering the CE1/CE2 case tree.

Lemma 3.12 (Master perimeter deficit)**.** *Put*

$$\theta = \frac{2}{\sqrt{3}} - 1.$$

Suppose a branch has a center contribution L_C, n supercritical Vd0 vertex triangles, $m \geq 1$ distinguished nonsupercritical vertex triangles with strict boundary caps $q_1, \ldots, q_m < 1$, and $6 - n - m$ remaining vertex triangles whose boundary contributions are at most 1. If

$$L_C + n\theta \leq \sum_{j=1}^{m} (1 - q_j), \tag{27}$$

then the seven triangles do not cover ∂H.

Proof. The supercritical Vd0 cap is $2/\sqrt{3}$. Hence the total boundary contribution is strictly less than

$$\begin{aligned} L_C + n\frac{2}{\sqrt{3}} + \sum_{j=1}^{m} q_j + (6 - n - m) &= 6 + L_C + n\theta - \sum_{j=1}^{m} (1 - q_j) \\ &\leq 6. \end{aligned}$$

The strictness follows from $m \geq 1$ and the strict distinguished-vertex triangle caps. A cover of the perimeter would give the opposite weak inequality by subadditivity. □

Lemma 3.13 (Small center slacks in the one-Vd1/Vd2 branch)**.** *Assume there is exactly one supercritical Vd0 vertex triangle, exactly one Vd1/Vd2 vertex triangle, and four nonsupercritical Vd0 vertex triangles. Then every candidate surviving the perimeter budget is CE2 and satisfies*

$$\boxed{\alpha + \delta < \frac{1}{24}, \qquad \alpha + \delta < \frac{\min\{R, W\}}{6}.} \tag{28}$$

Proof. The Vd1/Vd2 boundary cap is strict and equals 1/2. If the perimeter were covered, Lemma 3.12 would force

$$L_C + \theta > \frac{1}{2}.$$

Put
$$\kappa = \frac{1}{2} - \theta = \frac{3}{2} - \frac{2}{\sqrt{3}}.$$
Thus $L_C > \kappa$. The CE1 cap in Corollary 2.16 gives
$$L_C + \theta \le \left(\frac{\sqrt{3}}{2} - \frac{3}{4}\right) + \left(\frac{2}{\sqrt{3}} - 1\right) < \frac{1}{2},$$
so every survivor is CE2.

Set $T = \alpha + \delta$. The CE2 formula gives
$$T < M(E) := \frac{RW}{E^2}\left(\frac{E}{1+E} - \kappa\right).$$
Since $RW = 1 - E^2$,
$$M(E) = \frac{1-E}{E^2}\big(E - \kappa(1+E)\big),$$
and
$$M'(E) = -\frac{E - 2\kappa}{E^3} < 0 \qquad \left(\frac{\sqrt{3}}{2} \le E < 1\right).$$
Here $2\kappa < \sqrt{3}/2$, because it is equivalent to $6\sqrt{3} < 11$, whose square is $108 < 121$. Therefore
$$T < M\left(\frac{\sqrt{3}}{2}\right) = \frac{8\sqrt{3}}{9} - \frac{3}{2} < \frac{1}{24}.$$
The last inequality is equivalent to $64\sqrt{3} < 111$, and follows after squaring from $12288 < 12321$.

For the second estimate, use $E/(1+E) < 1/2$:
$$T < \frac{RW}{E^2}\left(\frac{1}{2} - \kappa\right) = \frac{RW}{E^2}\theta.$$
Since
$$E^2 = W + R^2 = R + W^2,$$
one has $RW/E^2 \le R$ and $RW/E^2 \le W$. Also $\theta < 1/6$, equivalently $12 < 7\sqrt{3}$, whose square is $144 < 147$. This proves (28). □

Lemma 3.14 (CE2 initial endpoints dominate the total slack)**.** *Assume* $\Delta_R > 0$ *and* $\Delta_L > 0$. *Put*
$$T = \alpha + \delta, \qquad x = \frac{\eta + T}{W}, \qquad y = \frac{\eta + T}{R}.$$
Then
$$\boxed{T < \frac{1}{2}\min\{x, y\}.} \tag{29}$$

Proof. Equation (18) gives $T < \eta/E$. Moreover
$$E^2 - (2R-1)^2 = 3RW > 0,$$
so $|2R-1| < E$ and therefore $|2R-1|T < \eta$. Hence
$$x - 2T = \frac{\eta - (1-2R)T}{W} > 0,$$

and

$$y - 2T = \frac{\eta - (2R-1)T}{R} > 0.$$

This is (29). ☐

Theorem 3.15 (Three short vertex triangles)**.** *Call a vertex triangle short if it is supercritical or has positive-length intersection with an adjacent radial arm. If an exact-M_0 CE1/CE2 branch contains at least three short vertex triangles, then the seven triangles do not cover the full skeleton S.*

Proof. The center skeleton cap is

$$L_S(T_C) < \frac{3}{2}.$$

Every short triangle also has skeleton contribution strictly less than 3/2: this is the supercritical cap for a supercritical triangle and the translated center-triangle cap for an original positive-support triangle. Every remaining triangle is nonsupercritical and has no adjacent-ray support, so its skeleton contribution is at most 2.

If there are $q \geq 3$ short triangles, then

$$\begin{aligned} L_S(T_C) + \sum_{i=0}^{5} L_S(T_i) &< \frac{3}{2} + q\frac{3}{2} + (6-q)2 \\ &= \frac{27-q}{2} \\ &\leq 12. \end{aligned}$$

The full skeleton has length 12, so subadditivity for a cover gives a contradiction. ☐

Lemma 3.16 (Short-triangle count)**.** *Let m be the number of vertex triangles of type Vd1, Vd2, or T3-like, and let q be the number of short vertex triangles. Then*

$$\boxed{q = N_+ + m.} \tag{30}$$

Consequently, every exact-M_0 CE1/CE2 branch with $N_+ + m \geq 3$ is impossible.

Proof. The exhaustive vertex classification says that Vd0 is exactly the class with no positive-length adjacent-arm support, while Vd1, Vd2, and T3-like are exactly the three classes with such support. Proposition 3.6 gives $A_i + B_i < 1/2$ for Vd1/Vd2, and Lemma 2.8 gives $A_i + B_i \leq 1$ for T3-like. Thus none of these m triangles is supercritical. Conversely, every supercritical triangle is Vd0, so it has no positive adjacent-arm support. The two disjoint sources of short triangles are therefore the N_+ supercritical triangles and the m non-Vd0 triangles. This proves (30); Theorem 3.15 gives the last assertion. ☐

Corollary 3.17 (Length branches closed by the common budgets)**.** *The following branches are impossible.*

1. *CE0, $N_+ = 1$, and at least one Vd1/Vd2 vertex triangle;*
2. *$N_+ = 0$ and at least one Vd1/Vd2 vertex triangle, for either CE1 or CE2;*
3. *CE1, $N_+ = 1$, and at least one Vd1/Vd2 vertex triangle;*
4. *CE2, $N_+ = 1$, and at least two Vd1/Vd2 vertex triangles;*
5. *CE2, $N_+ = 1$, with a Vd2 vertex triangle containing a neighboring midpoint;*

(6) *CE1 or CE2 with* $N_+ + m \geq 3$, *where* m *is the number of Vd1, Vd2, or T3-like triangles;*
(7) *CE1 or CE2 with* $N_+ \geq 2$.

Proof. For item 1, take $L_C = 0$, one supercritical cap $2/\sqrt{3}$, and one strict Vd1/Vd2 cap $1/2$ in Lemma 3.12; the needed inequality is

$$\frac{2}{\sqrt{3}} - 1 < \frac{1}{2}.$$

For item 2, the center contribution is strictly less than $1/2$ and the chosen Vd1/Vd2 vertex triangle has strict cap $1/2$; the other five vertex triangles contribute at most 1 each. This is Lemma 3.12 with $n = 0$ and $q_1 = 1/2$.

For item 3, the unique supercritical vertex triangle is Vd0, while the CE1 center cap, the Vd1/Vd2 cap, and the four remaining unit caps give

$$\left(\frac{\sqrt{3}}{2} - \frac{3}{4}\right) + \frac{1}{2} + \frac{2}{\sqrt{3}} + 4 < 6.$$

For item 4, the corresponding CE2 sum is strictly less than

$$\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{2}{\sqrt{3}} + 3 < 6.$$

For item 5, the neighboring-midpoint Vd2 cap is $1/3$, and the total is strictly less than

$$\frac{1}{2} + \frac{1}{3} + \frac{2}{\sqrt{3}} + 4 < 6.$$

Item 6 is Lemma 3.16 and Theorem 3.15.

Finally suppose $N_+ \geq 2$. Choose a supercritical index $s \neq 0$. Its own midpoint M_s is not covered by that vertex triangle, and the exact-M_0 center does not cover it. Diameter locality leaves only vertex triangles $s-1, s, s+1$ as possible vertex triangles containing M_s. Thus one of the adjacent vertex triangles contains M_s and, because it is the closure of an original open triangle, has positive-length support on the adjacent radial arm. By Lemma 3.16, this adjacent exceptional triangle is nonsupercritical and distinct from the two supercritical vertex triangles. Thus there are three short triangles, and the theorem applies. □

4. Boundary-Reach Propagation

The second mechanism converts a backward boundary reach and a radial reach into an upper bound for the forward reach of the same vertex triangle. Coverage of the next edge then produces a lower bound for the next backward reach. This section develops the common equilateral-enclosure calculus, the center-assisted path budget, the exceptional T3-like and Vd placements, and the complete branchwise propagation proof. Formalization-only optimization registries are kept in the repository but are not repeated in the printed article.

4.1. **Universal Enclosure and Transfer Calculus.** This section collects the geometry shared by the transfer and direct-witness arguments. The same equilateral enclosure functional governs the local admissible set of Method 2 and the forced witness set of Method 4. The transfer statements below distinguish three situations that must not be conflated: a center-free edge, an edge with a specified center interval, and a free strict-supercritical outgoing envelope.

4.1.1. *The equilateral enclosure gauge.*

Let R denote counterclockwise rotation through $2\pi/3$. For a nonempty compact set $K \subset \mathbb{R}^2$, put

$$h_K(n) = \max_{x \in K} \langle x, n \rangle$$

and

$$\Lambda(K) = \frac{2}{\sqrt{3}} \min_{\|n\|=1} \sum_{j=0}^{2} h_K(\mathsf{R}^j n). \tag{31}$$

Proposition 4.1 (Equilateral enclosure gauge)**.** *The number $\Lambda(K)$ is the least side length of a closed equilateral triangle containing K. It is monotone under inclusion, translation invariant, and positively homogeneous. If K is a finite polygonal hull, a minimizing normal may be chosen perpendicular to one of its hull edges.*

Proof. For a fixed unit normal n, the three supporting half-planes with outward normals $n, \mathsf{R}n, \mathsf{R}^2 n$ form an equilateral triangle. Its side is $2/\sqrt{3}$ times the sum of the three support numbers, proving (31). Monotonicity and homogeneity are immediate, while translation invariance follows from

$$n + \mathsf{R}n + \mathsf{R}^2 n = 0.$$

On an open normal-direction cell with three fixed support points, the support sum has the form $A\cos\theta + B\sin\theta$ and has second derivative equal to its negative. In the nondegenerate case an interior critical point is a maximum. Hence a minimum occurs at a cell boundary, where one support line contains a hull edge. □

For the anchor hull

$$K(a, b, c) = \operatorname{conv}\{0, au, bv, c(u + v)\},$$

where u and v are the two unit boundary directions meeting at 120°, the admissible set is

$$\mathcal{A} = \{(a, b, c) \in [0, 1]^3 : \Lambda(K(a, b, c)) \le 1\}. \tag{32}$$

The complete support calculation is given in Appendix 4.4.

4.1.2. *One equilateral radical.*

For $0 \le x \le 1$, define

$$\omega(x) = \sqrt{1 - x + x^2}, \qquad \sigma(x) = 1 - \omega(x). \tag{33}$$

Then

$$\omega(1 - x) = \omega(x), \qquad \sigma(x) = \frac{x(1 - x)}{1 + \omega(x)}, \tag{34}$$

and direct differentiation gives

$$\sigma''(x) = -\frac{3}{4\omega(x)^3} < 0. \tag{35}$$

Thus σ is strictly concave. In the signed center calculus,

$$E = \omega(R), \qquad \eta = \sigma(R).$$

For $0 < x < 1$, put $z = \sigma(x)/x$. Eliminating the square root gives

$$x = \frac{1 - 2z}{1 - z^2}, \qquad \sigma(x) = \frac{z(1 - 2z)}{1 - z^2}, \qquad 0 < z < \frac{1}{2}. \tag{36}$$

4.1.3. *One g-family.*

For $0 \le x, c \le 1$, define

$$g_c(x) = \max\{y \in [0,1] : (1-x, y, c) \in \mathcal{A}\}. \tag{37}$$

For any map $f : [0,1] \to [0,1]$, put

$$f^\vee(a) = 1 - f(1-a). \tag{38}$$

The exact appendix aliases are

$$B_c(a) = g_c(1-a), \qquad F_c(a) = \begin{cases} 1-a, & 0 \le c \le 1/2, \\ B_c(a), & 1/2 < c \le 1, \end{cases} \qquad G_c(a) = 1 - F_c(a). \tag{39}$$

The symbols F_c, G_c are retained only in the exact contact-cell calculations; the reader-facing transfer is stated directly in the two radial branches.

Lemma 4.2 (Midpoint threshold for the raw map)**.** *For $0 \le x \le 1$,*

$$g_c(x) \le x \qquad \left(\frac{1}{2} \le c \le 1\right), \qquad g_{1/2}(x) = x.$$

Consequently $g_c^\vee(a) \ge a$ whenever $c \ge 1/2$.

Proof. If $(1-x, y, c) \in \mathcal{A}$ and $c \ge 1/2$, the containing triangle also contains the radial midpoint $(u+v)/2$. Lemma 2.10 gives $(1-x) + y \le 1$, hence $y \le x$.

For the reverse inequality at $c = 1/2$, set $a = 1-x$ and $b = x$. In the exact cell description of Proposition 4.31, one has $s = a + b = 1$, $d = a^2 + ab + b^2 \le 1$, $q = ab \ge 0$, and, with $M = \max\{a, b\} \ge 1/2$,

$$F_T(a, b, 1/2) = M(1/2 - M) \le 0, \qquad 1/2 \le 2M.$$

Thus $(a, b, 1/2) \in \mathcal{A}$, so $g_{1/2}(x) \ge x$. The preceding upper bound gives equality. □

Lemma 4.3 (Raw and branchwise nonsupercritical transfer)**.** *Let T be an actual vertex triangle satisfying $A(T) \ge a$ and $C(T) \ge c$, and let T_{next} be the next incident actual vertex triangle. Then*

$$B(T) \le g_c(1-a).$$

If no center interval intervenes on the next edge, then

$$A(T_{\text{next}}) \ge g_c^\vee(a).$$

If T is nonsupercritical, then

$$B(T) \le \begin{cases} 1-a, & 0 \le c \le 1/2, \\ g_c(1-a), & 1/2 < c \le 1, \end{cases} \tag{40}$$

and, on a center-free next edge,

$$A(T_{\text{next}}) \ge \begin{cases} a, & 0 \le c \le 1/2, \\ g_c^\vee(a) \ge a, & 1/2 < c \le 1. \end{cases} \tag{41}$$

Proof. The closure of T realizes $(a, B(T), c)$, so the definition of g_c gives the raw outgoing bound. On a center-free edge, coverage gives $A(T_{\text{next}}) \ge 1 - B(T)$, and complementation proves the raw transfer. Under nonsupercriticality,

$$B(T) \le 1 - A(T) \le 1 - a.$$

This is the low-radial branch. In the high-radial branch the raw bound is already at most $1-a$ by Lemma 4.2. Taking complements gives (41). □

Proposition 4.4 (Zero-radial warning)**.** *For* $0 < x < 1$,

$$g_0(x) > x, \qquad g_0^{\vee}(a) < a \quad (0 < a < 1).$$

Thus a nonsupercritical zero-radial handoff uses the direct inequality $B \le 1 - A$, *not the raw map* g_0.

Proof. The exact diameter formula gives

$$g_0(x) = \frac{x - 1 + \sqrt{1 + 6x - 3x^2}}{2}.$$

Since

$$1 + 6x - 3x^2 - (1 + x)^2 = 4x(1 - x) > 0,$$

one has $g_0(x) > x$. Complementation gives the second assertion. □

4.1.4. *Center-assisted transfers and the free envelope.*

Let $J \subseteq [0, 1]$ be empty or a closed interval. For $0 \le p \le 1$, put

$$e_J(p) = \sup\{x \in [0, 1] : [0, x] \subseteq [0, p] \cup J\}, \qquad \mathcal{R}_J(p) = 1 - e_J(p).$$

If $J = [L, U]$, then

$$\mathcal{R}_{[L,U]}(p) = \begin{cases} 1 - p, & p < L, \\ 1 - \max\{p, U\}, & p \ge L, \end{cases} \qquad \mathcal{R}_{\varnothing}(p) = 1 - p. \tag{42}$$

The map is nonincreasing in p and under enlargement of J.

Lemma 4.5 (Generalized boundary handoff)**.** *Suppose a unit boundary edge is covered by a left trace contained in* $[0, p]$, *a center trace* J, *and a right trace contained in* $[1 - q, 1]$. *Then*

$$q \ge \mathcal{R}_J(p).$$

Define only the raw center-assisted transfer

$$g_{c,J}^{\vee}(a) = \mathcal{R}_J(g_c(1 - a)).$$

Apply it with $p = B(T)$ *and* $q = A(T_{\mathrm{next}})$, *where* T *is an actual vertex triangle satisfying* $A(T) \ge a$ *and* $C(T) \ge c$. *Then*

$$A(T_{\mathrm{next}}) \ge g_{c,J}^{\vee}(a). \tag{43}$$

If T *is nonsupercritical, the sharper branchwise statement is*

$$A(T_{\mathrm{next}}) \ge \begin{cases} \mathcal{R}_J(1 - a), & 0 \le c \le 1/2, \\ g_{c,J}^{\vee}(a), & 1/2 < c \le 1. \end{cases} \tag{44}$$

Proof. The component of $[0, p] \cup J$ containing 0 is $[0, e_J(p)]$. If $1 - q > e_J(p)$, the intervening open interval is uncovered. Since $\mathcal{R}_J$ is nonincreasing, the raw and branchwise outgoing bounds from Lemma 4.3 give the two transfer statements. □

For $0 \le c < 1/2$, define

$$g_c^{\mathrm{sc}} = \sup_{\{x : g_c(x) > x\}} g_c(x) = \frac{c + \sqrt{c^2 - 8c + 4}}{2}. \tag{45}$$

The supremum is not attained. Hence an actual strict-supercritical vertex triangle T with $C(T) \ge c$ satisfies

$$B(T) < g_c^{\mathrm{sc}}. \tag{46}$$

If, in addition, its outgoing edge is center-free and T_{next} is the next incident actual vertex triangle, then coverage gives

$$A(T_{\text{next}}) > 1 - g_c^{\text{sc}}. \tag{47}$$

The former notation was $B_{\text{sc}}(c) = g_c^{\text{sc}}$ and $A_{\text{sc}}(c) = 1 - g_c^{\text{sc}}$.

4.1.5. *Relaxed composition and path budgets.*

For maps listed in geometric vertex-triangle order, write

$$[\Phi_1 \mid \cdots \mid \Phi_r](x) = (\Phi_r \circ \cdots \circ \Phi_1)(x). \tag{48}$$

The leftmost slot acts first. Write $\mathrm{I}(x) = x$.

Lemma 4.6 (Relaxed composition)**.** *Suppose actual reach lower bounds satisfy*

$$x_j \geq \Phi_j(x_{j-1}) \qquad (1 \leq j \leq r),$$

where every Φ_j is nondecreasing. If $\underline{\Phi}_j \leq \Phi_j$, then

$$x_r \geq [\underline{\Phi}_1 \mid \cdots \mid \underline{\Phi}_r](x_0).$$

Proof. Set $y_0 = x_0$ and $y_j = \underline{\Phi}_j(y_{j-1})$. Monotonicity gives $x_j \geq \Phi_j(x_{j-1}) \geq \Phi_j(y_{j-1}) \geq y_j$ inductively. □

Lemma 4.7 (Boundary-path budget)**.** *Let $E_0, \dots, E_k$ be consecutive boundary edges and let $T_1, \dots, T_k$ be the triangles at their intermediate vertices. Suppose:*

(1) *on E_0, the component covered from the endpoint opposite T_1 by all triangles other than T_1 has extent at most u;*
(2) *on E_k, the analogous component opposite T_k has extent at most v;*
(3) *on every middle edge, no triangle other than the two incident path triangles has positive-length trace;*
(4) *the chain is covered.*

Then

$$\sum_{j=1}^{k}(A_j + B_j) \geq k + 1 - u - v. \tag{49}$$

Consequently, if all k internal vertex triangles are nonsupercritical, the chain is impossible whenever $u + v < 1$.

Proof. The first and last component hypotheses give $A_1 \geq 1 - u$ and $B_k \geq 1 - v$. On every middle edge the center and all nonincident triangles have been excluded, so $B_j + A_{j+1} \geq 1$. Adding proves the formula. □

4.1.6. *Low roots, selected Q_+ curves, and thresholds.*

For the high-radial maps define

$$e(d) = \frac{1-d}{2}\left(1 - \sqrt{4(1-d)^2 - 3}\right) \qquad \left(0 < d < 1 - \frac{\sqrt{3}}{2}\right), \tag{50}$$

and, for the low-radial order transition,

$$h(c) = \frac{2c}{1 + \sqrt{16c^2 - 3}} \qquad \left(\frac{1}{2} < c \leq \frac{\sqrt{3}}{2}\right). \tag{51}$$

Proposition 4.8 (Universal selected-Q_+ curve)**.** *Fix $0 < d < 1/2$, put $c = 1 - d$, and suppose an input p belongs to a selected Q_+ arc of $g_c^\vee$. Write*

$$q = g_c^\vee(p), \qquad \nu = q - p, \qquad x = \frac{q-d}{1-d}.$$

Then

$$\nu = \sigma(x) = 1 - \sqrt{1 - x + x^2}, \tag{52}$$

and

$$q = d + (1-d)x, \qquad p = d + (1-d)x - \sigma(x). \tag{53}$$

Consequently the selected arc is increasing and strictly concave.

Proof. The selected quadratic is

$$(1-q)(q-d) = (1-d)^2\nu(2-\nu).$$

Substitution gives $x(1-x) = \nu(2-\nu)$, and the selected component uses the smaller root $\nu = \sigma(x)$. The input function is increasing and strictly convex because σ is strictly concave; inversion gives the assertion. □

The corresponding chord bounds are

$$g_{1-d}^\vee(p) \geq p + \frac{d}{e(d)-d}(p-d), \tag{54}$$

and, with $t = h(1-d)$,

$$g_{1-d}^\vee(p) \geq p + \frac{1-2t}{t-d}(p-d). \tag{55}$$

For every coefficient λ proved valid on the relevant selected arc, write

$$g_{1-d}^{\vee,\lambda}(p) := p + \lambda(p-d) \leq g_{1-d}^\vee(p). \tag{56}$$

The rational form (36) gives, in the historical notation $x = 1 - \beta$,

$$\beta = \frac{z(2-z)}{1-z^2}, \qquad m_\beta = \frac{1-z+z^2}{1-z^2}. \tag{57}$$

For $0 < d < 1 - \sqrt{3}/2$, the exact high-reach lower bound threshold is

$$z > e(d) \implies g_{1-d}^\vee(z) \geq 1 - e(d). \tag{58}$$

Define

$$g_{1-d}^{\vee,\mathrm{th}}(z) = \begin{cases} z, & z \leq e(d), \\ 1 - e(d), & z > e(d), \end{cases} \qquad g_{1-d}^{\vee,\mathrm{th}} \leq g_{1-d}^\vee. \tag{59}$$

Lemma 4.9 (Threshold routing)**.** *Let $z_j = \Phi_j(z_{j-1})$, where every Φ_j is nondecreasing and extensive. If $\Phi_k = g_{1-d}^\vee$ and $z_{k-1} > e(d)$, then $z_j \geq 1 - e(d)$ for all $j \geq k$. If a chain contains later occurrences of $g_{1-d_1}^\vee$ and $g_{1-d_2}^\vee$ and*

$$z_0 > e(d_1), \qquad \min\{e(d_1), e(d_2)\} < Q,$$

then at least one of those two vertex triangles forces every subsequent output above $1 - Q$.

Proof. The first assertion is (58) followed by extensivity. If $e(d_1) < Q$, use the first threshold. Otherwise $z_0 > e(d_1) \geq Q > e(d_2)$, so the second threshold fires. The conclusion is "at least one", not uniqueness; both thresholds may be available. □

4.2. **Exceptional Local Profiles and Vd Placements.** This appendix proves the local profile estimates used by the one-special-triangle branches. A T3-like adjacent exceptional triangle and a Vd1 adjacent exceptional triangle satisfy the same two scalar inequalities and therefore share one center-hiding theorem. The remaining Vd1/Vd2 placements use the interval residuals from Section 4.1, one global radial envelope, and two immediate consequences of the corner normal form.

For $0 \le c < 1/2$, put

$$B_{\mathrm{sc}}(c) = \frac{c + \sqrt{c^2 - 8c + 4}}{2}, \qquad A_{\mathrm{sc}}(c) = 1 - B_{\mathrm{sc}}(c). \tag{60}$$

At $c = 1/2$, the same symbols denote the continuous endpoint values $B_{\mathrm{sc}}(1/2) = A_{\mathrm{sc}}(1/2) = 1/2$; this is a convention, not the supremum of a nonempty strict-supercritical feasible set. The free strict-supercritical envelope says that a strict supercritical vertex triangle with radial lower bound at least c has forward reach strictly below $B_{\mathrm{sc}}(c)$, and B_{sc} is strictly decreasing.

4.2.1. *A global quarter radial envelope.*

For $(p, h) \in [0, 1]^2$, let $c_{\max}(p, h)$ be the maximal radial coordinate such that (p, h, c) belongs to the local admissible set.

Lemma 4.10 (Quarter radial envelope)**.** *If*

$$0 \le h \le p, \qquad p + h < 1,$$

then

$$c_{\max}(p, h) \le 1 - \frac{h}{4}. \tag{61}$$

The inequality is strict on the selected $\mathcal{A}_T$ component.

Proof. Put $s = p+h$ and $q = s^4 - s^2 + ph$. Since $s < 1$, only the two nonsupercritical cells of Proposition 4.31 occur.

On $\mathcal{A}_L$, the smaller boundary coordinate is h, and the selected radial frontier is the unique root in $[\sqrt{3}/2, 1]$ of

$$P_h(c) = c^4 - c^2 + hc - h^2 = 0.$$

At $c_0 = 1 - h/4$,

$$P_h(c_0) = \frac{h}{256}(h^3 - 16h^2 - 240h + 128).$$

The cubic decreases on $[0, 1/2]$ and has value $33/8$ at $1/2$, so this is positive for $h > 0$. Moreover

$$P_h\left(\frac{\sqrt{3}}{2}\right) = -\left(h - \frac{\sqrt{3}}{4}\right)^2 \le 0, \qquad P_h'(c) > 0 \quad (c \ge \sqrt{3}/2).$$

Hence the selected root is below c_0; for $h = 0$ equality gives $c_{\max} = 1$.

On $\mathcal{A}_T$, put

$$r = \sqrt{4s^2 - 3}, \qquad \gamma = \frac{p - h}{s}.$$

The exact identity

$$\frac{q}{s^2} = \frac{r^2 - \gamma^2}{4}$$

gives $0 \le \gamma < r$. Write $\gamma = wr$, $0 \le w < 1$. Then

$$p = \frac{s(1+wr)}{2}, \qquad h = \frac{s(1-wr)}{2}, \qquad c_{\max} = \frac{s(1+wr)}{1+r}.$$

For

$$\Psi(w) = c_{\max} + \frac{h}{4}$$

one has

$$\Psi'(w) = \frac{sr(7-r)}{8(1+r)} > 0,$$

so

$$\Psi(w) < \Psi(1) = \frac{s(9-r)}{8}.$$

Since $4s^2 = r^2 + 3$,

$$256 - (r^2+3)(9-r)^2 = (1-r)(r^3 - 17r^2 + 67r + 13) > 0$$

for $0 \le r < 1$; the cubic is increasing and positive on this interval. Thus $s(9-r) < 8$ and $c_{\max} + h/4 < 1$. □

4.2.2. *The rational T3-like adjacent-exception profile.*

Lemma 4.11 (T3-like adjacent-exception profile)**.** *Let a translated T3-like triangle at V_0 contain M_1, have boundary trace $[0,a]$ on $e_{5,0}$, and have adjacent-radial trace $[c,u]$ on r_1 measured from V_1 toward O. Then*

$$a \le A_{\mathrm{sc}}(c), \qquad \frac{a}{a+1-u} \le A_{\mathrm{sc}}(c). \tag{62}$$

Proof. The translated T3-like normal form supplies

$$1 \le D \le \frac{2}{\sqrt{3}}, \qquad E = \sqrt{4-3D^2}, \qquad \varrho = \frac{D+E}{2},$$

and

$$\frac{E}{2D} \le a \le \frac{4-3D+E}{4D}.$$

Its radial interval is

$$c = \frac{D(1+a)-1}{\varrho}, \qquad u = 1 - \frac{\varrho}{D} + a.$$

Put

$$x = \frac{aD}{\varrho}, \qquad \theta = \frac{D-1}{\varrho}.$$

The relation $\varrho^2 - D\varrho + D^2 = 1$ gives

$$\varrho = \frac{1-2\theta}{1-\theta+\theta^2}, \qquad D = \frac{1-\theta^2}{1-\theta+\theta^2}, \qquad E = \frac{1-4\theta+\theta^2}{1-\theta+\theta^2},$$

with $0 \le \theta \le 2-\sqrt{3}$. Moreover

$$c = x + \theta, \qquad \frac{1-4\theta+\theta^2}{2(1-2\theta)} \le x \le \frac{1}{2} - \theta. \tag{63}$$

Let

$$\Phi(z) = \frac{z(2-z)}{1+z}.$$

It is increasing on $[0,1/2]$ and $\Phi(A_{\rm sc}(c)) = c$. Hence it is enough to prove $\Phi(x) \le x+\theta$, or equivalently

$$Q_\theta(x) = 2x^2 + (\theta-1)x + \theta \ge 0.$$

The vertex is $x_* = (1-\theta)/4$. If $0 \le \theta \le 1/5$, the lower endpoint x_L in (63) satisfies

$$x_L - x_* = \frac{1-5\theta}{4(1-2\theta)} \ge 0,$$

and

$$Q_\theta(x_L) = \frac{\theta(1-5\theta+11\theta^2-\theta^3)}{2(1-2\theta)^2} \ge 0.$$

If $1/5 \le \theta \le 2-\sqrt{3}$, the vertex lies in the feasible interval and

$$Q_\theta(x) \ge Q_\theta(x_*) = \frac{10\theta - 1 - \theta^2}{8} > 0.$$

Thus $x \le A_{\rm sc}(c)$. Since $D \ge \varrho$, one has $a \le x$, and

$$\frac{a}{a+1-u} = \frac{aD}{\varrho} = x.$$

This proves both inequalities. □

4.2.3. *The Vd1 adjacent-exception profile.*

Lemma 4.12 (Vd1 adjacent-exception profile)**.** *Let a Vd1 triangle at V_0 contain M_1, have boundary trace $[0,a]$ on $e_{5,0}$, and have adjacent-radial trace $[c,u]$ on r_1. Assume it has no positive support on r_5. Then*

$$a \le A_{\rm sc}(c), \qquad \frac{a}{a+1-u} \le A_{\rm sc}(c). \tag{64}$$

Proof. The Vd corner normal form gives $t>0$, $d=\sqrt{t^2+t+1}$, and

$$c = \frac{t(1-b)}{t+1}, \qquad u = \frac{d-a-tb-1}{t}.$$

The midpoint condition implies

$$tb = t - c(t+1), \qquad a + tb \le d - 1 - \frac{t}{2}.$$

One-sided support forces $t \ge 1$. Indeed, if $t<1$, then

$$d - a - tb - t \ge 1 - \frac{t}{2} > \frac{1}{t+1} > \frac{1-a}{t+1},$$

so the raw r_5 interval has positive length, contrary to the Vd1 type.

Put

$$F(t,c) = \sqrt{t^2+t+1} - 1 - \frac{3t}{2} + c(t+1).$$

Then $a \le F(t,c)$. Since $c \le 1/2$,

$$\frac{\partial F}{\partial t} = \frac{2t+1}{2\sqrt{t^2+t+1}} - \frac{3}{2} + c \le \frac{2t+1}{2\sqrt{t^2+t+1}} - 1 < 0,$$

where the last inequality follows from $(2t+1)^2 < 4(t^2+t+1)$. Thus F decreases for $t \ge 1$, and

$$a \le F(t,c) \le L(c) := \sqrt{3} - \frac{5}{2} + 2c.$$

Every feasible vertex triangle has $L(c) \geq 0$. The inequality $2L(c) \leq A_{\mathrm{sc}}(c)$ is equivalent to

$$\sqrt{c^2 - 8c + 4} \leq 12 - 4\sqrt{3} - 9c.$$

The right side is positive on $[0, 1/2]$. Squaring is therefore legitimate, and the squared difference is $4Q(c)$, where

$$Q(c) = 20c^2 + (18\sqrt{3} - 52)c + 47 - 24\sqrt{3}.$$

The polynomial is decreasing on $[0, 1/2]$ and $Q(1/2) = 26 - 15\sqrt{3} > 0$. Hence $2L(c) \leq A_{\mathrm{sc}}(c)$, and therefore $a \leq A_{\mathrm{sc}}(c)$.

Put $\varepsilon = 1 - u > 0$. The endpoint formula gives

$$\varepsilon = \varepsilon_0 + \frac{a}{t}, \qquad \varepsilon_0 = \frac{2t + 1 - c(t+1) - d}{t}.$$

Since $c \leq 1/2$ and $t \geq 1$,

$$2t + 1 - c(t+1) - d \geq \frac{3t+1}{2} - d > 0;$$

the last inequality follows after squaring from $(3t+1)^2/4 - d^2 = (5t^2 + 2t - 3)/4 > 0$. Thus $\varepsilon_0 > 0$. Moreover,

$$\varepsilon_0 + \frac{F(t,c)}{t} = \frac{1}{2}.$$

Since $z/(z + \varepsilon_0 + z/t)$ increases in z,

$$\frac{a}{a+\varepsilon} \leq \frac{F(t,c)}{F(t,c) + 1/2} \leq \frac{L(c)}{L(c) + 1/2} \leq 2L(c) \leq A_{\mathrm{sc}}(c).$$

□

4.2.4. *A common adjacent-exception theorem.*

Lemma 4.13 (Common adjacent-exception obstruction)**.** *Let a local adjacent exceptional triangle T_0 have trace $[0, a]$ on $e_{5,0}$ and adjacent-radial trace $[c, u]$ on r_1, measured from V_1 toward O. Put*

$$\varepsilon = 1 - u > 0, \qquad \Theta = \frac{a}{a+\varepsilon}.$$

Assume

$$a \leq A_{\mathrm{sc}}(c), \qquad \Theta \leq A_{\mathrm{sc}}(c), \tag{65}$$

and assume radial isolation forces the center to cover the O-side gap of length ε. If T_1 is the unique strict supercritical vertex triangle and T_2, T_3, T_4, T_5 are nonsupercritical, then the perimeter and r_1 are not covered.

Proof. The center exit on r_1 is δ/R, so radial isolation gives

$$\delta \geq R\varepsilon.$$

Let h be the reach forced on T_5 along $e_{5,0}$. If the companion center trace does not hide the gap after coordinate a, then

$$h \geq 1 - a \geq B_{\mathrm{sc}}(c).$$

In the only hiding order,

$$\frac{k}{W} \leq a < R + \alpha.$$

The left endpoint inequality and radial isolation give

$$k \le Wa, \qquad R\varepsilon \le Wa,$$

so $R(a+\varepsilon) \le a$ and $R \le \Theta$. Also

$$\alpha \le Wa - R\varepsilon - \eta,$$

whence

$$R + \alpha \le a + R(u - a).$$

The hiding order forces $u > a$, and therefore

$$R + \alpha \le a + \Theta(u - a) = \Theta \le A_{\mathrm{sc}}(c).$$

Thus in every ordering

$$h \ge B_{\mathrm{sc}}(c).$$

Coverage of r_1 gives the supercritical vertex triangle radial lower bound at least c, so its forward reach satisfies

$$b_1 < B_{\mathrm{sc}}(c) \le h.$$

Lemma 4.7, applied to T_2, T_3, T_4, T_5, gives

$$\sum_{i=2}^{5} (a_i + b_i) \ge 4 + h - b_1 > 4,$$

contrary to their four nonsupercritical caps. $\square$

Corollary 4.14 (T3-like and Vd1 adjacent rescue)**.** *The common adjacent-adjacent exceptional triangle lemma applies both to the exactly-one-T3-like branch and to the normalized Vd1 vertex triangle containing the neighboring midpoint.*

Proof. Lemmas 4.11 and 4.12 give the two local inequalities. In both branches, midpoint and support isolation force the center to cover the O-side radial gap before the adjacent exceptional triangle begins. $\square$

4.2.5. *Adjacent Vd1/Vd2 placement.*

Lemma 4.15 (Adjacent Vd1/Vd2 placement)**.** *Assume a reduced CE2 exact-M_0 candidate has $N_+ = 1$, vertex triangle T_0 uniquely supercritical, vertex triangle T_1 the unique Vd1/Vd2 vertex triangle, and vertex triangles T_2, T_3, T_4, T_5 nonsupercritical Vd0. Then the perimeter together with r_2 is not covered. The reflected T_5 placement is also impossible.*

Proof. Let

$$I_L = \left[\frac{k}{W}, R + \alpha\right], \qquad I_R = \left[\frac{k}{R}, W + \delta\right].$$

For the supercritical reaches (a_0, b_0), put

$$A = \mathcal{R}_{I_R}(b_0), \qquad H = \mathcal{R}_{I_L}(a_0).$$

The endpoint-distance inequality is

$$a_0^2 + a_0 b_0 + b_0^2 \le 1.$$

It implies $a_0, b_0 < 1$. The signed center bounds give $R + \alpha < 1$, so $H > 0$. Boundary propagation gives

$$a_1 \ge A, \qquad b_i \ge H \quad (1 \le i \le 5).$$

The Vd cap gives

$$A + H < \frac{1}{2}. \tag{66}$$

The common perimeter budget gives

$$T := \alpha + \delta < \frac{1}{24}, \qquad T < \frac{\min\{R, W\}}{6}.$$

We claim

$$\delta < \frac{H}{4}. \tag{67}$$

If $H = W - \alpha$, then

$$4\delta + \alpha \le 4T < \frac{2W}{3} < W.$$

Suppose $H = 1 - a_0$. The other residual cannot be $1 - b_0$, since otherwise (66) would give $a_0 + b_0 > 3/2$, whereas the endpoint-distance inequality gives $a_0 + b_0 \le 2/\sqrt{3} < 3/2$. Hence

$$A = 1 - (W + \delta), \qquad b_0 \ge y := \frac{k}{R},$$

and $W + \delta > 1/2 + H$. Diameter transfer gives

$$H > \frac{y}{2} > \frac{\eta}{2R}.$$

Thus

$$\delta > J(R) := R - \frac{1}{2} + \frac{\eta}{2R}.$$

Writing $\eta/R = W/(1 + E)$, exact differentiation gives

$$4E(1 + E)^2 J'(R) = 2E(1 + E)(1 + 2E) - W(2R - 1).$$

If $R \le 1/2$, the second term is nonnegative after subtraction. If $R \ge 1/2$, then $W(2R - 1) \le 1/8$, while the first term is larger than $1/8$. Thus $J'(R) > 0$. Since $J(3/8) = 1/24$, the bound $\delta < 1/24$ forces $R < 3/8$. Then

$$(2 - 3R)^2 - E^2 = (3 - 8R)(1 - R) > 0,$$

so $\eta/R > 1/3$ and $H > 1/6$. Hence $4\delta < 1/6 < H$, proving (67).

The center interval on r_2 begins from the vertex side at $1 - \delta$. If T_1 supports r_2, its exact endpoint satisfies

$$u_2 < 1 - b_1 - \frac{a_1}{t} \le 1 - H < 1 - \delta.$$

Thus it does not bridge to the center. For T_2, boundary handoff gives

$$a_2 > \frac{1}{2} + A, \qquad b_2 \ge H.$$

Put $p = 1/2 + A$. Equation (66) gives $0 \le H \le p$ and $p + H < 1$. Lemma 4.10 gives

$$c_2 \le c_{\max}(p, H) \le 1 - \frac{H}{4} < 1 - \delta.$$

Thus neither possible radial bridge reaches the center interval, and r_2 is uncovered. $\square$

4.2.6. *Nonadjacent Vd1/Vd2 placements.*

Lemma 4.16 (Nonadjacent Vd1/Vd2 placements)**.** *Under the same reduced CE2 branch, let the unique Vd1/Vd2 vertex triangle be T_τ with $\tau \in \{2,3,4\}$. Then the perimeter together with r_τ is not covered.*

Proof. Put
$$T = \alpha + \delta, \qquad x = \frac{\eta + T}{W}, \qquad y = \frac{\eta + T}{R}.$$
Lemma 3.14 gives
$$T < \frac{1}{2}\min\{x, y\}.$$
Let (a_0, b_0) be the reaches of the unique supercritical vertex triangle. Then
$$a_0 + b_0 > 1, \qquad a_0^2 + a_0 b_0 + b_0^2 \le 1.$$
Let
$$B = e_{[y, W+\delta]}(b_0), \qquad U = e_{[x, R+\alpha]}(a_0), \qquad A = 1 - B, \qquad H = 1 - U.$$
Boundary propagation and the Vd cap give
$$a_\tau \ge A, \qquad b_\tau \ge H, \qquad A + H < \frac{1}{2}.$$

Let $\beta(q) = (-q + \sqrt{4 - 3q^2})/2$. The component endpoint B is either b_0 or $W + \delta$. In the latter case the center diameter gives $B \le \beta(x)$. In the former case, if $a_0 < x$, then $U = a_0$ and $A + H < 1/2$ would force $a_0 + b_0 > 3/2$, contrary to $a_0 + b_0 \le 2/\sqrt{3}$. Hence $a_0 \ge x$, and $B = b_0 \le \beta(a_0) \le \beta(x)$. Reflection gives $U \le \beta(y)$. Consequently
$$A > \frac{x}{2} > T, \qquad H > \frac{y}{2} > T.$$
Thus
$$d_\tau^C < \min\{A, H\} \qquad (\tau = 2, 3, 4),$$
because the three exits are δ, α, and $\min\{\alpha/R, \delta/W\} \le T$.

The Vd corner normal form gives the radial margin
$$c_\tau < 1 - \min\{A, H\}.$$
But the center interval begins from the vertex side at
$$1 - d_\tau^C > 1 - \min\{A, H\}.$$
The two radial traces are disjoint, and all other triangles are excluded by Vd0 locality and diameter locality. ∎

4.2.7. *Placement interfaces.*

The local profile and radial-separation lemmas above are assembled exactly once in Section 4.8. This appendix contains no independent CE2 one-Vd placement assembly.

4.3. **Propagation statements and geometric routing.** The detailed proof below repeatedly uses the same endpoint, return, and radial separation conclusions. We state them once in geometric form. Their full branchwise proofs occupy the subsequent subsections; no independent optimization layer is repeated in the printed article.

Theorem 4.17 (One-gap endpoint inequality)**.** *For every CE1 or CE2 one-gap endpoint pair* $(a_L, c_L), (a_R, c_R)$ *obtained from the signed center normal form,*

$$F_{c_L}(a_L) + F_{c_R}(a_R) < 1.$$

Theorem 4.18 (Paired-gap endpoint inequality)**.** *For every CE2 paired-gap endpoint pair obtained from the signed center normal form,*

$$F_{c_L}(a_L) + F_{c_R}(a_R) < 1.$$

Theorem 4.19 (CE1 return inequality)**.** *In the CE1,* $N_+ = 1$*, all-Vd0 one-gap configuration, the exact five-triangle propagation returns a backward-reach lower bound strictly larger than the remaining forward capacity of the unique supercritical triangle.*

Theorem 4.20 (CE2 return inequality)**.** *The same strict return conclusion holds in the CE2 exactly-one-gap configuration. The threshold alternative is inclusive: at least one of the two threshold locations fires, and both may fire.*

Definition 4.21 (Support-isolated T3-like endpoint state)**.** Assume $N_+ = 0$, no vertex triangle is Vd1 or Vd2, and, after normalization and reflection, T_0 is T3-like with selected adjacent support on r_1. Include every center interval and every nonincident positive trace in the two endpoint suffixes, and denote the resulting boundary/radial pairs by

$$(a_5^*, c_5^*), \qquad (a_1^*, c_1^*).$$

For $c_i^* > 1/2$ the endpoint capacity is $F_{c_i^*}(a_i^*)$; for $c_i^* \le 1/2$ it is the direct non-supercritical value $1 - a_i^*$.

Theorem 4.22 (Support-isolated T3-like endpoint inequality)**.** *For every support-isolated T3-like endpoint configuration, including the direct low-radial linear case,*

$$F_{c_5^*}(a_5^*) + F_{c_1^*}(a_1^*) < 1.$$

Theorem 4.23 (Adjacent exceptional-triangle inequality)**.** *The normalized adjacent T3-like and Vd1 profiles satisfy the two local inequalities required by the common center-hiding obstruction.*

Theorem 4.24 (Adjacent Vd separation)**.** *In the reduced CE2 exactly-one-Vd branch, every adjacent Vd1/Vd2 placement covered by the radial calculation leaves a strict gap between the possible vertex-side radial trace and the center-side radial trace.*

Theorem 4.25 (Nonadjacent Vd separation)**.** *The corresponding strict radial separation holds for each nonadjacent placement of the unique Vd1/Vd2 triangle.*

The first four theorems are proved in the all-Vd0 propagation and endpoint subsections below. The T3-like theorem is the complete four-branch endpoint calculation. The final three theorems are proved by the explicit local-profile and Vd-placement calculations immediately preceding and following the all-Vd0 analysis.

Proposition 4.26 (Common CE2 two-gap application)**.** *Assume both CE2 center traces contain boundary gaps and $T_1, \dots, T_5$ are nonsupercritical Vd0 triangles. Then the perimeter is not covered. No hypothesis on the criticality of T_0 is required.*

Proof. Theorem 4.18 makes the two endpoint capacities sum to less than one. The three intervening vertex triangles form the center-free path required by Lemma 4.7, giving the contradiction. □

Proposition 4.27 (CE1 one-gap obstruction)**.** *The CE1, $N_+ = 1$, all-Vd0 one-gap state is impossible.*

Proof. The exact five-triangle interface gives a returning lower bound and a final upper capacity. Theorem 4.19 puts the former strictly above the latter. □

Corollary 4.28 (CE2 one-gap obstruction)**.** *The CE2, $N_+ = 1$, all-Vd0 exactly-one-gap state is impossible in either orientation.*

Proof. Apply Theorem 4.20. Reflection reverses the five-triangle order and exchanges the two signed center slacks. □

Proposition 4.29 (Cases excluded by boundary-reach propagation)**.** *Boundary-reach propagation closes*

1. *the $N_+ = 0$ all-Vd0 and T3-like-without-Vd1/Vd2 branches for CE1 and CE2;*
2. *every nonempty-gap $N_+ = 1$ all-Vd0 state;*
3. *the exactly-one-T3-like $N_+ = 1$ state with no Vd1/Vd2 triangle;*
4. *the complementary propagation placements in the CE2, $N_+ = 1$, exactly-one-Vd1/Vd2 branch.*

Singleton boundary gaps are included.

Proof. Use Theorems 4.17–4.25, the strict-cycle case, and the complete CE2 positional assembly in Proposition 4.49. The no-gap $N_+ = 1$ all-Vd0 cell is reserved for the nine-point obstruction in Section 6. □

4.4. **Detailed Local Reach Calculus.** This section develops the local calculus for configurations in which a coarse length sum does not by itself retain enough information. The mechanism is always the same. A center trace fixes boundary inputs and complementary radial lower bounds; an exact local map propagates these data through nonsupercritical vertex triangles; and the returning reach lower bound exceeds the remaining boundary or radial capacity.

Actual maximal reaches continue to be denoted by (A_i, B_i, C_i), while lowercase letters denote prescribed reach lower bounds. Every estimate in this section is proved analytically. In particular, no empirical branch catalogue, machine-assisted certificate, or formal root from a discarded algebraic component is used.

4.4.1. *The exact local feasible set.* Put

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \qquad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right).$$

For $0 \le a, b, c \le 1$ define

$$K(a, b, c) = \operatorname{conv}\{0, au, bv, c(u + v)\}.$$

Thus a, b are lower bounds on the two incident boundary rays and c is a lower reach lower bound on the inward radial ray.

Definition 4.30. The local admissible set is

$$\mathcal{A} = \{(a, b, c) \in [0, 1]^3 : K(a, b, c) \text{ is contained in a closed unit equilateral triangle}\}.$$

For $a, c \in [0, 1]$ put

$$B_c(a) = \max\{b : (a, b, c) \in \mathcal{A}\}.$$

The distinction between reach lower bounds and maximal reaches is important. If a triangle has maximal reaches (A, B, C) and $A \geq a$, $C \geq c$, then (a, B, c) is a reach triple realized by the same triangle. The nonsupercritical restriction will be imposed below by two direct radial branches, not by a second extremal-map family.

Proposition 4.31 (Exact admissible set and radial envelope)**.** *Put*

$$s = a + b, \qquad d = a^2 + ab + b^2, \qquad m = \min\{a, b\}, \qquad M = \max\{a, b\},$$

and

$$q = s^4 - s^2 + ab.$$

Define

$$\begin{aligned}
F_L(a, b, c) &= c^4 - c^2 + mc - m^2,\\
F_T(a, b, c) &= (s^2 - 1)c^2 + Mc - M^2,\\
F_S(a, b, c) &= (m^2 - 1)c^2 + (2mM^2 + M)c + M^4 - M^2.
\end{aligned}$$

Then $\mathcal{A} = \mathcal{A}_L \cup \mathcal{A}_T \cup \mathcal{A}_S$, where

$$\begin{aligned}
\mathcal{A}_L &= \{d \leq 1, s \leq 1, q \leq 0, F_L \leq 0\},\\
\mathcal{A}_T &= \{d \leq 1, s \leq 1, q \geq 0, F_T \leq 0, c \leq 2M\},\\
\mathcal{A}_S &= \{d \leq 1, s > 1, F_S \leq 0, c \leq \tfrac{1}{2}\}.
\end{aligned}$$

All variables in these three sets are also constrained to $[0, 1]$.

For $d \leq 1$, the maximal radial lower bound is

$$c_{\max}(a, b) = \begin{cases} 1, & a = b = 0,\\ C_L(m), & s \leq 1, q \leq 0, (a, b) \neq (0, 0),\\ C_T(m, M), & s \leq 1, q > 0,\\ C_S(m, M), & s > 1, \end{cases} \tag{68}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0,$$

and

$$C_T(m, M) = \frac{2M}{1 + \sqrt{4s^2 - 3}}, \qquad C_S(m, M) = \frac{M(2mM + 1) - M\sqrt{4d - 3}}{2(1 - m^2)}.$$

If $d > 1$, the radial envelope is undefined. At $q = 0$ the two subcritical formulas agree and equal s.

Proof. If $s > 0$ and $cs \le ab$, then $c/a + c/b \le 1$ (with the zero-coordinate cases obtained by continuity), so $c(u+v)$ lies in $\mathrm{conv}\{0, au, bv\}$. Its minimum enclosing equilateral side is the diameter

$$\sqrt{a^2 + ab + b^2}.$$

Assume now that $cs > ab$. The four points occur cyclically as $0, bv, c(u+v), au$. For three outward unit normals separated by $120°$, the required side equals $2/\sqrt{3}$ times the sum of the three support values. On an angular cell with fixed support points this sum has second derivative equal to its negative. Hence an angular minimum occurs when a normal is perpendicular to one of the four hull edges. The four resulting side lengths are

$$L_{0A} = b + \max\{a, c\}, \qquad L_{B0} = a + \max\{b, c\},$$

$$L_{AC} = \begin{cases} (a^2 + bc)/\sqrt{a^2 - ac + c^2}, & a \ge c, \\ c(a+b)/\sqrt{a^2 - ac + c^2}, & a < c \le a + b, \\ c^2/\sqrt{a^2 - ac + c^2}, & c > a + b, \end{cases}$$

and L_{CB} is obtained by interchanging a, b. Thus the minimum side is the minimum of these four expressions.

Ordering a, b as $m \le M$, squaring the positive support equations gives $F_L = 0, F_T = 0, F_S = 0$ in the three respective contact regimes; the inactive support inequalities give $q \le 0, q \ge 0, s > 1$. Squaring creates one spurious component in each of the last two cells. In the T cell the two roots are

$$\frac{2M}{1 + \sqrt{4s^2 - 3}}, \qquad \frac{2M}{1 - \sqrt{4s^2 - 3}},$$

and $c \le 2M$ selects the first. In the S cell the smaller root lies below $1/2$ and the other above $1/2$; hence $c \le 1/2$ is exactly the geometric selector. The constant-branch fiber is the interval from zero to its unique selected root. Solving the selected equations gives (68). This also proves that $\mathcal{A}$ is coordinatewise down-closed. □

4.4.2. *Branchwise nonsupercritical output.* For later contact-cell calculations retain the technical aliases

$$F_c(a) = \begin{cases} 1 - a, & 0 \le c \le 1/2, \\ B_c(a), & 1/2 < c \le 1, \end{cases} \qquad G_c(a) = 1 - F_c(a). \tag{69}$$

Proposition 4.32 (Exact midpoint split)**.** *For every $a, c \in [0, 1]$,*

$$F_c(a) = \min\{B_c(a), 1 - a\}.$$

If an actual triangle has reaches (A, B, C) with

$$A \ge a, \qquad C \ge c, \qquad A + B \le 1,$$

then

$$B \le F_c(a). \tag{70}$$

Moreover,

$$F_c(a) = 1 - a \quad (c \le 1/2), \qquad F_c(a) = B_c(a) \quad (c > 1/2).$$

Proof. First take $c \geq 1/2$. Any triangle realizing (a, b, c) also contains the radial midpoint $(u+v)/2$, so Lemma 2.10 gives $a + b \leq 1$. Hence $B_c(a) \leq 1 - a$.

At $c = 1/2$, put $b = 1 - a$. In Proposition 4.31, one has $s = 1$, $d \leq 1$, $q = ab \geq 0$, and

$$F_T(a, b, 1/2) = M(1/2 - M) \leq 0, \qquad M = \max\{a, b\} \geq 1/2.$$

The selector $1/2 \leq 2M$ also holds, so $(a, 1-a, 1/2) \in \mathcal{A}$. Downward closure in the radial coordinate therefore gives $B_c(a) \geq 1 - a$ for every $c \leq 1/2$. This proves the displayed formula for F_c and the minimum identity.

Finally, nonsupercriticality gives $B \leq 1 - A \leq 1 - a$, while admissibility gives $B \leq B_c(a)$. Thus $B \leq F_c(a)$. □

4.4.3. *The exact high-radial map.* For $1/2 < c < 1$ put

$$\ell(c) = \frac{c}{2}\left(1 - \sqrt{4c^2 - 3}\right), \qquad r(c) = c - \ell(c)$$

when $c \geq \sqrt{3}/2$, and use the order-transition function $h(c)$ from (51). Also define

$$Q_-(a, c) = \frac{\sqrt{a^2 - ac + c^2}}{c} - a,$$
$$Q_+(a, c) = \frac{2(1 - a^2)c^2}{2ac^2 + c + \sqrt{(2ac^2 + c)^2 - 4(1 - c^2)(1 - a^2)c^2}}.$$

Proposition 4.33 (Four-branch high-radial map). *For $1/2 < c \leq \sqrt{3}/2$,*

$$F_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ Q_+(a, c), & 1 - c < a < h(c), \\ h(c), & a = h(c), \\ Q_-(a, c), & h(c) < a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases} \tag{71}$$

For $\sqrt{3}/2 < c < 1$,

$$F_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ Q_+(a, c), & 1 - c < a \leq \ell(c), \\ \ell(c), & \ell(c) < a < r(c), \\ Q_-(a, c), & r(c) \leq a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases} \tag{72}$$

Thus there are exactly four branches

$$\text{Lin}, \qquad \text{Const}, \qquad Q_-, \qquad Q_+.$$

At $a = \ell(c)$ in (72), $F_c(a) = r(c)$; immediately to its right the value is $\ell(c)$. This downward jump is genuine. The formal other root of the Q_+ quadratic is not feasible because it violates $c \leq 2\max\{a, F_c(a)\}$.

The maps F_c are nonincreasing, the maps $G_c = 1 - F_c$ are nondecreasing and extensive, and

$$G_c(a) \leq z \iff G_c(1 - z) \leq 1 - a. \tag{73}$$

Proof. Because $c > 1/2$, the midpoint threshold forces $a + b \le 1$ and eliminates the S cell of Proposition 4.31. On the cap $b = 1 - a$, the T inequality reduces to $M(c - M) \le 0$; hence the linear face is feasible exactly for $a \le 1 - c$ or $a \ge c$. Off that face a maximal point lies on $F_L = 0$ or $F_T = 0$. The constant-branch equation has roots $\ell(c), r(c)$ and the selector $q \le 0$ keeps only the smaller coordinate. The ordered T halves give respectively the displayed lower quadratic root Q_+ and the root Q_-. Their order boundary is $a = b = h(c)$, while their $q = 0$ boundaries are $(\ell(c), r(c))$ and $(r(c), \ell(c))$. These observations give (71)–(72), including all equality assignments.

Down-closedness gives monotonicity, while $F_c(a) \le 1 - a$ gives $G_c(a) \ge a$. Finally,

$$G_c(a) \le z \iff F_c(a) \ge 1 - z \iff (a, 1 - z, c) \in \mathcal{A}, \quad a + 1 - z \le 1.$$

Reflection $a \leftrightarrow b$ gives (73). ☐

Put

$$\mu = \frac{2\sqrt{3} - 3}{4},$$

and, for $0 < X < 1 - \sqrt{3}/2$, write

$$e(X) = \ell(1 - X).$$

The root equation gives

$$\begin{aligned} & X < e(X) && \left(0 < X < 1 - \frac{\sqrt{3}}{2}\right), \\ & e(X) < \frac{2X}{1 - 2X} && (0 < X \le \mu), \\ & e(X) \le 2X + 5X^2 && (0 < X \le 1/8). \end{aligned} \tag{74}$$

Moreover,

$$a > e(X) \implies G_{1-X}(a) \ge 1 - e(X) \qquad \left(0 < X < 1 - \frac{\sqrt{3}}{2}\right). \tag{75}$$

Indeed, for

$$p_X(z) = z^2 - (1 - X)z + (1 - X)^2 - (1 - X)^4,$$

one has

$$p_X(X) = X(1 - 3X + 4X^2 - X^3) > 0.$$

Since $X < (1 - X)/2$, this puts X strictly below the smaller root $e(X)$. For $U = 2X/(1 - 2X)$, direct substitution gives

$$p_X(U) = -\frac{X^2}{(1 - 2X)^2}(4X^4 - 20X^3 + 37X^2 - 28X + 3).$$

The polynomial in parentheses is decreasing on $[0, \mu]$ and at the right endpoint equals $8217/64 - 74\sqrt{3} > 0$. Indeed, its derivative is at most $16/512 + 74/8 - 28 < 0$, and $(8217/64)^2 - 3 \cdot 74^2 = 230001/4096 > 0$. Hence $p_X(U) < 0$, which puts U strictly between the roots. Finally,

$$p_X(2X + 5X^2) = X^2(3X + 4)(8X - 1) \le 0,$$

and $X < 2X + 5X^2 < (1 - X)/2$ on the stated range. Thus the trial point lies between the two roots, proving the third comparison. The implication (75) is read directly from (72). We shall also use $e(X) < 2X + 3X^2$ for $0 < X \leq 1/10$, obtained from $p_X(2X + 3X^2) = X^2(8X^2 + 19X - 2) < 0$; here too the trial point lies strictly between X and $(1 - X)/2$.

4.5. **Cyclic Propagation in the All-**Vd0 **Cases.**

4.5.1. *The exactly-one-supercritical all-*Vd0 *branch.* For a center interval, call the nonempty closed set missed by its two open endpoint triangles a *V-gap*. It may be a singleton. If the maximal closed center trace is $[s, t]$, open coverage gives the weak bounds $B_i \geq s$ and $A_{i+1} \geq 1 - t$; thus none of the arguments below discards a singleton gap.

Proposition 4.34 (CE1 one-gap five-map obstruction)**.** *Assume that every vertex triangle is* Vd0*, $N_+ = 1$, and the center triangle is CE1. If its boundary trace contains a V-gap, the triangles do not cover H.*

Proof. Use the signed center variables of Proposition 2.14, and put

$$w = W, \qquad A = \alpha, \qquad D = \delta, \qquad X = R - D, \qquad H = \frac{\eta + A + D}{2R}.$$

The CE1 sign gives $D + RA \geq P$ and $RD > wA$, so

$$m_3 = \min\left\{\frac{A}{R}, \frac{D}{w}\right\} = \frac{A}{R}.$$

The exact strict domain needed below is

$$\begin{gathered} A > 0, \quad D > 0, \quad A + wD < P, \quad D + RA \geq P, \\ A < \frac{w}{2}, \quad D < \frac{R}{2}, \quad RD - wA > 0, \end{gathered} \tag{76}$$

Proposition 2.18 contains the five actual-vertex triangle handoffs, the three duality steps, and the final diameter comparison. It remains only to prove its scalar target

$$(G_{1-D} \circ G_{1-m_3} \circ G_{1-A})\,(H) > 1 - X. \tag{77}$$

Here $A < P \leq \mu$, because $P = E(1 - E)$ and $E \geq \sqrt{3}/2$. We first prove the two input comparisons used at vertex triangle 4. Since $D = R - X$, the inequality $A + wD < P$ and the identity $Rw = \eta + P$ give

$$A < wX - \eta.$$

Put

$$\varphi(q) = wq - \frac{q}{2(1 + q)}.$$

The bound $0 < D < R/2$ gives $R/2 < X < R$, and φ is convex. At the endpoints,

$$\varphi(R/2) < \frac{Rw}{2} < \eta,$$

while $\varphi(R) < \eta$ is equivalent to

$$E(1 - 2R^2) < 1,$$

which follows from $E < 1$. Convexity therefore gives $\varphi(X) < \eta$, and hence

$$A < \frac{X}{2(1 + X)}.$$

Using the rational low-root bound in its valid range,

$$e(A) < \frac{2A}{1-2A} < X. \tag{78}$$

We also have

$$2R(H-A) = \eta + D + (1-2R)A.$$

This is positive if $R \le 1/2$. If $R > 1/2$, then $A < P = \eta E$ gives

$$2R(H-A) > \eta\big(1-(2R-1)E\big) > 0.$$

Thus $H > A$. The positive signed right trace gives

$$\frac{\eta + A + D}{R} < w + D < 1,$$

so $H < 1/2 < 1-A$.

Now consult the exact high-radial catalog for $c_4 = 1-A$. The lower and upper linear ranges would require $H \le A$ and $H \ge 1-A$, respectively, so both are impossible. On the the constant or Q_- branch range,

$$F_{1-A}(H) \le e(A),$$

and therefore

$$G_{1-A}(H) \ge 1 - e(A) > 1 - X$$

by (78). Thus only the Q_+ range remains. Its selector gives

$$H \le e(A) \le 2A + 5A^2.$$

We claim that this range forces $X > 1/2$. Indeed, if $X \le 1/2$, the selector and the lower center inequality give

$$2RH = \eta + A + D \ge \eta + P + wA = w(R+A).$$

Together with $H < 2A/(1-2A)$, this implies

$$f_R(A) < 0, \qquad f_R(z) := w(R+z)(1-2z) - 4Rz.$$

But $f_R''(z) = -4w < 0$ and $f_R(0) = Rw > 0$. We check the other endpoint of the possible interval for A.

If $R \le 1/2$, then $0 < A < P$. Direct simplification gives

$$f_R(P) = (1-E)\left(4E^3 - 2E^2 + 1 - R(2E^3 - 2E^2 + 5E)\right).$$

The coefficient of R is positive, so

$$f_R(P) \ge \frac{1-E}{2}\left(6E^3 - 2E^2 - 5E + 2\right).$$

For $\phi(E) = 6E^3 - 2E^2 - 5E + 2$ on $[\sqrt{3}/2, 1)$,

$$\phi'(E) = 18E^2 - 4E - 5 \ge \frac{17}{2} - 2\sqrt{3} > 0,$$

and

$$\phi\left(\frac{\sqrt{3}}{2}\right) = \frac{2-\sqrt{3}}{4} > 0.$$

Hence $f_R(P) > 0$. Strict concavity puts f_R above its endpoint chord, so $f_R(A) > 0$ for $0 < A < P$, a contradiction.

If $R > 1/2$, the assumption $X \le 1/2$ gives $D \ge R - 1/2$, and therefore

$$0 < A < A_0, \qquad A_0 := P - w\left(R - \frac{1}{2}\right) = \frac{w}{2} - \eta.$$

Direct simplification gives

$$f_R(A_0) = \frac{1-E}{2}(E + 11R - 3ER - 5).$$

Since $11 - 3E > 0$ and $R > 1/2$,

$$E + 11R - 3ER - 5 > \frac{11-3E}{2} + E - 5 = \frac{1-E}{2} > 0.$$

Thus $f_R(A_0) > 0$. Concavity again gives $f_R(A) > 0$ on $(0, A_0)$, a contradiction. Both ranges are impossible, proving $X > 1/2$.

Put $p_1 = G_{1-A}(H)$. If vertex triangle 3 is not on the Q_+ branch, then $p_1 \le A + e(A) < 3A + 5A^2 < 29/64 < 1/2$, and $2R(H - m_3) = \eta + D - A > \eta - P > 0$. Extensivity gives $p_1 \ge H > m_3$, while $p_1 < 1/2 < 1 - m_3$; thus both linear ranges are impossible. On the constant, Q_-, or the low-c tie one has $F_{1-m_3}(p_1) \le p_1$, so

$$G_{1-m_3}(p_1) \ge 1 - p_1 > 1 - X.$$

Extensivity of the final reverse map G_{1-D} preserves this strict bound and proves (77). It remains to treat the selected vertex triangle-3 Q_+ branch. Proposition 4.8 proves once for all that every selected Q_+ arc is increasing and strictly concave.

When $d < 1-\sqrt{3}/2$, the arc endpoints are (d, d) and $(e(d), d+e(d))$, so concavity gives

$$q \ge p + \frac{d}{e(d) - d}(p - d). \tag{79}$$

For $0 < d \le \mu$, the valid rational low-root bound gives

$$\frac{d}{e(d) - d} > \frac{1-2d}{1+2d} > 1 - 4d > 1 - 5d.$$

This covers vertex triangle 4, because its parameter is $d = A < P \le \mu$. For the vertex triangle-3 parameter $d = m_3$, which need not be at most μ, the remaining high-radial range is handled without the rational estimate. If $\mu < d \le 1/8$, then

$$\frac{d}{e(d) - d} \ge \frac{1}{1+5d} > 1 - 5d$$

by $e(d) \le 2d + 5d^2$. If $1/8 < d < 1 - \sqrt{3}/2$, then $e(d) < (1-d)/2$ and hence

$$\frac{d}{e(d) - d} > \frac{2d}{1-3d} > 1 - 5d.$$

The last comparison is equivalent to $15d^2 - 10d + 1 < 0$; its smaller root $(5-\sqrt{10})/15$ is less than $1/8$. In the low-c vertex triangle-3 range $1 - \sqrt{3}/2 \le d < 1/5$, the other endpoint is $(h(1-d), 1 - h(1-d))$, and the chord coefficient is

$$\frac{1 - 2h(1-d)}{h(1-d) - d}.$$

On $3/4 \le c \le \sqrt{3}/2$, direct differentiation of the formula for h shows that $h(c)$ is nonincreasing. Moreover,

$$h\left(\frac{3}{4}\right) = \frac{3}{2(\sqrt{6}+1)} < \frac{7}{16}.$$

Here low c gives $d \ge 1 - \sqrt{3}/2 > 2/15 > 1/16$, and hence

$$h(1-d) < \frac{7}{16} < \frac{3+d}{7}.$$

Thus the displayed chord coefficient is greater than $1/3$, while $1 - 5d < 1/3$. For $d \ge 1/5$, extensivity alone is stronger than the vertex triangle-3 estimate.

Applying these chord bounds first with $d = A$ and then with $d = m_3$ proves, without an omitted selected branch,

$$p_1 > L_1 := (2-4A)H - (1-4A)A,$$
$$p_2 := G_{1-m_3}(p_1) > L_2 := (2-5m_3)L_1 - (1-5m_3)m_3.$$

Finally let

$$D_0 = P - RA, \qquad D_h = A(4R-1) + 10RA^2 - \eta, \qquad x = Rw.$$

Feasibility puts $D_0 \le D \le D_h$. We first prove that the actual feasible value satisfies $D < 1/10$. Since

$$P = \frac{xE}{1+E} < \frac{x}{2}, \qquad \eta = \frac{x}{1+E} > \frac{x}{2},$$

the contrary assumption $D \ge 1/10$ and $A + wD < P$ give

$$A < \overline{A} := \frac{w(5R-1)}{10}.$$

The function $D_h(A)$ is increasing, so

$$D \le D_h(A) < D_h(\overline{A}) < U(R) := \overline{A}(4R-1) + 10R\overline{A}^2 - \frac{x}{2}.$$

Direct expansion yields

$$\frac{1}{10} - U(R) = -\frac{R}{10}P_4(R), \qquad P_4(R) = 25R^4 - 60R^3 + 26R^2 + 22R - 14.$$

For $y = 2R - 1 \in [0,1]$,

$$16P_4(R) = 25y^4 - 20y^3 - 106y^2 + 124y - 39 \le -101y^2 + 124y - 39 < 0.$$

The last quadratic has discriminant -380. Hence $U(R) < 1/10$, a contradiction, and therefore $D < 1/10$.

It remains to compare L_2 with $2D + 3D^2$. Put

$$J(D) = L_2(D) - \frac{23}{10}D.$$

Since

$$L_2 - 2D - 3D^2 - J(D) = 3D\left(\frac{1}{10} - D\right) > 0,$$

it is enough to prove $J(D) > 0$. Exact differentiation gives

$$J'(D) = \frac{S(R,A)}{10R^2}, \qquad S(R,A) = 100A^2 - (40R+50)A + R(20-23R).$$

The inequality $D_0 \le D_h$ is equivalent to

$$x \le A(5R - 1 + 10RA).$$

Because $A < P < x/2 \le 1/8$,

$$A > L := \frac{4x}{25R-4}.$$

Also $\partial_A S = 200A - 40R - 50 < -45$, and direct substitution gives

$$S(R, \mathrm{Const}) = -\frac{Rq_3(R)}{(25R-4)^2},$$

where

$$q_3(R) = 8775R^3 - 14260R^2 + 7928R - 1120.$$

For $y = 2R - 1$,

$$8q_3(R) = 8775y^3 - 2195y^2 + 997y + 3007 \ge 812 > 0.$$

Thus $J'(D) < 0$ and $J(D) \ge J(D_h)$.

At $D = D_h$ one has $H = 2A + 5A^2$. Write $L_{2,h} = L_2(D_h)$. Expansion gives

$$L_{2,h} - A(1+4R) = \frac{A}{R^2}\mathcal{Q}(R, A),$$

where

$$\begin{aligned}\mathcal{Q}(R, A) = {} & 100A^3R - 40A^2R^2 - 30A^2R + 12AR^2 - 15AR + 5A \\ & - 4R^3 + 5R^2 - R.\end{aligned}$$

On $0 \le A \le x/2$,

$$\partial_A^2 \mathcal{Q} = 20R(30A - 4R - 3) < 0.$$

Its two endpoint values are

$$\mathcal{Q}(R, 0) = x(4R-1) > 0, \qquad \mathcal{Q}\left(R, \frac{x}{2}\right) = \frac{x}{2}p(R),$$

where

$$p(R) = 25R^5 - 30R^4 + 20R^3 - 3R^2 - 7R + 3.$$

For $y = 2R - 1$,

$$32p(R) = 25y^5 + 65y^4 + 90y^3 + 106y^2 - 35y + 5 > 0,$$

because the terms of degree at least three are nonnegative and $106y^2 - 35y + 5$ has discriminant -895. Concavity therefore gives $L_{2,h} > A(1 + 4R)$.

On the other hand, $A < P = \eta E$ and $A < P < x/2$ imply

$$\frac{D_h}{A} = 4R - 1 + 10RA - \frac{\eta}{A} < C_0 := 4R - 2 + 5Rx - \frac{x}{2}.$$

Here we used

$$\frac{1}{E} > 1 + \frac{x}{2}, \qquad (1-x)\left(1 + \frac{x}{2}\right)^2 = 1 - \frac{x^2(3+x)}{4} < 1.$$

Finally,

$$1 + 4R - \frac{23}{10}C_0 = \frac{h_3(R)}{20}, \qquad h_3(R) = 230R^3 - 253R^2 - 81R + 112.$$

Since $h_3''(R) = 46(30R - 11) \ge 184$, strong convexity at $R_0 = 20/23$ gives

$$h_3(R) \ge \frac{788}{529} - \frac{(17/23)^2}{368} = \frac{289695}{194672} > 0.$$

Consequently

$$J(D_h) > A\left(1 + 4R - \frac{23}{10}C_0\right) > 0,$$

and hence

$$p_2 > L_2 > 2D + 3D^2.$$

The low root $e(D)$ is the smaller root of

$$H_D(z) = z^2 - (1-D)z + (1-D)^2 - (1-D)^4,$$

and direct substitution gives $H_D(2D+3D^2) = D^2(8D^2 + 19D - 2) < 0$. Hence $e(D) < 2D + 3D^2 < p_2$. The one-hit part of Lemma 4.9, applied to the vertex triangle-2 map, gives

$$G_{1-D}(p_2) \geq 1 - e(D) > 1 - X,$$

because

$$e(D) < 2D + 3D^2 < \frac{23}{100} < \frac{1}{2} < X.$$

This is (77). The actual-vertex triangle and final inequalities in Proposition 2.18 now give the contradiction. □

Proposition 4.35 (nonempty gap all-Vd0, $N_+ = 1$)**.** *If the center triangle is CE1 or CE2, every vertex triangle is* Vd0*, and $N_+ = 1$, then no cover of H can have a boundary gap in the actual open vertex-triangle traces.*

Proof. The midpoint argument used above makes T_0 uniquely supercritical. In an assumed cover, every boundary gap in the vertex-triangle traces must lie in a center trace. CE1 therefore has the one-gap state excluded by Proposition 4.34. For CE2, the exactly-one-gap state is Proposition 2.20; the two-gap state is the common application in Proposition 4.26. This exhausts the one- and two-gap states, including singleton gaps. The zero-gap state is intentionally owned by the center-independent obstruction of Method 4. □

4.6. **Nonsupercritical, T3-like, and Vd1/Vd2 Reach Branches.**

4.6.1. *The exactly-one-T3 branch.*

Proposition 4.36 (Exactly one T3-like vertex triangle)**.** *Assume the center triangle is CE1 or CE2, $N_+ = 1$, exactly one vertex triangle is T3-like, and no triangle is Vd1 or Vd2. Then the triangles do not cover H.*

Proof. Normalize the center midpoint to M_0. A T3-like vertex triangle misses its own midpoint, a supercritical vertex triangle misses its local midpoint, and a Vd0 vertex triangle cannot rescue a neighboring midpoint. Consequently, after reflection,

$$T_0 \text{ is T3-like}, \qquad M_1 \in T_0, \qquad T_1 \text{ is uniquely supercritical},$$

and $T_2, \ldots, T_5$ are nonsupercritical Vd0 vertex triangles. The rational local calculation is Lemma 4.11. It verifies the two local inequalities in (65), while midpoint and support isolation force the center to cover the O-side radial gap. The common adjacent-exception obstruction, Corollary 4.14, therefore gives the contradiction. □

4.6.2. *The nonsupercritical CE1/CE2 boundary packages.* Figure 7 gives a common schematic for the package proved below.

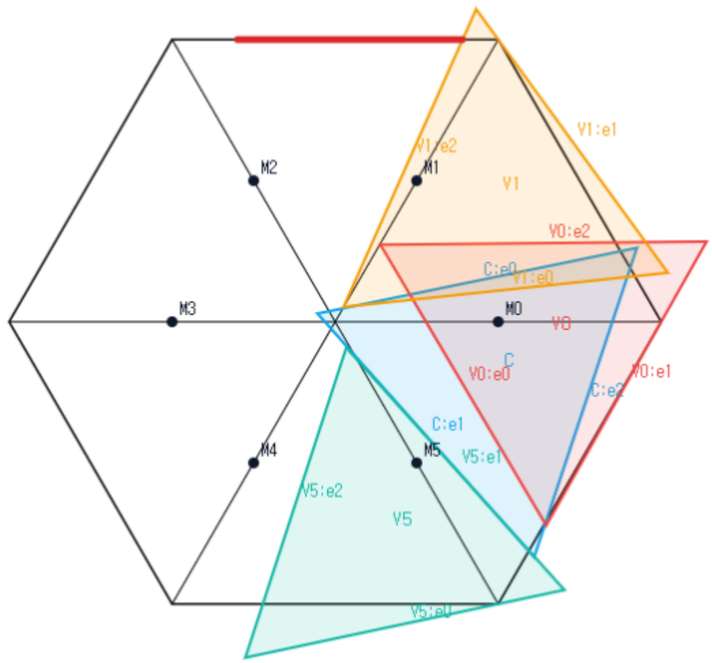


FIGURE 7. Schematic view of the common CE1/CE2, $N_+ = 0$, all-Vd0 boundary package, not to scale. The center triangle and three representative Vd0 vertex triangles are shown; the remaining three triangles are omitted for readability. The simultaneous placement is not a candidate cover, and no proof depends on visual inspection.

Proposition 4.37 (All-Vd0 skeleton-data obstruction)**.** *Let the center triangle be CE1 or CE2. Assume all six vertex triangles are Vd0,*

$$A_i + B_i \leq 1 \qquad (i = 0, \ldots, 5),$$

and the seven open triangles cover the full skeleton

$$S = \partial H \cup \bigcup_{i=0}^{5} r_i.$$

Then no such configuration exists. Consequently the same distinguished-index data cannot cover all of H.

Proof. Let d_i^C be the center reach on r_i, measured from O, and set $c_i^C = 1 - d_i^C$. Since every vertex triangle is Vd0, no adjacent vertex triangle has a positive-length trace on r_i; diameter locality excludes nonlocal triangles. Skeleton coverage therefore forces

$$(*) \qquad C_i \geq c_i^C \qquad (i = 0, \ldots, 5).$$

Let gr be the number of positive center traces containing a nonempty V-gap. Singleton V-gaps are included. The center classification gives $\mathrm{gr} \in \{0, 1, 2\}$, with $\mathrm{gr} \leq 1$ in CE1.

If $\mathrm{gr} = 0$, the six open vertex traces cover every boundary edge. Thus

$$B_i > 1 - A_{i+1}.$$

Nonsupercriticality gives $1 - B_i \geq A_i$, while open boundary coverage gives

$$A_{i+1} > 1 - B_i \geq A_i,$$

which produces the impossible strict cycle

$$A_0 < A_1 < \cdots < A_5 < A_0.$$

Suppose $\mathrm{gr} = 1$. Normalize the active trace to

$$I_R = \left[\frac{k}{R}, W + \delta\right] \subset e_{0,1}$$

and put

$$s = \frac{k}{R}, \qquad q = R - \delta.$$

Gap containment gives $B_0 \geq s$ and $A_1 \geq q$. The companion edge is absent or gap-free, so $A_0 + B_5 \geq 1$; since $A_0 + B_0 \leq 1$, one has $B_5 \geq s$. By $(*)$,

$$C_1 \geq 1 - \frac{\delta}{R}, \qquad C_5 \geq 1 - \frac{\alpha}{W}.$$

The exact one-side endpoint theorem gives

$$g_{1-\alpha/W}(1 - s) + g_{1-\delta/R}(1 - q) < 1.$$

The three internal vertex triangles T_2, T_3, T_4 lie on a center-free boundary path. The path budget forces their total boundary contribution above three, contrary to their three nonsupercritical caps.

If $\mathrm{gr} = 2$, the center is CE2. The paired endpoint theorem, together with $(*)$, gives

$$p = W - \alpha, \qquad q = R - \delta, \qquad F_{p/W}(p) + F_{q/R}(q) < 1.$$

The same center-free three-vertex-triangle path gives the contradiction.

The three ranks are exhaustive. $\square$

We next treat the additional T3-like vertex triangles in the $N_+ = 0$ branch. The following local facts also reappear later.

Lemma 4.38 (T3-like midpoint and side tradeoff)**.** *Let T_i be T3-like. Then $M_i \notin T_i$. After applying the closed-trace domination of Proposition 2.6, suppose the selected adjacent support is on r_{i+1} and M_{i+1} belongs to the dominated trace. If b is its boundary reach toward V_{i+1}, c its exit on its own arm, and ℓ its entry on r_{i+1}, all measured from the relevant vertex, then*

$$b + c < 1, \qquad b + \ell > 1. \tag{80}$$

Consequently two adjacent T3-like traces that cover each other's midpoint, whose selected supports point toward one another, cannot cover both outer half-arms if no other triangle has positive support there.

Proof. Apply Proposition 2.6 and reflect if necessary. Its translated Type-II majorant has parameters

$$0 < t < 1, \qquad z = \sqrt{1 - t + t^2}, \qquad \alpha = 0, \qquad 0 < \beta < z.$$

Its wedge vertex has

$$Q_x = \frac{z - t\beta}{1 - t + t^2} > 1.$$

Hence

$$t\beta < z - z^2.$$

The majorant's radial exit is $c = \beta/(1-t)$. Since

$$(1 - z)(1 + z) = t(1 - t),$$

we obtain

$$c < \frac{z - z^2}{t(1 - t)} = \frac{z}{1 + z} < \frac{1}{2}.$$

Here $z < 1$ because $0 < t < 1$. Thus the translated majorant misses M_0, and its closed-trace domination implies that the original T3-like triangle misses M_0 as well.

For the tradeoff, the translated T3-like trace has the exhaustive Type-II form

$$\begin{aligned} (1-t)x + ty &\geq 0, \\ \beta + tx - y &\geq 0, \qquad 0 < t < 1, \quad z = \sqrt{1 - t + t^2}, \quad 0 < \beta < z. \\ \beta + x - (1-t)y &\leq z, \end{aligned}$$

For the selected support on r_1, substitution in the Type-II half-planes gives

$$b = z - \beta, \qquad c = \frac{\beta}{1 - t}, \qquad \ell = \frac{\beta + 1 - z}{1 - t}.$$

The condition $M_1 \in T$ is $\beta \leq z - (1 + t)/2$. Therefore

$$b + c \leq z + \frac{t}{1 - t}\left(z - \frac{1 + t}{2}\right) = 1 - \frac{(1 - z)^2}{2(1 - t)} < 1,$$

whereas

$$b + \ell - 1 = \frac{t(1 - z + \beta)}{1 - t} > 0.$$

For a crossed adjacent pair, coverage forces $c_i \geq \ell_{i+1}$ and $c_{i+1} \geq \ell_i$. Combining these two inequalities with (80) gives both $b_i < b_{i+1}$ and $b_{i+1} < b_i$, a contradiction. □

Lemma 4.39 (The support-isolated four-branch inequality)**.** *Assume T_C is CE1 or CE2, $N_+ = 0$, no triangle is Vd1 or Vd2, and, after normalization and reflection, T_0 is T3-like with selected adjacent support on r_1. Then the two endpoint vertex triangles in the isolated configuration satisfy*

$$F_{c_5^*}(a_5^*) + F_{c_1^*}(a_1^*) < 1, \tag{81}$$

where a_1^*, a_5^* are the exact boundary inputs and c_1^*, c_5^* the exact radial lower bounds defined below.

Proof. Use the CE1 side variables $0 < \lambda < 1$, $\rho = \sqrt{1 - \lambda + \lambda^2}$ and center slacks

$$X = F_0(O), \qquad Y = F_2(O).$$

Then

$$a_C = \lambda - Y, \qquad \lambda s = X + Y + 1 - \rho,$$

and Proposition 2.14 gives

$$\rho - X - Y > \frac{1}{2}, \qquad \frac{Y}{\lambda} < \frac{1}{2}, \qquad \frac{X}{1-\lambda} < \frac{1}{2}.$$

Consequently the apparent minima in the two radial exits collapse to

$$\gamma_1 = \frac{Y}{\lambda}, \qquad \gamma_5 = \frac{X}{1-\lambda}.$$

For CE2 the second boundary interval is $[S, T]$, where

$$S = \frac{X + Y + 1 - \rho}{1 - \lambda}, \qquad T = X + \lambda.$$

The T3-like normal form supplies $D > 1$, $0 < R < 1$, and $\alpha, p \geq 0$ with

$$R^2 - DR + D^2 = 1, \qquad \alpha + p = 1/D, \qquad q = 1 - p = \alpha + (D-1)/D < 1/2.$$

Its interval on r_1, measured from V_1, is

$$[c, u], \qquad c = Dq/R, \qquad u = q + (1-R)/D.$$

For an initial trace $[0, p]$ and a center interval $J = [L, U]$, define

$$\Theta(p, J) = \begin{cases} 1 - p, & J = \varnothing \text{ or } p < L, \\ 1 - \max\{p, U\}, & p \geq L. \end{cases}$$

Then

$$a_1^* = \Theta(p, [s, t]), \qquad a_5^* = \Theta(\alpha, [S, T]),$$

with the second interval empty in CE1, and

$$c_5^* = 1 - \gamma_5, \qquad c_1^* = \begin{cases} c, & 1 - u \leq \gamma_1 \leq 1 - c, \\ 1 - \gamma_1, & \text{otherwise.} \end{cases}$$

If $a_1^* + a_5^* > 1$, the two caps already imply $F_{c_5^*}(a_5^*) + F_{c_1^*}(a_1^*) < 1$. Assume the hard region $a_1^* + a_5^* \leq 1$. The following table is an exhaustive case analysis of the four possible first branches; the middle column states the exact decisive inequality.

first branch	right branches	conclusion
Const	Lin	$e(\gamma_5) < a_1^*$
Const	Const, Q_-	$F_{c_5^*}(a_5^*) + F_{c_1^*}(a_1^*) < 1$
Const	Q_+	$e(\gamma_5) < a_1^*$
Q_-	Const, Q_-	$F_{c_5^*}(a_5^*) + F_{c_1^*}(a_1^*) < 1$
Q_-	Lin, Q_+	$F_{c_5^*}(a_5^*) < a_1^*$
Lin	all	impossible
Q_+	Lin, Q_+	impossible
Q_+	Const, Q_-	$F_{c_5^*}(a_5^*) + F_{c_1^*}(a_1^*) < 1$.

We verify the nonformal entries. Put $\kappa = (1-\rho)/(1-\lambda)$. From $X + (1-\lambda)Y < \rho(1-\rho)$ one obtains

$$\gamma_1 \geq a_C \Longrightarrow e(\gamma_5) < a_C, \tag{82}$$

$$S \leq Y/\lambda \Longrightarrow e(\gamma_5) < S. \tag{83}$$

For (82), one has $\gamma_5 < a_C - \kappa$ and $a_C - \kappa < 1/8$; then (74) applies, and the remaining comparison is

$$(a_C - \kappa) + 5(a_C - \kappa)^2 \leq \kappa.$$

After squaring its positive sides, the difference is $\lambda(23\lambda^3+58\lambda^2+7\lambda+72)>0$. For (83), combine the premise with the center inequality and put $x=\lambda$. This gives $0<x<3/8$ and

$$X<X_*=\frac{(1-\rho)(\rho(1-2x)-x(1-x))}{1-x-x^2}.$$

Set

$$\delta=\frac{1-\rho}{1-x}=\frac{x}{1+\rho},\qquad \eta=\frac{X}{1-x},\qquad f=\rho(1-2x)-x(1-x),\qquad d=1-x-x^2.$$

Then $\gamma_5\le\eta<\delta f/d$. Moreover

$$\frac{f}{d}<1-2x,$$

because, after multiplication by positive denominators, this is just

$$\rho<d+\frac{x(1-x)}{1-2x}=1+\frac{2x^3}{1-2x}.$$

Hence $\eta<\delta(1-2x)<1/8$. If $\eta=0$, the assertion is immediate. Otherwise,

$$S\ge\frac{X+1-\rho}{1-2x}=\frac{1-x}{1-2x}(\eta+\delta)>2\eta\left(\frac{1-x}{1-2x}\right)^2.$$

On the other hand,

$$5\eta<5\delta(1-2x)<2\left[\left(\frac{1-x}{1-2x}\right)^2-1\right].$$

For the middle sign use $\rho>1-x$ and

$$2(2-3x)(2-x)-5(1-2x)^3=(4x-3)(10x^2-6x-1)>0\qquad(0<x<3/8).$$

Indeed both factors are negative; for the quadratic use $10x^2\le 15x/4<6x+1$. The low-root estimate (74) now gives

$$e(\gamma_5)\le 2\gamma_5+5\gamma_5^2\le 2\eta+5\eta^2<S,$$

proving (83) without any polynomial coefficient list. The constant-branch separation follows from its exact polynomial: if its output is $z=e(\eta)$ at input $1-\beta$, then

$$\beta\ge z+\eta.$$

Indeed the selector polynomial $P(x,z)=x^4-4x^3+5x^2-(2+z)x+z-z^2$ is strictly decreasing on the low range and vanishes at $x=\eta$. Equations (82)–(4.6.2), with the three possibilities for Θ, prove every first-constant entry. For a first Q_- branch the exact root test

$$Q_-(a,c)<z\iff c^2(a+z)^2-(a^2-ac+c^2)>0$$

is increasing in z. Substitution of a_1^*, followed by the center inequality, proves $F_{c_5^*}(a_5^*)<a_1^*$ except in the low–low entries, which are immediate from $Q_-(a,c)\le\min\{a,1-a\}$ and the constant-branch output is below $1/2$. First-coordinate linear would force either $X\ge(1-\lambda)\alpha$ and $Y\ge\lambda-\alpha$, contradicting $s\le p$, or, in the CE2 overlap case, $S\ge 1/2$, contradicting $S\le\alpha<1/2$.

For a first Q_+ branch put $r=1-\lambda$, $y=Y/\lambda$ and choose β so that

$$m=\sqrt{\beta^2-\beta+1},\quad 1-\gamma_5=\frac{m}{r+\beta},\quad F_{c_5^*}(a_5^*)=\frac{\beta m}{r+\beta}.$$

The exact selector is

$$\beta \geq \max\left\{r, \frac{1}{2}, \frac{1-r^2}{1+2r}\right\}, \qquad r \geq (1-\beta)(r+\beta)^2. \tag{84}$$

The center inequalities give $y < 1/10$ and, with

$$y_* = \frac{r}{1-r}\left(\frac{m}{r+\beta} - \frac{1}{2} - \frac{1-r}{1+\sqrt{r^2-r+1}}\right),$$

give $y < y_*$ and

$$3y_* \leq 1 - \frac{\beta m}{r+\beta}.$$

This proves the right-constant entry by $e(y) < 3y$. It excludes right linear and right high by the exact high-input filter. For right Q_- with input a_C, direct substitution gives a sum below one. For input q, put $c = 1-y$, $q = tc$, and let $\widehat{b}_1 = F_{c_1^*}(q)$. The selected Q_- equations give

$$\widehat{b}_1 \leq \kappa := \frac{\sqrt{13}-1}{6}, \qquad q \leq \tau < \frac{93}{200},$$

where $\tau \in (0, 1/2)$ is the unique root of $\tau^4 - 3\tau^3 + 3\tau^2 - 3\tau + 1 = 0$. In this CE2 overlap subcase the support ordering gives $q > S$. The common left-high identities are

$$S = S_0 + \frac{1-r}{r}y, \qquad S_0 = 1 - \frac{m}{r+\beta} + \frac{1-r}{1+\sqrt{r^2-r+1}}.$$

Since $y > 0$, we have the complete chain

$$S_0 < S < q \leq \tau < \frac{93}{200}.$$

We now replace the former interval subdivision by a one-variable comparison. Write $s = r+\beta$ and $\rho_r = \sqrt{r^2-r+1}$. The selected Cell 2 inequality factors as

$$r - (1-\beta)(r+\beta)^2 = (s-1)(\beta^2 + r\beta - r).$$

The second factor is nonnegative because $r \leq \beta$ and $\beta \geq 1/2$; if it vanishes, then $r = \beta = 1/2$. Hence $s \geq 1$. It follows that $\rho_r \leq m$, since $\rho_r^2 - m^2 = (r-\beta)(s-1) \leq 0$.

Define the one-variable lower bound

$$\overline{S}(\beta) = 1 - m + \frac{\beta}{1+m}.$$

Then

$$S_0 \geq 1 - \frac{m}{s} + \frac{1-r}{1+m} \geq \overline{S}(\beta).$$

Indeed, the second difference is

$$(s-1)\left(\frac{m}{s} - \frac{1}{1+m}\right) \geq 0.$$

For the last sign use $s \leq 2\beta \leq m(1+m)$: if $g = 3\beta - \beta^2 - 1$, then $g > 0$ and

$$m^2 - g^2 = -\beta(\beta-1)(\beta^2 - 5\beta + 5) \geq 0, \qquad m^2 + g = 2\beta.$$

Put $B_0 = 5657/10000$, $T = 93/200$, and $\beta_* = 161/250$. If $\beta m/s \geq B_0$, then $\beta m \geq B_0$. The function $\phi(\beta) = \beta m$ is strictly increasing, and the exact comparison

$$B_0^2 - \beta_*^2 m(\beta_*)^2 = \frac{22782769}{62500000000} > 0$$

therefore gives $\beta > \beta_*$. On $[\beta_*, 1]$ put

$$H(\beta) = \frac{107}{200} - Tm - (1-\beta)^2.$$

Since $H'' = -3T/(4m^3) - 2 < 0$, the function H is concave. Its endpoint values are positive: $H(1) = 7/100$, while, for $a_* = 107/200 - (1-\beta_*)^2 = 51033/125000$,

$$a_*^2 - T^2 m(\beta_*)^2 = \frac{1693881}{62500000000} > 0.$$

Thus $H > 0$ throughout the interval. The identity

$$(\overline{S}(\beta) - T)(1+m) = H(\beta)$$

shows that $\beta m/s \geq B_0$ would force $S_0 > T$, a contradiction. Consequently

$$F_{c_5^*}(a_5^*) < \frac{5657}{10000} < \frac{7-\sqrt{13}}{6}.$$

The final strict comparison follows by squaring $\sqrt{13} < 18029/5000$, whose squared difference is $44841/25000000 > 0$. Thus the final sum is also below one. The table is exhaustive, proving (81). $\square$

Proposition 4.40 (The $N_+ = 0$ T3-like branch)**.** *Assume that T_C is CE1 or CE2, $N_+ = 0$, no vertex triangle is Vd1 or Vd2, and at least one vertex triangle is T3-like. Then the seven triangles do not cover H.*

Proof. Normalize the unique center midpoint to M_0 and simultaneously apply the closed-trace domination of Proposition 2.6 to every T3-like triangle. If T_0 were not T3-like, a maximal consecutive block of T3-like indices among $1, \ldots, 5$ would have to match its triangles bijectively to the same block of uncovered midpoints. The matching uses only adjacent indices, because a T3-like triangle misses its own midpoint and Vd0 triangles cannot cover adjacent midpoints. Its first two triangles therefore form a crossed pair forbidden by Lemma 4.38. Hence T_0 is T3-like; reflect so that it supports r_1.

There are at most two T3-like triangles. Indeed, with $N_+ = 0$ and no Vd1/Vd2 triangle, the short-triangle identity gives $q = k$, where k is the number of T3-like triangles. If $k \geq 3$, Theorem 3.15 gives an immediate full-skeleton contradiction. Midpoint locality then forces T_2, T_4, T_5 to be Vd0. Consequently r_1 has positive support only from T_C, T_0, T_1, and r_5 only from T_C, T_5.

The actual forward reaches of T_1, T_5 are bounded by the two safe maps in Lemma 4.39; an extra adjacent-ray reach lower bound on a possible T3-like T_1 only shrinks its feasible set. Thus their sum is less than one. vertex triangles T_2, T_3, T_4 must cover more than three units of the four intervening boundary edges, while their three nonsupercritical caps sum to at most three. This contradiction proves the proposition. $\square$

Lemma 4.41 (Free strict-supercritical envelope)**.** *For $0 \leq c < 1/2$,*

$$\sup\{b : \exists a, (a,b,c) \in \mathcal{A}, a+b > 1\} = \mathcal{B}(c) := \frac{c + \sqrt{c^2 - 8c + 4}}{2}.$$

The supremum is not attained. For $c \geq 1/2$ the strict feasible set is empty. The function $\mathcal{B}$ is strictly decreasing on $[0, 1/2]$.

Proof. Lemma 2.10 excludes $c \geq 1/2$. On the boundary $a = 1 - b$, a support rotation has strict slack precisely when

$$c < \frac{1-b^2}{2-b}.$$

The right side decreases from $1/2$ to zero. Solving equality gives the displayed larger root. Equality has no strict support slack, which explains nonattainment. Differentiation gives

$$\mathcal{B}'(c) = \frac{1}{2}\left(1 + \frac{c-4}{\sqrt{c^2-8c+4}}\right) < 0,$$

since $(4-c)^2 - (c^2 - 8c + 4) = 12$. □

4.6.3. *Two reusable high-radial endpoint inequalities.*

Lemma 4.42 (One-side endpoint loss)**.** *Let* $0 < R < 1$*,* $S = \sqrt{1 - R + R^2}$*, and let* $s, u, w > 0$ *satisfy* $s + u + w = 1$ *and* $0 < u < R$*. Put*

$$\delta = \frac{R}{1+S}, \qquad \gamma = u - \delta - \frac{R}{1-R}w,$$

and assume $\gamma > 0$*. Set*

$$c_1 = u/R, \qquad c_5 = 1 - \gamma.$$

Then

$$F_{c_5}(s) + F_{c_1}(u) < 1. \tag{85}$$

Proof. Put $\eta = (1-R)\delta = 1 - S$ and

$$\widehat{b}_5 = F_{c_5}(s), \qquad \widehat{b}_1 = F_{c_1}(u).$$

The center identity may be written

$$c_5 = \frac{1 - u + \eta - Rs}{1 - R}. \tag{86}$$

The hypotheses imply $1/2 < c_1, c_5 < 1$: for example $u > \delta$ gives $c_1 > 1/(1+S) > 1/2$, and $\gamma < u - \delta < R - \delta = RS/(1+S) < 1/2$ gives $c_5 > 1/2$. Thus the four high-radial branches of the four-branch high-radial map in Proposition 4.33 are exhaustive.

Right low branches. If the right branch is Q_-, its exact frontier equation gives $(u + \widehat{b}_1)^2 = 1 - R + R^2 = S^2$, so $\widehat{b}_1 = S - u$. For a right constant branch write $\widehat{b}_1 = kc_1$, $0 \leq k \leq 1/2$, and $C = 1 - k + k^2 = c_1^2$. With $X_R = R + k$, the constant-branch selector factors as

$$q = c_1^2(X_R - 1)G_k(X_R), \qquad G_k(X) = CX^3 + CX^2 - k(1-k)X + k^2.$$

For $X \geq k$, $G_k(X) > 0$: when $k > 0$, $G_k'(X) \geq k(1 + 2k - k^2 + 3k^3) > 0$ and $G_k(k) = k^2(1 + k + k^3) > 0$. Hence $q \leq 0$ gives $X_R \leq 1$. If $h_R = 1 - X_R$, then

$$S^2 - (u + \widehat{b}_1)^2 = h_R(1 + 2k^2 + k(1-k)h_R) \geq 0.$$

Consequently either right low branch satisfies

$$\widehat{b}_1 \leq X := S - u. \tag{87}$$

A left low branch against right high or linear. For left Q_- put $z = s/c_5 = 1 - p$ and $h = \sqrt{1 - p + p^2}$. The component selector gives $0 \leq p \leq 1/2$, the frontier gives $s + \widehat{b}_5 = h$, and its selector gives $s \le (1-p)h$. Equation (86) becomes $c_5(1 - Rp) = 1 - u + \eta$, whence

$$u \ge \eta + (1-h) + Rph.$$

Put $q_0 = ph$. If $q_0 > R$, then $p(1-p) > R(1-R)$ and the preceding lower bound would give $u > R$. Thus $q_0 \le R$, and

$$\eta + 1 - h > \frac{R(1-R) + p(1-p)}{2} \ge (1-R)q_0.$$

Substitution in the exact formula

$$w = \frac{(1-R)(1-u)p - \eta(1-p)}{1 - Rp}$$

now gives $w < 1 - h$. A right high output has the form $H - u$, where $H = \sqrt{1 - \beta + \beta^2} \le 1$; for linear take $H = 1$. Hence

$$\widehat{b}_5 + \widehat{b}_1 = h + H - (s + u) = 1 + w - (1-h) - (1-H) < 1.$$

For a left constant branch write

$$c_5 = h(x) := \sqrt{1 - x + x^2}, \qquad \widehat{b}_5 = xh(x), \qquad 0 \le x \le \frac{1}{2},$$

and, along the center line,

$$s(x) = \frac{R - u + \eta + (1-R)(1 - h(x))}{R}.$$

If $y = s(x)/h(x)$, the exact selector factorization is

$$\frac{q}{h(x)^2} = (y - (1-x))P_x(y),$$

where

$$\begin{aligned} P_x(y) = {} & (1 - x + x^2)y^3 + (1 + 2x - 2x^2 + 3x^3)y^2 \\ & + (x + 2x^2 - x^3 + 3x^4)y + (x^2 + x^3 + x^5). \end{aligned}$$

Every coefficient is nonnegative on $0 \le x \le 1/2$, and the polynomial is positive at every nondegenerate point. Thus a selected constant-branch point has $s(x) \le (1-x)h(x)$. The two curves meet once in $(0, 1/2)$: at $x = 0$ the left side is smaller, while at $x = 1/2$ it is enough to use

$$u \le \frac{R}{1+R} \quad (\mathrm{Lin}), \qquad u \le \min\{RS, S/2\} \quad (Q_+).$$

Writing $e_0 = 1 - \sqrt{3}/2$, the corresponding endpoint differences are

$$\frac{(1-R)(R^2 + 2Re_0 + 2e_0)}{2(1+R)} > 0$$

for linear, and respectively $[2e_0(1-R) - R^3]/2 \ge 0$ for $R \le 1/2$ and $(1-R)(3R + 4e_0 - 2)/4 \ge 0$ for $R \ge 1/2$ in the high case. At the unique transition x_0, the constant frontier is the $q = 0$ endpoint of the preceding Q_- branch. Since $xh(x)$ increases, every selected left constant-branch output is no larger than that transition output. The preceding strict loss therefore also covers (Const, Q_+) and $(\mathrm{Const}, \mathrm{Lin})$.

linear and high exclusions. Left linear forces $s < 1/2$ and $\gamma \geq s$. Right linear or right high gives $u \leq 1/2$, but

$$\gamma - s = 2u - 1 - \delta + \frac{1-2R}{1-R} w < 0 :$$

for $R \geq 1/2$ this is immediate, while for $R < 1/2$ use $w < 1-u$ to bound it by $(u-R)/(1-R) - \delta < 0$. Thus left linear cannot pair with either branch. If the left branch is high and the right one linear, then $s < 1/2$ and $u \leq R/(1+R)$, whereas $\gamma > 0$ and $w > 1/2 - u$ force $u > R/2 + \eta > R/(1+R)$; the final inequality is equivalent to $2(1+R) > 1 + S$. This pair is also impossible.

A right low branch. Use (87). If $s > X$, the cap gives $\widehat{b}_5 + \widehat{b}_1 < 1$. Suppose $s \leq X$ and put

$$c_X = X + 2\delta, \qquad f(c, z) = c^4 - c^2 + zc - z^2.$$

At the limiting value $u = R$, with $\kappa = \delta$, one has

$$X_0 = \frac{1-2\kappa}{1+\kappa}, \qquad c_0 = \frac{1+2\kappa^2}{1+\kappa}.$$

Now $c_0 > \sqrt{3}/2$ because $16\kappa^4 + 13\kappa^2 - 6\kappa + 1 > 0$, and

$$f(c_0, X_0) = \frac{\kappa^2(1-2\kappa)^2(4\kappa^4 + 4\kappa^3 + 10\kappa^2 + 6\kappa + 5)}{(1+\kappa)^4} > 0.$$

Along the u-line, $\frac{d}{du} f(c_X, X) = -4c_X^3 + c_X + X < 0$, so the same two signs hold before the limiting endpoint. Therefore, for $\ell(c) = c(1 - \sqrt{4c^2 - 3})/2$,

$$\ell(c_X) < X < c_X - \ell(c_X), \qquad \ell(c_X) < 1 - X.$$

Write $c = h(z) = \sqrt{1 - z + z^2}$ and let z_X satisfy $h(z_X) = c_X$. The actual $c_5(s)$ corresponds to $0 < z \leq z_X < 1/2$, and

$$s(z) = s_0 + \frac{1-R}{R}(1 - h(z)), \qquad s_0 = \frac{R - u + \eta}{R} > 0.$$

The transition input $(1-z)h(z)$ decreases, and the preceding root ordering gives $s(z_X) < (1 - z_X)h(z_X)$, hence the same strict inequality for all $z \leq z_X$. To check the order of the two boundary coordinates put

$$\phi(z) = s(z) - zh(z), \qquad k_R = \frac{1-R}{R}.$$

Its endpoint values are $s_0 > 0$ and $X - \ell(c_X) > 0$. Moreover

$$\phi'(z) = \frac{(k_R + z)(1-2z)}{2h(z)} - h(z).$$

For $k_R \leq 2$ this is nonpositive; for $k_R > 2$ its zero is described by the strictly increasing function $K_0(z) = (2 - 3z + 4z^2)/(1-2z)$, so the minimum is again at an endpoint. Thus $zh(z) < s(z)$. Applying the already displayed polynomial P_z gives the strict constant-branch selector, and $s + zh(z) \leq X + \ell(c_X) < 1$. Since $\partial_b F_L = c(1-2z) > 0$ and each feasible fiber is an interval, the nonsupercritical maximum is

$$\widehat{b}_5 \leq zh(z) \leq z_X h(z_X) = \ell(c_X) < 1 - X.$$

Together with $\widehat{b}_1 \leq X$, this proves the loss for every right low branch.

High–high. Let $c_L(z)$ be the unique root in $[\sqrt{3}/2,1]$ of $f(c,z)=0$. With $D_c = 4c^3-2c+z>0$, implicit differentiation gives

$$c_L''(z) = \frac{2(c^2-cz+z^2)+16c^2z(c-z)}{D_c^3} > 0.$$

A selected left high branch requires $s<1/2$ and $c_5(s)\le c_L(s)$. Define

$$s_0 = \frac{R-u+\eta}{R};$$

then $0<s_0<s$ and $c_5(s_0)=1$. The explicitly named function $g(z)=c_5(z)-c_L(z)$ is concave, with $g(s_0)>0$. The right high bound $u\le\min\{RS,S/2\}$ gives $c_5(1/2)>7/8$. For $R\le 1/2$ this follows after squaring from $17-10R+9R^2-64R^3-64R^4>0$ (a decreasing polynomial with value $9/4$ at $1/2$); for $R\ge 1/2$ it follows from $(3+R)^2-16S^2=(1-R)(15R-7)>0$. Finally

$$f(7/8,1/2)=33/4096>0$$

and f increases in c on this range, so $c_L(1/2)<7/8$ and $g(1/2)>0$. Concavity contradicts $g(s)\le 0$. Thus high–high is impossible.

If both branches are low, the cap already gives $\widehat{b}_5+\widehat{b}_1\le s+u=1-w<1$. The preceding five paragraphs cover every other ordered pair of the four branches, proving (85). □

Lemma 4.43 (CE2 two-endpoint loss)**.** *Let $[x,U]$ and $[y,v]$ be the two exact CE2 traces in the legacy coordinates of Remark 2.15, where U denotes the endpoint called u there. Put*

$$p=1-U,\qquad q=1-v,\qquad r=\frac{x}{x+y},\qquad w=\frac{y}{x+y},$$

and reflect if necessary so that $0<w\le 1/2$ and $r=1-w$. Then

$$F_{p/w}(p)+F_{q/r}(q)<1. \tag{88}$$

Proof. Put

$$S_0=x+y,\quad K=\sqrt{1-wr},\quad z=\frac{w}{1+K},\quad \zeta=\frac{r}{1+K},\quad \alpha=\frac{p}{w},\quad \beta=\frac{q}{r}.$$

Then $D=S_0K$, $1-K=wr/(1+K)$, and the exact midpoint inequalities give $1/2<\alpha,\beta<1$. Dividing the coupling equation $(U+v)S_0-xy=D$ by S_0, and using successively $x<U$ and $y<v$, gives the strict endpoint inequalities

$$\beta+p>1+z,\qquad \alpha+q>1+\zeta. \tag{89}$$

Define

$$\Phi_w(\alpha)=F_\alpha(w\alpha),\qquad \Phi_r(\beta)=F_\beta(r\beta),\qquad h(t)=\sqrt{1-t+t^2}.$$

Thus the desired sum is $\Phi_w(\alpha)+\Phi_r(\beta)$.

Selected branch parametrizations. For a side of weight $\sigma\in\{w,r\}$, every constant branch has

$$c=h(k),\qquad F_c(\sigma c)=kh(k),\qquad 0\le k\le w, \tag{90}$$

on either side. On the large side the range follows from

$$Q_{\text{cell}}/h(k)^2=(k-w)G_k(r+k),$$

where $G_k(X) = h(k)^2X^3 + h(k)^2X^2 - k(1-k)X + k^2 > 0$ for $X \geq k$. The remaining selected parametrizations are

$$\alpha = \frac{h(a)}{w+a}, \qquad \Phi_w(\alpha) = a\alpha, \qquad r \leq a < 1, \tag{91}$$

$$\beta = \frac{h(b)}{r+b}, \qquad \Phi_r(\beta) = b\beta, \qquad r \leq b < 1, \tag{92}$$

$$\beta = \frac{K}{r+t}, \qquad \Phi_r(\beta) = t\beta = K - r\beta, \qquad w \leq t \leq r. \tag{93}$$

For example, the small-high selector factors as

$$h(a)^2 a \left(\frac{w}{(w+a)^2} - (1-a) \right), \qquad w - (1-a)(w+a)^2 = (a-r)(a^2 + wa - w),$$

which gives $a \geq r$; the other two factorizations are $(b-w)(b^2+rb-r)$ and $(t-w)(r^2-wt)$, respectively. These formulas specify the selected geometric components, not discarded algebraic roots.

The small-side Q_- branch is absent for $w < 1/2$. Large-side linear is incompatible with the first inequality in (89). Small-side linear can pair only with the constant branch: either high or Q_- on the large side has $\beta \leq K$, which would force $p > (1+r)z > w/(1+w)$, contrary to linear. Thus, up to the symmetric $w = r = 1/2$ tie, the exact list is

$$\begin{gathered}(\mathrm{Lin}, \mathrm{Const}),\ (Q_+, \mathrm{Const}),\ (\mathrm{Const}, \mathrm{Const}),\ (\mathrm{Const}, Q_-),\\ (\mathrm{Const}, Q_+),\ (Q_+, Q_-),\ (Q_+, Q_+).\end{gathered} \tag{94}$$

Low–low sums are at most $K < 1$: a constant-branch output is at most wK, while (93) is at most rK. For linear–constant, put

$$b_{\mathrm{thr}} = 1 + z - \frac{w}{1+w}, \qquad \kappa = \frac{w}{1 + (1+w)z}.$$

linear and (89) give $\beta > b_{\mathrm{thr}}$ and $(1+\kappa)b_{\mathrm{thr}} = 1 + z$. Direct reduction yields

$$b_{\mathrm{thr}}^2 - h(\kappa)^2 = \frac{w^2(A(w) + KC(w))}{D_0(w,K)} > 0,$$

where

$$\begin{aligned} A(w) &= 8 - 5w + 22w^2 + 6w^3 + 5w^4 + 5w^5 - w^6, \\ C(w) &= 8 - w + 14w^2 + 4w^3 - 2w^4 + w^5, \\ D_0(w,K) &= ((1+w)(1+K))^2(1 + K + w + w^2)^2. \end{aligned}$$

Both numerator polynomials are positive on $[0, 1/2]$. Since h decreases and $(1+k)h(k)$ increases there, $\beta = h(k) > b_{\mathrm{thr}} > h(\kappa)$ implies $(1+k)\beta < 1 + z$; combined with (89), this gives $k\beta < p$ and hence the strict linear–constant loss.

It remains to verify the four pairs containing a high branch on the small side.

(Const, Q_+). Use parameters $0 \leq k \leq w$ and $r \leq t < 1$:

$$\alpha = h(k), \quad \Phi_w = kh(k), \qquad \beta = \frac{h(t)}{r+t}, \quad \Phi_r = \frac{th(t)}{r+t}.$$

The first output and the last output increase in their parameters, while $h(t)/(r+t)$ decreases. If $w \leq 2/5$, then

$$\Phi_w + \Phi_r < wK + \frac{1}{2-w} < 1;$$

after using $K \le 1-wr/2$, the positive cleared gap is $P(w) = w^4-3w^3+4w^2-6w+2$, which decreases on $[0, 2/5]$ and has $P(2/5) = 46/625$. If $w \ge 2/5$, the first endpoint inequality gives $\beta > r + z \ge 3/4$. Since $h(3/4)/(r + 3/4) = \sqrt{13}/(7 - 4w) < 3/4$, monotonicity forces $t < 3/4$. Therefore

$$\Phi_w + \Phi_r < \frac{\sqrt{3}}{4} + \frac{3\sqrt{13}}{20} < 1.$$

(Q_+, Q_-). Use a, t from (91) and (93). Both reach lower bounds are at most K. The first endpoint inequality gives

$$\alpha > \frac{1 + z - K}{w} = \frac{1 + r}{1 + K} =: a_0.$$

Thus $K > 1-w^2$, which forces $w > 1/3$, and comparison at $a = 2/3$ forces $a < 2/3$. The function

$$G(a) = \frac{(1 + a)h(a)}{w + a}$$

decreases on $[r, 2/3]$, because its derivative numerator $2a^3 + (4w-1)a^2 + (1-w)a + w - 2$ is at most its value $-7/54$ at $(2/3, 1/2)$. Hence $G(a) \le (1 + r)K$. Using the second endpoint inequality and $\Phi_r = K - r\beta$ gives

$$\Phi_w + \Phi_r < (2 + r)K - 1 - \zeta < 1.$$

After multiplication by $1+K$, the last nonempty gap is $r(3w-w^2-K)$; its squared comparison is

$$(3w - w^2)^2 - K^2 = w^4 - 6w^3 + 8w^2 + w - 1 > 0$$

for $w > 1/3$ (value $1/81$ at $1/3$, positive derivative thereafter).

(Q_+, Q_+). Use parameters a, b from (91)– (92). The endpoint inequalities first force $w > 2/5$: for $w \le 2/5$, the asserted contrary inequality $K(w+1/(2r)) \le 1+z$ reduces, after multiplication by $2r(1 + K)$, to

$$K(2w^2 - 4w + 1) + 2w^4 - 4w^3 + w^2 - w + 1 > 0,$$

which is immediate according as the coefficient of K is nonnegative or, using $K \le 1$, from the decreasing polynomial $2w^4 - 4w^3 + 3w^2 - 5w + 2 \ge 172/625$.

For $F_c(t) = (1 + t)h(t)/(c + t)$, the derivative numerator $2t^3 + (4c - 1)t^2 + (1 - c)t + c - 2$ is increasing on the present rectangle, so F_c has at most one interior minimum. Endpoint comparison gives

$$\Phi_w + \alpha \le \frac{2}{1 + w}, \qquad \Phi_r + \beta \le \frac{2}{1 + r}.$$

The nontrivial endpoint signs are exactly

$$4 - (1 + r)^2(1 + w)^2K^2 = -w^2(w - 1)^2(w^2 - w - 3) > 0$$

and

$$16r^2 - (1 + r)^4h(r)^2 = (1 - r)(r^5 + 4r^4 + 7r^3 + 9r^2 - 4r - 1) > 0;$$

the bracket is increasing on $[1/2, 1]$ and equals $13/32$ at $1/2$. Weighting the two strict endpoint inequalities by w/K^2 and r/K^2 gives

$$\alpha + \beta > \frac{2K + 1}{K(1 + K)}.$$

Consequently

$$\Phi_w + \Phi_r < \frac{2}{1+w} + \frac{2}{1+r} - \frac{2K+1}{K(1+K)} < 1.$$

The numerator of the last gap is $3Kwr - Q(w)$, where $Q(w) = w^4 - 2w^3 + 7w^2 - 6w + 2 > 0$. Its squared difference is

$$D(w) = -w^8 + 4w^7 - 9w^6 + 13w^5 - 41w^4 + 65w^3 - 55w^2 + 24w - 4.$$

Here $D'(w) = (1-2w)H(w)$ with

$$H(w) = 4w^6 - 12w^5 + 21w^4 - 22w^3 + 71w^2 - 62w + 24 > 0$$

on $[2/5, 1/2]$; use $71w^2 - 62w + 24 \geq 743/71$ and the remaining cubic block at least $-11/4$. Thus D increases and $D(2/5) = 4904/390625 > 0$, proving the claim.

(Q_+, Const). Use the fully specified parameters

$$\alpha = \frac{h(t)}{w+t}, \quad p = w\alpha, \quad b = t\alpha, \quad \frac{1}{2} \leq t < 1,$$

and

$$\beta = H = h(k), \quad \widehat{b}_L = kH, \quad 0 \leq k \leq w.$$

Since $p + b = h(t)$, the first endpoint inequality gives

$$1 - (b + kH) > G := 2 + z - h(t) - (1+k)H. \tag{95}$$

Put

$$p_0 = 1 + z - K = \frac{w(2-w)}{1+K},$$

let $t_0 \in (1/2, 1)$ be the unique point with $p(t_0) = wh(t_0)/(w+t_0) = p_0$, and put $b_0 = t_0 p_0 / w$. The exact endpoint lemma is

$$b_0 < 1 - wK. \tag{96}$$

To verify it, set $b_* = 1 - wK$ and $T_* = wb_*/p_0$. One has $0 < T_* < 1$, and

$$p_0^2(w+T_*)^2 - w^2 h(T_*)^2 = -\frac{w^3(KE(w) + J(w))}{(1+K)^2(2-w)^2} > 0,$$

where

$$E(w) = 2w^5 + 3w^3 - 20w^2 + 23w - 6,$$
$$J(w) = w^6 + 8w^5 - 22w^4 + 12w^3 + 6w^2 + 2w - 6.$$

Here $J \leq -111/64$; if $E > 0$, then $KE + J < E + J < 0$, because $E + J$ is increasing and equals $-139/64$ at $w = 1/2$. Thus $p(T_*) < p(t_0)$. Since $p(t)$ decreases, $t_0 < T_*$ and (96) follows.

Finally put $P = 1 + z - p$. If $P < K$, then $t < t_0$ and $k \leq w$, so

$$G > 2 + z - h(t_0) - (1+w)K = 1 - b_0 - wK > 0.$$

If $P \geq K$, then $K \leq P \leq P_1 := 1 + z - w/(1+w) \leq 1$ and there is $a \in [0, w]$ with $h(a) = P$. Since $P < H$, one has $k < a$ and $(1+k)H < (1+a)P = P + \ell(P)$, where $\ell(P) = P(1 - \sqrt{4P^2 - 3})/2$. Hence

$$G > \mathcal{D}(P) := 1 - b(P) - \ell(P).$$

This function is concave. Indeed $\ell''(P) > 0$, and the high output b is convex in $p = 1 + z - P$: with $A_t = 2 + w - (1+2w)t > 0$ and $Q_t = 8wt^2 + 4t^2 - 8wt - 13t - w + 4 < 0$,

$$\frac{d^2 b}{dp^2} = -\frac{2h(t)(t+w)^3 Q_t}{w^2 A_t^3} > 0.$$

At $P = K$, (96) gives $\mathcal{D}(K) = 1-b_0-wK > 0$. At $P = P_1$, set $\kappa = w/(1+(1+w)z)$. The linear–constant comparison already proved gives $P_1 > h(\kappa)$, hence $\ell(P_1) < \kappa P_1 = w/(1+w)$, while $b(P_1) = 1/(1+w)$. Thus $\mathcal{D}(P_1) > 0$, and concavity gives $G > 0$ throughout. Equation (95) proves the final hard pair.

The exact list (94) is now exhausted, proving (88). □

4.6.4. *The exactly-one-Vd1/Vd2 placements.* We now assume CE2, $N_+ = 1$, and exactly one vertex triangle is Vd1 or Vd2. The corner normal form of Proposition 3.6 will be used in the following form: for some $t > 0$ and $d = \sqrt{t^2+t+1}$,

$$x-(t+1)y \le a, \qquad ty-(t+1)x \le tb, \qquad tx+y \le d-a-tb. \tag{97}$$

In particular $a + b < 1/2$ for a positive-support vertex triangle.

Lemma 4.44 (Vd1 adjacent rescue)**.** *Assume an original open-triangle skeleton cover in the CE2 exact-M_0 branch, with T_0 the unique Vd1 vertex triangle, $M_1 \in T_0$, T_1 uniquely supercritical, $T_2, \ldots, T_5$ nonsupercritical Vd0, and no other positive-adjacent-support triangle. Then the perimeter together with r_1 is not covered.*

Proof. Lemma 4.12 proves the two local inequalities in (65). Midpoint and support isolation force the center to cover the O-side radial gap before the Vd1 interval. The result is therefore Corollary 4.14. □

Lemma 4.45 (Vd1-pair replacement interface)**.** *In the remaining adjacent Vd1–supercritical placement, the pair can be replaced by two open nonsupercritical Vd0 triangles while preserving full skeleton coverage.*

Proof. This is Proposition 4.48, which uses separate local charts at the two distinguished vertices and proves all reach, triangle, and skeleton-preservation claims explicitly. □

4.6.5. *Final boundary-propagation cases.*

Proposition 4.46 (Cases excluded by boundary propagation)**.** *Boundary-reach propagation proves the following exclusions:*

1. *CE1 or CE2, $N_+ = 0$, all vertex triangles Vd0;*
2. *CE1 or CE2, $N_+ = 0$, with at least one T3-like triangle and no Vd1/Vd2;*
3. *CE1 or CE2, $N_+ = 1$, all vertex triangles Vd0, with at least one boundary gap;*
4. *CE1 or CE2, $N_+ = 1$, exactly one T3-like triangle and no Vd1/Vd2;*
5. *the CE2, $N_+ = 1$, exactly-one-Vd1/Vd2 branch.*

All gap alternatives include singleton gaps.

Proof. The five items are respectively Propositions 4.37, 4.40, 4.35, 4.36, and 4.49. The strictness at open handoffs appears in the weak endpoint bounds, the nonattained supercritical envelope, and the strict replacement margins; no positive-length assumption is made about a V-gap itself. □

4.7. **Exact Endpoint and Replacement Verification.** This section supplies two pieces that must not be hidden by the concise reader presentation. First, it formally incorporates the complete status-bearing $407X$ four-branch endpoint proof. Second, it gives the explicit Vd1–supercritical two-chart replacement in TeX. The incorporated proof-package objects are part of the proof at the exact Git blobs listed below.

4.7.1. *The incorporated* $407X$ *endpoint supplement.* All paths in the following table are relative to the numbered `407X_T3_like_no_Vd1Vd2` proof directory. linear blob identities are stored in `proof/407X_PROVENANCE.json`; the paper prints only stable proof identifiers and twelve-character prefixes.

proof object	blob prefix	triangle
`4073`	`910c7a8a1330`	residual reduction
`4074`	`7c72dfaf7eb3`	(Const, Lin)
`4075`	`ebe2b65f8840`	first Q_- family
`4078`	`f1323325b069`	remaining first-constant family
`4079`	`c66eea05277d`	first-linear exclusion
`407a`	`aa5cb1a6dc63`	first Q_+ family
`407c`	`c80243f67124`	analytic details
`407d`	`e22c85c95fdd`	final assembly

The branchwise endpoint rule used by these files is Proposition 4.32; on the high-radial branch its four-branch formula is Proposition 4.33. Its four branches are

$$\text{Lin}, \qquad \text{Const}, \qquad Q_-, \qquad Q_+.$$

The supplement retains the exact residual inputs a_1^*, a_5^* and radial reach lower bounds c_1^*, c_5^* from Lemma 4.39. It does not replace them by a symmetric or universal envelope.

first branch	right branches	incorporated final source
Const	linear	‘4074‘, low-root separation
Const	Const, Q_-	‘4078‘, exact low-sheet inequalities
Const	Q_+	‘4078‘ and ‘407c‘
Q_-	Const, Q_-	‘4075‘, exact root ordering
Q_-	linear, Q_+	‘4075‘, strict endpoint separation
linear	all branches	‘4079‘, first-linear exclusion
Q_+	linear, Q_+	‘407a‘, high-input exclusion
Q_+	Const, Q_-	‘407a‘ and ‘407c‘, analytic threshold and right-Q_- bounds

TABLE 2. The exhaustive four-branch endpoint case analysis. The listed exact source objects contain every polynomial selector, domain restriction, and strict inequality used in the corresponding entry.

Theorem 4.47 (Complete support-isolated T3-like endpoint certificate)**.** *For every support-isolated state of Definition 4.21,*

$$F_{c_5^*}(a_5^*) + F_{c_1^*}(a_1^*) < 1.$$

Proof. If $a_1^* + a_5^* > 1$, the direct bounds $F_c(a) \le 1 - a$ already give the strict inequality. Assume $a_1^* + a_5^* \le 1$. If an endpoint radial lower bound is at most $1/2$, Proposition 4.32 makes that endpoint the linear value $1 - a$. If it is larger than $1/2$, Proposition 4.33 assigns one of the same four branches. Thus Table 2 remains an exhaustive partition of the ordered branch pairs, including the low-radial right-linear case. The incorporated source in each row proves either that the pair is impossible or that its two exact branchwise outputs have sum strictly below one. File ‘407d‘ checks that the geometric residual reduction in ‘4073‘ reaches exactly these branches and no others. Therefore the displayed endpoint inequality holds throughout the complete support-isolated domain. $\square$

4.7.2. *The explicit Vd1–supercritical replacement.*

Proposition 4.48 (Vd1–supercritical two-chart replacement)**.** *Assume the CE2 branch in which an adjacent Vd1 triangle and the unique supercritical Vd0 triangle lie away from the vertex triangle at the center's unique midpoint. Assume the shared boundary edge is center-free, all other vertex triangles are nonsupercritical Vd0, and no additional triangle has positive support on the two radial arms of the pair. Then the pair can be replaced by two open nonsupercritical Vd0 triangles while preserving coverage of the full skeleton.*

Proof. After a fresh cyclic renumbering, let the Vd1 triangle be based at V_0, the supercritical triangle at V_1, and let the Vd1 triangle contain M_1. Write a, b for the Vd1 boundary reaches and use the Vd corner parameter $t \geq 1$, with $d = \sqrt{t^2+t+1}$. Put

$$c = \frac{d-a-tb}{t+1}, \qquad \lambda = \frac{t(1-b)}{t+1}, \qquad u_{\mathrm{adj}} = \frac{d-a-tb-1}{t}.$$

The strict midpoint and one-sided-support inequalities give

$$a+c<1, \qquad a<\lambda \leq \frac{1}{2}, \qquad b<\frac{1}{2}, \qquad u_{\mathrm{adj}}<1-a. \tag{1}$$

Let (A_1, B_1, C_1) be the maximal reaches of the supercritical triangle. The center-free shared edge and radial coverage give

$$A_1 \geq 1-b > \frac{1}{2}, \qquad C_1 \geq \lambda.$$

The selected supercritical admissible cell has $C_1 \leq 1/2$. The half-square lemma

$$(x,y,z) \in \mathcal{A}, \qquad x \geq \frac{1}{2}, \qquad z \leq \frac{1}{2} \quad \Longrightarrow \quad y \leq 1-z$$

therefore gives $B_1 \leq 1-C_1 \leq 1-\lambda$, and hence

$$a + B_1 < 1. \tag{2}$$

If $c_1^{\mathrm{req}} = 1 - d_1^C$ is the vertex-side beginning of the center trace on r_1, open skeleton coverage gives

$$c_1^{\mathrm{req}} < \max\{C_1, u_{\mathrm{adj}}\} \leq \max\{1-B_1, 1-a\}. \tag{3}$$

Choose $p_2 \in (a, 1-B_1)$ with

$$\max\{p_2, 1-p_2\} > c_1^{\mathrm{req}};$$

such a point exists because the supremum of the left side on that interval is $\max\{1-B_1, 1-a\}$. Then choose

$$a<p_1<p_2, \qquad p_1<\frac{1}{2}, \qquad 1-p_1>c.$$

Finally choose $\varepsilon > 0$ smaller than

$$p_1-a, \quad p_2-p_1, \quad 1-B_1-p_2, \quad 1-p_1-c, \quad \max\{p_2, 1-p_2\}-c_1^{\mathrm{req}}. \tag{4}$$

Use two distinct local charts:

$$X_0(x,y) = V_0 + x(V_5-V_0) + y(V_1-V_0),$$
$$X_1(x,y) = V_1 + x(V_0-V_1) + y(V_2-V_1).$$

For $p \leq 1/2$ put

$$\Delta_p^- = \operatorname{conv}\{(0,1-p),(1,1-p),(0,-p)\},$$

and for $p > 1/2$ put
$$\Delta_p^+ = \operatorname{conv}\{(p,0),(p,1),(p-1,0)\}.$$
Define
$$U_0' = X_0\big(\operatorname{int}(\Delta_{p_1}^-) + (-\varepsilon,0)\big)$$
and
$$U_1' = \begin{cases} X_1\big(\operatorname{int}(\Delta_{p_2}^-) + (-\varepsilon,0)\big), & p_2 \le 1/2, \\ X_1\big(\operatorname{int}(\Delta_{p_2}^+) + (0,-\varepsilon)\big), & p_2 > 1/2. \end{cases}$$

In its own chart the shifted minus template contains the origin and has reach triple
$$(p-\varepsilon, 1-p, 1-p),$$
while the shifted plus template contains the origin and has reach triple
$$(p, 1-p-\varepsilon, p).$$
Their closures stay strictly below both adjacent-arm lines $x = 1$ and $y = 1$. Thus $V_0 \in U_0'$, $V_1 \in U_1'$, both triangles are Vd0, and both boundary sums equal $1-\varepsilon < 1$.

The first replacement reaches beyond a on $e_{5,0}$ and beyond c on r_0. On the shared edge its reach is $1-p_1$, while the second replacement reaches at least $p_2-\varepsilon$ from V_1; their sum exceeds one by (4). The second replacement reaches beyond B_1 on $e_{1,2}$ and has radial reach $\max\{p_2, 1-p_2\} > c_1^{\mathrm{req}}$ on r_1. Hence every boundary and radial component formerly supplied by the special pair remains covered, and all other skeleton components are unchanged.

The modified triangles therefore cover the full skeleton and have six nonsupercritical Vd0 vertex triangles. Proposition 4.37 gives the contradiction. □

4.8. **Authoritative CE2 One-Vd Placement Assembly.** The following proposition is the sole CE2 exactly-one-Vd placement assembly used by the final proof. It cites only complete placement lemmas and the explicit two-chart replacement in Appendix 4.7.

Proposition 4.49 (Complete CE2 exactly-one-Vd1/Vd2 assembly)**.** *Assume the center has type CE2, $N_+ = 1$, exactly one vertex triangle has type Vd1 or Vd2, and no T3-like triangle is present. Then the seven triangles do not cover H.*

Proof. Normalize the unique center midpoint to M_0. If an additional vertex triangle has positive adjacent-arm support, then the unique supercritical vertex triangle, the Vd1/Vd2 vertex triangle, and that additional triangle are three short triangles; the skeleton contradiction of Theorem 3.15 applies. Hence assume the no-additional-positive-support complement.

First suppose the unique supercritical vertex triangle is T_0. If the Vd1/Vd2 vertex triangle is adjacent, it is eliminated by Lemma 4.15; the proof treats both possible supported arms and its reflection. If it is nonadjacent, its index lies in $\{2,3,4\}$ after reflection, and Lemma 4.16 gives a strict radial separation.

Next suppose the unique Vd1/Vd2 vertex triangle is T_0. It must rescue the midpoint of the unique supercritical vertex triangle, so midpoint locality makes that vertex triangle adjacent. If T_0 is Vd2, the neighboring-midpoint cap Lemma 3.7, combined with the center and supercritical perimeter caps, is Proposition 3.10. If T_0 is Vd1, the exact Vd1 local profile and the common center-hiding theorem give Corollary 4.14.

Finally suppose neither special vertex triangle is T_0. Since the exact-M_0 center covers no other radial midpoint, the Vd1/Vd2 vertex triangle must rescue the midpoint missed by the unique supercritical vertex triangle. Diameter locality makes the two vertex triangles adjacent. A Vd2 rescue is again eliminated by the neighboring-midpoint cap. For a Vd1 rescue, the shared edge has no center trace in this placement and the no-additional-support hypotheses are exactly those of Proposition 4.48. Replace the pair by two open nonsupercritical Vd0 triangles while preserving every boundary and radial reach lower bound used by Proposition 4.37. All other vertex triangles were already nonsupercritical Vd0, so that all-Vd0 theorem gives the contradiction.

These alternatives exhaust the relative positions of the center-midpoint vertex triangle, the unique supercritical vertex triangle, and the unique Vd1/Vd2 vertex triangle. □

5. Area-Loss Estimate

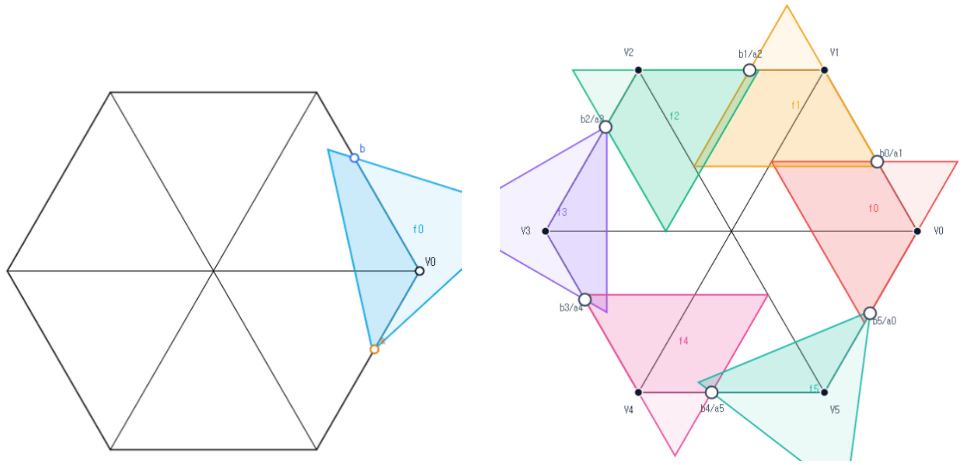
The third mechanism measures the normalized area forced outside H by a vertex triangle with prescribed boundary reaches. The local square-loss and T3-like estimates are proved first, then summed around the six-cycle to close the two CE0 area branches.

5.1. **Detailed Area-Loss Estimates.** This section proves the local inequalities and cyclic estimates used in the area-loss argument.

Let

$$A_{\triangle} = \frac{\sqrt{3}}{4}$$

be the area of a unit equilateral triangle. Throughout this section areas are divided by $A_{\triangle}$. Thus a unit triangle has normalized area one and H has normalized area six.



(a) local retained area (b) global retained areas

FIGURE 8. Schematic views of Method 3, not to scale. The local panel shows one vertex triangle with boundary reaches a, b, and the global panel shows the six cyclic triangles. Only the darker portions represent area retained inside H; the proof uses the inequalities below, not the picture.

5.1.1. *The local square-loss inequalities.* Fix a vertex V_i. Let P_a and P_b be the points at distances a and b from V_i on its incoming and forward boundary edges. For a closed unit equilateral triangle T containing V_i, P_a, P_b, put

$$\mathcal{L}(T) = \frac{\operatorname{area}(T \setminus H)}{A_\triangle}.$$

Equivalently, if

$$f(a,b) = \max_T \frac{\operatorname{area}(T \cap H)}{A_\triangle}, \qquad \mathcal{L}(a,b) = 1 - f(a,b),$$

then $\mathcal{L}(a,b)$ is the least possible normalized loss. Reflection in the radial axis interchanges a and b, and increasing either reach lower bound can only decrease f.

Lemma 5.1 (Local wedge reduction)**.** *After rotating indices to $i = 0$, write*

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0), \qquad W = \{(x,y) : x \geq 0,\ y \geq 0\}.$$

If a closed unit equilateral triangle T contains V_0, then

$$T \cap H = T \cap W.$$

Normalized Euclidean area is twice the corresponding (x,y)-coordinate area.

Proof. The two basis vectors have length one and angle $120°$, so

$$\|(x,y)\|^2 = x^2 + y^2 - xy, \qquad |\det(V_1 - V_0, V_5 - V_0)| = \frac{\sqrt{3}}{2}.$$

The hexagon is

$$H = \{0 \leq x, y \leq 2,\ |x - y| \leq 1\} \subset W.$$

Conversely, if $(x,y) \in T \cap W$, then the diameter-one condition gives $x^2+y^2-xy \le 1$. Hence

$$|x-y|^2 \le x^2+y^2-xy \le 1, \qquad \frac{3}{4}x^2 \le x^2+y^2-xy,$$

and the same estimate holds for y. Thus $|x-y| \le 1$ and $0 \le x, y \le 2/\sqrt{3} < 2$, proving $(x,y) \in H$. The determinant identity proves the area assertion. □

Theorem 5.2 (Unconditional local square loss)**.** *For every feasible pair $0 \le a, b \le 1$ and every realizing closed unit equilateral triangle T,*

$$\mathcal{L}(T) \ge \min(a,b)^2. \tag{98}$$

If $a+b>1$, then

$$\mathcal{L}(T) \ge \max(a,b)^2. \tag{99}$$

Consequently the same inequalities hold for $\mathcal{L}(a,b) = 1 - f(a,b)$.

Proof. In the coordinates of Lemma 5.1, physical rotation through $60°$ is $R(x,y) = (x-y, x)$. Put

$$0 \le t \le 1, \qquad D = 1 - t + t^2, \qquad z = \sqrt{D}.$$

After choosing one of the six directed sides, every orientation has one of the following two forms:

$$\begin{array}{c|cc} & d & Rd \\ \hline \mathrm{I} & z^{-1}(1,t) & z^{-1}(1-t,1) \\ \mathrm{II} & z^{-1}(t,-(1-t)) & z^{-1}(1,t). \end{array} \tag{100}$$

Indeed these forms cover, respectively, the physical direction sectors $[120°, 180°]$ and $[60°, 120°]$; their endpoints cover the common boundary orientations.

For Type I introduce

$$U = \alpha + y - tx, \qquad V = \beta + x - (1-t)y.$$

Then T is the simplex $U, V \ge 0$, $U+V \le z$. Containment of $(0,0), (0,a), (b,0)$ gives

$$\alpha \ge tb, \quad \beta \ge (1-t)a, \quad \alpha+\beta+(1-t)b \le z, \quad \alpha+\beta+ta \le z. \tag{101}$$

The two inequalities defining W become

$$(1-t)U + V \ge (1-t)\alpha + \beta, \qquad U + tV \ge \alpha + t\beta.$$

For fixed t, lowering α or β therefore enlarges $T \cap W$. The outside loss is minimized at $\alpha = tb$, $\beta = (1-t)a$. At that placement the portion outside W has vertices, in the (U,V)-plane,

$$(0,0), \quad (a+tb, 0), \quad (tb, (1-t)a), \quad (0, b+(1-t)a).$$

Since the Jacobian of $(x,y) \mapsto (U,V)$ has absolute value D,

$$\mathcal{L}(T) \ge \mathcal{L}_{\mathrm{I}}(a,b;t) = \frac{(1-t)a^2 + tb^2 + 2t(1-t)ab}{D}. \tag{102}$$

If $a \le b$, then

$$D(\mathcal{L}_{\mathrm{I}} - a^2) = t\{b^2 - a^2 + (1-t)(a^2+2ab)\} \ge 0;$$

the reflected factorization proves $\mathcal{L}_{\mathrm{I}} \ge b^2$ when $b \le a$. This proves (98) in Type I.

Suppose now that $a+b>1$ and $a\le b$. Set

$$r=\frac{a}{b},\qquad t_0=\frac{1-r^2}{1+2r}.$$

Since $b>1/(1+r)$, feasibility in (102) and (101) excludes $0\le t<t_0$: for $0<t<t_0$,

$$\left(\frac{1+r(1-t)}{1+r}\right)^2-D=\frac{t(1+2r)(t_0-t)}{(1+r)^2}>0,$$

so $b+(1-t)a>z$, while $t=0$ would give $a+b\le 1$. Hence $t\ge t_0$, and

$$D(\mathcal{L}_{\mathrm{I}}-b^2)=(1-t)b^2\{r^2-1+t(2r+1)\}\ge 0.$$

If $b\le a$, put $r=b/a$ and $t_1=(r^2+2r)/(1+2r)$. The identical reflected calculation excludes $t>t_1$ and gives

$$D(\mathcal{L}_{\mathrm{I}}-a^2)=ta^2\{r^2+2r-t(1+2r)\}\ge 0.$$

Thus (99) holds in Type I.

For Type II put

$$U=\alpha+(1-t)x+ty,\qquad V=\beta+tx-y.$$

Again T is the simplex $U,V\ge 0$, $U+V\le z$. Its exact reaches on the positive y- and x-axes are

$$A=\beta,\qquad B=z-\alpha-\beta,\qquad A+B\le z\le 1. \tag{103}$$

In particular Type II is impossible when $a+b>1$. Its wedge intersection is the quadrilateral with vertices

$$(0,0),\quad (B,0),\quad \left(\frac{B+(1-t)A}{D},\frac{A+tB}{D}\right),\quad (0,A).$$

Hence

$$\mathcal{L}(T)=\mathcal{L}_{\mathrm{II}}(A,B;t)=1-\frac{(1-t)A^2+tB^2+2AB}{D}. \tag{104}$$

Let $S=A+B$. If $A\le B$, write $A=rS$, $B=(1-r)S$ with $0\le r\le 1/2$. Since $S^2\le D$, and with

$$C=(1-t)r^2+t(1-r)^2+2r(1-r),$$

we have $\mathcal{L}_{\mathrm{II}}\ge 1-C$. The exact factorization

$$1-C-r^2D=(1-t)(1-2r+tr^2)\ge 0$$

gives $\mathcal{L}_{\mathrm{II}}\ge r^2D\ge A^2$. Reflection gives $\mathcal{L}_{\mathrm{II}}\ge B^2$ when $B\le A$. Since $A\ge a$ and $B\ge b$, this proves (98). The two orientation types are exhaustive, so the theorem follows. ☐

5.1.2. *The direct* T3-like *loss.* The next theorem uses the classification data (o,n) from Section 2.1: o counts triangle vertices outside H, and n counts adjacent radial arms met in positive length. Thus a T3-like triangle has $(o,n)=(2,1)$.

Theorem 5.3 (Direct T3-like loss)**.** *Let T be a closed unit V_i-triangle of* T3-like *type, and let a,b be lower bounds for its adjacent boundary reaches. Then*

$$a+b\le 1. \tag{105}$$

Moreover, for $0\le m\le 1/2$,

$$a,b\ge m\quad\Longrightarrow\quad \mathcal{L}(T)\ge 2m-4m^2. \tag{106}$$

Proof. The first assertion is Lemma 2.8. For the loss bound apply the closed-trace domination in Proposition 2.6. It gives a translated Type-II triangle $\widehat{T}$ of the same area with

$$T \cap H \subset \widehat{T} \cap H, \qquad \mathcal{L}(T) \geq \mathcal{L}(\widehat{T}).$$

In the notation of that proposition, $0 < t < 1$, $D = 1 - t + t^2$, $z = \sqrt{D}$, and after reflection the translated majorant has exact reaches

$$A = \beta, \qquad B = z - \beta$$

and unique wedge vertex

$$Q = \left(\frac{B + (1-t)A}{D}, \frac{A + tB}{D} \right), \qquad Q_x > 1, \quad Q_y \leq 1.$$

Thus $z - tA > D$, so

$$A < L := \frac{z - D}{t} = \frac{z(1-z)}{t}. \tag{107}$$

The exact normalized loss is

$$\mathcal{L}(\widehat{T}) = \mathcal{L}(z, A) = \frac{tA^2 + (1-t)(z-A)^2}{D}.$$

This decreases for $A < L < (1-t)z$, and therefore

$$\mathcal{L}(T) \geq \mathcal{L}(\widehat{T}) > \mathcal{L}(z, L). \tag{108}$$

Set $r = (1-z)/t$. Then $0 < r < 1/2$, and eliminating t, z gives

$$t = \frac{1 - 2r}{1 - r^2}, \qquad z = \frac{r^2 - r + 1}{1 - r^2},$$

$$L = A_0(r) := \frac{r(r^2 - r + 1)}{1 - r^2}, \qquad z - L = B_0(r) := \frac{r^2 - r + 1}{1 + r}.$$

The limiting normalized inside area and loss are

$$F_0(r) = \frac{(r^2 - r + 1)^2}{1 - r^2}, \qquad \mathcal{L}_0(r) = 1 - F_0(r).$$

Since the exact reach A is at least the required reach lower bound $a \geq m$, (107) gives $A_0(r) > m$. Direct simplification yields

$$\mathcal{L}_0(r) - \{2A_0(r) - 4A_0(r)^2\} = \frac{r^2(5r^4 - 8r^3 + 13r^2 - 8r + 2)}{(1 - r^2)^2} \geq 0,$$

because the last polynomial is at least $9r^2 - 8r + 2 \geq 2/9$ on $[0, 1/2]$. Also $A_0(r) \leq r$, and

$$\mathcal{L}_0(r) - \frac{1}{4} = \frac{(1 - 2r)(2r^3 - 3r^2 + 6r - 1)}{4(1 - r^2)} \geq 0 \quad \text{when } A_0(r) \geq \frac{1}{4};$$

the cubic is increasing and has value $11/32$ at $r = 1/4$.

Finally $\psi(u) = 2u - 4u^2$ is increasing on $[0, 1/4]$ and never exceeds $1/4$ on $[0, 1/2]$. If $A_0(r) \leq 1/4$, then $\mathcal{L}_0(r) \geq \psi(A_0(r)) \geq \psi(m)$; if $A_0(r) \geq 1/4$, then $\mathcal{L}_0(r) \geq 1/4 \geq \psi(m)$. Together with (108), this proves (106). The reflected crossing is identical. $\square$

5.1.3. *The cyclic area certificates.* The selected handoffs of Proposition 2.9 have the form

$$(a_i, b_i) = (1 - \xi_{i-1}, \xi_i), \qquad 0 < \xi_i < 1,$$

with cyclic indices. Vertex triangle i is selected supercritical exactly when $\xi_i > \xi_{i-1}$.

Lemma 5.4 (Cyclic area-loss certificate)**.** *Let the six actual vertex triangles realize selected handoffs*

$$(a_i, b_i) = (1 - \xi_{i-1}, \xi_i), \qquad 0 < \xi_i < 1.$$

Their total normalized loss is strictly greater than one if either

(1) *at least two selected vertex triangles are supercritical; or*
(2) *exactly one selected vertex triangle is supercritical, every triangle is* Vd0 *or* T3-like*, and at least one triangle is* T3-like*.*

Proof. Put $m = \min_i \xi_i$ and $M = \max_i \xi_i$. The reflection $y_i = 1 - \xi_{-i-1}$ swaps each vertex triangle's two coordinates and preserves both alternatives and the total loss. Hence assume $m \le 1 - M$. Every selected vertex triangle coordinate is then at least m, so (98) gives loss at least m^2 for every vertex triangle.

In case (1), choose an ascent out of a minimum plateau, $\xi_{p-1} = m < \xi_p$. Its vertex triangle contains the coordinate $1 - m$ and loses at least $(1 - m)^2$ by (99). A second ascent has one coordinate strictly larger than $1/2$ and hence loses strictly more than $1/4$. The other four vertex triangles lose at least m^2 each, so the total loss is strictly larger than

$$4m^2 + (1 - m)^2 + \frac{1}{4} = 5\left(m - \frac{1}{5}\right)^2 + \frac{21}{20} > 1.$$

In case (2), rotate so that the unique ascent leaves the minimum: $\xi_5 = m < \xi_0 = M$. The supercritical vertex triangle loses at least $(1-m)^2$. Lemma 2.8 shows that a T3-like vertex triangle is different from it, and Theorem 5.3 gives that vertex triangle loss at least $2m - 4m^2$. The remaining four vertex triangles lose at least m^2 each. Their total loss is at least

$$(1 - m)^2 + (2m - 4m^2) + 4m^2 = 1 + m^2 > 1.$$

□

Proposition 5.5 (Branches closed by area)**.** *The following branches are impossible:*

(1) T_C *is* CE0*,* $N_+ \ge 2$*, with arbitrary vertex-triangle types;*
(2) T_C *is* CE0*,* $N_+ = 1$*, no triangle is* Vd1 *or* Vd2*, and at least one triangle is* T3-like*.*

Proof. In the CE0 class, the center triangle has no positive-length boundary trace. If the six vertex triangles missed a boundary point, openness of the center triangle would give such a trace. Hence the vertex triangles cover ∂H, so Proposition 2.9 applies.

If $N_+ \ge 2$, its at-least-two clause selects handoffs with at least two supercritical vertex triangles. Lemma 5.4 makes the six vertex triangles' total normalized inside area strictly less than five. The center triangle contributes at most one, so the total is strictly less than six.

If $N_+ = 1$, the exact-one clause preserves the unique actual supercritical index for every strict selection. The exhaustive vertex-type hypothesis gives case (2) of Lemma 5.4; the six vertex triangles contribute less than five normalized area, and the center triangle contributes at most one. Again the sum is strictly below the normalized area six of H. $\square$

6. Nine-Point Enclosure Obstruction

The fourth mechanism applies when all six vertex triangles are Vd0, the boundary is covered, and exactly one vertex triangle is supercritical. Six radial points and three asymmetric frontier points are forced into the center triangle. The detailed construction is followed by the support-arc covering argument and the final enclosure contradiction.

6.1. **Detailed Nine-Point Enclosure Argument.**

This section supplies the exact frontier, witness, sign, and cap-overlap calculations for the direct nine-point obstruction.

This section proves a center-class-independent theorem. Its hypotheses are that all six vertex triangles are Vd0, that they cover the full boundary, and that exactly one actual boundary vertex triangle is supercritical. The theorem will therefore apply both to the CE0 branch and to the zero-gap subcases of CE1 and CE2. Six common radial points and three asymmetric points are excluded directly from all six actual vertex triangles and are therefore forced into the center triangle. Only the unique supercritical vertex triangle uses an AB-union; no nonsupercritical model core is introduced.

6.1.1. *The finite caliper criterion.* We first state the finite caliper criterion used below. It enters the proof through the exact statement and verification argument given here; neither a plot nor an unrecorded numerical convention is used.

Let R and J denote counterclockwise rotations through $2\pi/3$ and $\pi/2$, respectively. For a nonempty finite set $S \subset \mathbb{R}^2$ and a vector N, put

$$W_S(N) = \sum_{k=0}^{2} \max_{z\in S} \langle z, R^k N\rangle. \tag{109}$$

Theorem 6.1 (Finite caliper certificate)**.** *If S has at least two distinct points, then S is contained in a closed unit equilateral triangle if and only if there are distinct $p, q \in S$ and $\varepsilon \in \{-1, 1\}$ for which*

$$W_S\big(\varepsilon J(q-p)\big) \le \frac{\sqrt{3}}{2}\|p-q\|. \tag{110}$$

If S consists of one point, it is contained in equilateral triangles of arbitrarily small positive side length.

For $S_x = \{O, A, B, x\}$, every clause of (110) *is a finite polynomial clause of degree at most four in the coordinates of A, B, x; at most $12 \cdot 4^3$ clauses are required, and repeated point labels merely delete zero-length pairs.*

Proof. For a unit vector n, write

$$G_S(n) = \sum_{k=0}^{2} h_S(R^k n), \qquad h_S(n) = \max_{z\in S}\langle z, n\rangle.$$

The three supporting half-planes with normals n, Rn, R^2n form an equilateral triangle of side $2G_S(n)/\sqrt{3}$. Hence S fits in a closed unit equilateral triangle precisely when $\min_{\|n\|=1} G_S(n) \le \sqrt{3}/2$.

On an open normal-direction cell on which the three support points are fixed, $G_S(\theta) = c\cos\theta + d\sin\theta$ and $G_S'' = -G_S < 0$ unless all points coincide. Thus a minimum occurs at a cell boundary, where some support line contains two distinct points p, q. After a cyclic rotation of the three normals, its normal is $\varepsilon J(q-p)/\|p-q\|$. Homogeneity yields (110). A nondegenerate segment is checked directly with its two normals.

For the polynomial expansion fix (p, q, ε), put $N = \varepsilon J(q-p)$ and $N_k = R^k N$, and choose support labels $z_0, z_1, z_2 \in S_x$. The support cell is

$$\langle z_k, N_k\rangle \geq \langle z, N_k\rangle \quad (z \in S_x,\ k = 0, 1, 2), \tag{111}$$

with $\|p-q\|^2 > 0$. Since $W_S(N) \geq 0$, the remaining test is

$$4\left(\sum_{k=0}^{2}\langle z_k, N_k\rangle\right)^2 \leq 3\|p-q\|^2. \tag{112}$$

The points are affine-linear in the parameters, so (111)–(112) have degree at most four. Six unordered pairs, two signs, and 4^3 support-label triples give the asserted finite exhaustive list. □

6.1.2. *Corner-clipped AB-unions.* Let e_A, e_B be unit vectors with angle 120°, put

$$O = 0, \qquad A = ae_A, \qquad B = be_B, \qquad C = \{ue_A + ve_B : u, v \geq 0\},$$

and define

$$\mathcal{R}(a,b) = C \cap \bigcup\{T : T \text{ is a closed unit equilateral triangle and } O, A, B \in T\}. \tag{113}$$

The finite description in Theorem 6.1 applies to the four-point set $\{O, A, B, x\}$ and makes membership in (113) an exact finite semialgebraic condition of degree at most four.

The strict supercritical branch admits a simpler closed form. In the next statement write $x = ue_A + ve_B$ and $\|x\|^2 = u^2 + v^2 - uv$.

Theorem 6.2 (Strict supercritical AB-union)**.** *Assume*

$$a + b > 1, \qquad \rho = a^2 + ab + b^2 < 1,$$

put $h = \sqrt{3}/2$, $D = \sqrt{4\rho - 3}$, and define

$$\alpha = \frac{h(a + 2b - aD)}{2\rho}, \qquad \beta = \frac{h(a - b + (a+b)D)}{2\rho},$$

$$\gamma = \frac{h(-a + b + (a+b)D)}{2\rho}, \qquad \delta = \frac{h(2a + b - bD)}{2\rho}.$$

Then all four coefficients are positive and

$$\mathcal{R}(a,b) = (C \cap \mathcal{D}_A \cap \mathcal{H}_A) \cup (C \cap \mathcal{D}_B \cap \mathcal{H}_B), \tag{114}$$

where

$$\mathcal{D}_A = \{(u,v) : (u-a)^2 + v^2 - (u-a)v \leq 1\},$$
$$\mathcal{D}_B = \{(u,v) : u^2 + (v-b)^2 - u(v-b) \leq 1\},$$
$$\mathcal{H}_A = \{(u,v) : \alpha(u-a) + \beta v \leq 0\},$$
$$\mathcal{H}_B = \{(u,v) : \gamma u + \delta(v-b) \leq 0\}.$$

Its non-axis frontier consists, in counterclockwise order, of the A-circle, the A-line, the B-line, and the B-circle.

Proof. The identities

$$4\rho - 3(a+b)^2 = (a-b)^2, \qquad (a+b)^2 D^2 - (a-b)^2 = 4\rho((a+b)^2 - 1)$$

give $0 < D < 1$ and positivity of the four coefficients.

We apply the caliper criterion of Theorem 6.1. For a point $x = (u, v)$ in the relative interior of C but outside $\triangle OAB$, the four hull vertices occur in the order O, B, x, A. The two cone-axis hull edges cannot certify a unit triangle: their support sums, divided by h, are respectively

$$\max\{a, u\} + \max\{b, v-u\}, \qquad \max\{b, v\} + \max\{a, u-v\},$$

and are at least $a + b > 1$. It remains to test the Ax and Bx calipers.

For the Ax caliper put $p = u - a$ and $r^2 = p^2 + v^2 - pv$. Exact evaluation of the three supports gives

(115)
$$\frac{G_A}{h} = \frac{\max\{r^2, -ap + (a+b)v\} + N_A}{r}, \quad N_A = \begin{cases} 0, & p \le 0, \\ ap, & 0 \le p \le v, \\ (a+b)p - bv, & p \ge v. \end{cases}$$

No support label is omitted in these three sectors. If $p \le 0$, $G_A \le h$ is equivalent to

$$r \le 1, \qquad -ap + (a+b)v \le r.$$

The second inequality is the half-plane condition in (114), because

$$r^2 - (-ap + (a+b)v)^2 = \frac{(cp + dv)(c'p + d'v)}{4\rho},$$

where

$$c = a + 2b - aD, \quad d = a - b + (a+b)D, \quad c' = a + 2b + aD, \quad d' = a - b - (a+b)D,$$

and $c, c', d > 0 > d'$. Thus the low sector is exactly $\mathcal{D}_A \cap \mathcal{H}_A$. In the middle sector $0 < p \le v$ one has $r \le v$, whereas the numerator in (115) is at least $(a+b)v > r$, so no fit is possible.

In the high sector $p \ge v$, the inequalities $G_A \le h$ imply

$$r^2 + (a+b)p - bv \le r, \qquad bp + av \le r. \tag{116}$$

Write $x - A = ry(\theta)$, $0 \le \theta \le 60°$, with cone coordinates

$$y(\theta) = te_A + we_B, \quad t = \frac{2}{\sqrt{3}}\cos(\theta - 30°), \quad w = \frac{2}{\sqrt{3}}\sin\theta.$$

Then (116) gives

$$r \le f(\theta) = 1 - (a+b)t + bw, \qquad S(\theta) = bt + aw \le 1.$$

Let n_B be the unit vector with dual coordinates (γ, δ). The identities

$$b\gamma + (a+b)\delta = h, \quad (a+b)\gamma + a\delta = \frac{h}{2}(1+D), \quad b\delta - a\gamma = \frac{h}{2}(1-D)$$

show that the unique angle θ_B of this normal satisfies $S(\theta_B) = 1$ and $f(\theta_B) = (1-D)/2$. The function S has at most one critical point, a maximum, so $S(\theta) \le 1$

implies $\theta \le \theta_B$. On the feasible part $f, f' \ge 0$; hence $f(\theta)\cos(\theta_B - \theta)$ is increasing. Consequently

$$\langle n_B, x - B\rangle \le a\gamma - b\delta + \frac{h}{2}(1-D) = 0.$$

The diameter bound also gives $\|x - B\| \le 1$. Thus every high sector fit belongs to $\mathcal{D}_B \cap \mathcal{H}_B$. Reflection exchanges a, u with b, v and supplies the corresponding result for the Bx caliper. This proves (114) in the cone interior; the axes follow by taking limits, since both sides are closed.

For completeness, the line–circle junctions are obtained by solving the two displayed line and circle equations. The line normals satisfy

$$\alpha^2 + \alpha\beta + \beta^2 = h^2, \qquad \gamma^2 + \gamma\delta + \delta^2 = h^2,$$

and the line determinant is

$$\alpha\delta - \gamma\beta = \frac{3(1-D)(D+3)}{16\rho} > 0.$$

The successive cone slopes are

$$+\infty, \quad \frac{2}{1-D}, \quad \frac{a(1-D)+2b}{2a+b(1-D)}, \quad \frac{1-D}{2}, \quad 0;$$

the two middle comparisons reduce to $a(4-(1-D)^2) > 0$ and $b(4-(1-D)^2) > 0$. This proves the asserted four-piece order. □

6.1.3. *Exact-one handoffs and common reach bounds.*

Lemma 6.3 (Exact-one handoff chain)**.** *Assume the six vertex triangles cover ∂H and exactly one actual vertex triangle is supercritical. Rotate its index to* 4*. The strict handoffs supplied by Proposition 2.9 satisfy*

$$\xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4.$$

With

$$a = 1 - \xi_3, \qquad b = \xi_4, \qquad p = 1 - b, \qquad q = 1 - a,$$

one has

$$0 < a, b < 1, \qquad a + b > 1, \qquad a^2 + ab + b^2 < 1, \tag{117}$$

and every selected vertex triangle obeys

$$a_i \ge p, \qquad b_i \ge q.$$

Proof. Every selected vertex triangle other than 4 is strictly nonsupercritical, so $\xi_i < \xi_{i-1}$ for $i \ne 4$, which is precisely (6.3). In particular ξ_3 is the global minimum and ξ_4 the global maximum. The two anchors of vertex triangle 4 lie in its open triangle; their mutual squared distance is $a^2 + ab + b^2$, strictly below the diameter squared of the closed unit triangle. This proves (117). Finally every $\xi_j \in [\xi_3, \xi_4]$, so $1 - \xi_{i-1} \ge 1 - \xi_4 = p$ and $\xi_i \ge \xi_3 = q$. □

6.1.4. *Six directly forced radial points.* Put $h = \sqrt{3}/2$ and

$$s = p + q, \qquad m = \min(p, q), \qquad M = \max(p, q), \qquad \chi = s^4 - s^2 + pq.$$

The strict domain (117) gives

$$pq > (1 - s)(2 - s), \qquad m > 1 - s, \qquad s > 1 - m. \tag{118}$$

Let $c_* = c_{\max}(p, q)$ be the exact radial envelope of Proposition 4.31. On the present subcritical domain, equation (68) becomes

$$c_* = \begin{cases} C_L(m), & \chi \le 0, \\ \dfrac{2M}{1 + \sqrt{4s^2 - 3}}, & \chi > 0, \end{cases} \tag{119}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

At $\chi = 0$ the two expressions agree and equal s.

Lemma 6.4 (Direct radial forcing)**.** *One has*

$$0 < c_* < 1, \qquad c_* > 1 - m. \tag{120}$$

Assume that every U_i is Vd0 *and that $U_C, U_0, \dots, U_5$ cover H. Then, for every i,*

$$D_i = (1 - c_*)V_i \in U_C.$$

Consequently U_C contains the centered closed disk

$$\mathcal{D}_\eta = \{x : \|x\| \le \eta\}, \qquad \eta = h(1 - c_*). \tag{121}$$

Proof. On the C_L branch, $C_L(m) < 1$ because the defining polynomial is positive at 1; if $s < \sqrt{3}/2$, then $c_* \ge \sqrt{3}/2 > s > 1 - m$, while if $s \ge \sqrt{3}/2$ its value at s is $\chi \le 0$, so $c_* \ge s > 1 - m$. On the other branch put $E = \sqrt{4s^2 - 3}$. The selector gives $sE > M - m$, whence $c_* < s < 1$. Moreover, with

$$A_0 = \frac{2M}{1 - m} - 1,$$

one has

$$A_0^2 - E^2 = \frac{4(1 - s)\{(1 - m)(1 - 2m) + m(2 - m)(1 - s)\}}{(1 - m)^2} > 0.$$

Thus $A_0 > E$ and $c_* > 1 - m$. This proves (120).

Fix i. If $D_i \in U_i$, openness supplies $\varepsilon > 0$ with $c_* + \varepsilon < 1$ such that $(1 - c_* - \varepsilon)V_i \in U_i$. The same triangle contains its two selected handoff anchors. Since their incoming and forward-reach lower bounds dominate p and q, convexity and passage to the closure give a closed unit triangle realizing $(p, q, c_* + \varepsilon)$. This contradicts the definition of $c_* = c_{\max}(p, q)$. Hence $D_i \notin U_i$.

If D_i belonged to U_{i-1} or U_{i+1}, a small plane ball inside that open triangle would meet the adjacent radial arm r_i in a positive-length interval. This contradicts the Vd0 definition. For the remaining triangles put $r = 1 - c_* > 0$. Directly,

$$\|rV_i - V_{i\pm 2}\|^2 = 1 + r + r^2 > 1, \qquad \|rV_i - V_{i+3}\| = 1 + r > 1.$$

Since $V_j \in U_j$ and two points of an open unit triangle are at distance strictly below 1, these inequalities exclude D_i from the other three vertex triangles. Thus no vertex triangle contains D_i. The point lies in $\mathrm{int}(H)$ by (120), so the assumed cover forces $D_i \in U_C$.

The six points D_i form a regular hexagon of circumradius $1 - c_*$. Their convex hull contains its centered incircle of radius $h(1 - c_*)$, and convexity of U_C gives (121). $\square$

We next define the three asymmetric witnesses. At V_4 use local coordinates

$$P = V_4 + u(V_5 - V_4) + v(V_3 - V_4), \qquad \mathsf{q}(u, v) = u^2 + v^2 - uv.$$

The forward-reach lower bound is b and the backward-reach lower bound is a. Accordingly, set

$$\alpha = \frac{h(b + 2a - bD)}{2\rho}, \qquad \beta = \frac{h(b - a + (a + b)D)}{2\rho},$$
$$\gamma = \frac{h(-b + a + (a + b)D)}{2\rho}, \qquad \delta = \frac{h(2b + a - aD)}{2\rho},$$

where now $\rho = a^2 + ab + b^2$ and $D = \sqrt{4\rho - 3}$. The two line pieces of Theorem 6.2 are

$$L_-(\lambda) = (b - \beta\lambda, \alpha\lambda), \qquad L_+(\mu) = (\delta\mu, a - \gamma\mu).$$

They meet at $\lambda_* = \mu_* = 8h\rho/(3(D + 3))$ in

$$J = \left(\frac{2b + a - aD}{D + 3}, \frac{b + 2a - bD}{D + 3}\right). \tag{122}$$

Put

$$E_- = (1 - b, 2 - b), \qquad E_+ = (2 - a, 1 - a).$$

The actual adjacent handoffs have local centers

$$C_5(t) = (1 + t, t), \qquad 1 - a \le t \le b, \quad \text{and} \quad C_3(y) = (y, 1 + y), \qquad 1 - b \le y \le a.$$

The restrictions of the two unit-circle equations are

$$g_-(\lambda) = \frac{3}{4}\lambda^2 - 3(\alpha + b\beta)\lambda + 3b^2 - 3b + 2,$$
$$g_+(\mu) = \frac{3}{4}\mu^2 - 3(\delta + a\gamma)\mu + 3a^2 - 3a + 2. \tag{123}$$

Let λ_- and μ_+ be their first roots; explicitly,

$$\lambda_- = 2(\alpha + b\beta) - \frac{2}{3}\sqrt{9(\alpha + b\beta)^2 - 9b^2 + 9b - 6},$$
$$\mu_+ = 2(\delta + a\gamma) - \frac{2}{3}\sqrt{9(\delta + a\gamma)^2 - 9a^2 + 9a - 6}.$$

Lemma 6.5 (Moving adjacent-triangle signs)**.** *Throughout* (117),

$$\lambda_* < \lambda_- < h^{-1}, \qquad \mu_* < \mu_+ < h^{-1}. \tag{124}$$

Moreover $J_u, J_v < 1/2$. *The point* $L_+(\mu_+)$ *satisfies*

$$(L_+(\mu_+))_u + (L_+(\mu_+))_v < 3 - 2a,$$

and the reflected bound holds for $L_-(\lambda_-)$. *Writing* $F(P, t) = \mathsf{q}(P - C_5(t)) - 1$, *one has, for every* $t \in [1 - a, b]$,

$$F(L_+(\mu_+), t) \ge 0, \qquad F(J, t) > 0, \qquad F(L_-(\lambda_-), t) > 0. \tag{125}$$

Under the reflection $(u, v, a, b) \leftrightarrow (v, u, b, a)$, *the analogous signs hold for every* $y \in [1 - b, a]$*: the first-root point* $L_-(\lambda_-)$ *has nonnegative sign against* $C_3(y)$*, and the other two witnesses have strictly positive sign.*

Proof. Put

$$U = a + b - 1, \qquad z = a - b, \qquad D^2 = z^2 + 6U + 3U^2.$$

Then $0 < U < 1/6$ and $z^2 < \Omega := 1 - 6U - 3U^2$. We give the fixed-line calculation for the circle centered at $E_+ = (2-a, 1-a)$; reflection gives the calculation for E_-. Put

$$F(P,t) = \mathsf{q}(P - (1+t,t)) - 1, \qquad t_0 = 1 - a,$$

so this circle is $F(P,t_0) = 0$. At the junction, exact expansion gives

$$(D+3)^2 F(J,t_0) = 3D(z^2 + Uz + 1 - 3U) + P_J,$$

where

$$P_J = z^4 + 5z^2 + 6Uz^2 + 3U^2z^2 + 9Uz + 3U + 15U^2 > 0;$$

indeed $z^2 + Uz + 1 - 3U \geq 1 - 3U - U^2/4 > 0$, and $5z^2 + 9Uz + 15U^2$ is positive definite. Thus $F(J,t_0) > 0$. Also

$$J_u + J_v = \frac{(1+U)(3-D)}{3+D} < 1.$$

Indeed this inequality is $3U < D(2+U)$, which follows from $D^2 \geq 3U(2+U)$ and $(2+U)^3 > 3U$. Since

$$\frac{\partial F}{\partial t}(P,t) = 1 + 2t - P_u - P_v,$$

$F(J,t)$ is strictly increasing for $t \geq 0$, and therefore is positive throughout $[1-a, b]$.

On the opposite line L_-, the derivative at the junction, after multiplication by a positive factor, has at $t = 1 - a$ the numerator

$$N_0 = D(9 + 3z^2 + 6Uz - 9U^2) + P_0,$$

where

$$P_0 = 18U^3 + 6U^2z^2 + 63U^2 + 18Uz^2 + 18Uz + 54U + 2z^4 + 9z^2 + 9.$$

The coefficient of D is at least $9 - 6U - 9U^2 > 0$, while $P_0 \geq 9 + 54U - 18U > 0$. Hence $N_0 > 0$. At the other endpoint $t = b$, the corresponding numerator is

$$N_1 = D(9 + 3z^2 + 6U - 3U^2) + P_1,$$

where

$$P_1 = 9U^4 - 3U^3z + 45U^3 + 9U^2z^2 - 6U^2z + 72U^2 - Uz^3 + 21Uz^2 + 9Uz + 45U + 2z^4 + 9z^2 + 9.$$

Its D-coefficient is positive and, using $|z| < 1$,

$$P_1 \geq 9 + 45U - 9U - U - 6U^2 - 3U^3 > 0.$$

The junction derivative is affine in t, so it is positive throughout $[1-a, b]$. Since the restriction of F to L_- is a convex quadratic, is positive at the junction, and is increasing there, it has no intersection on the opposite line side for any actual adjacent handoff.

On the correct line L_+, the value at its outer endpoint h^{-1} has, after multiplication by a positive factor, numerator

$$A_0 = P_2 - 4DU(1 + U + z),$$

where

$$P_2 = 3U^4 + 6U^3z + 6U^3 + 4U^2z^2 + 12U^2z + 12U^2 \\ + 2Uz^3 + 6Uz^2 + 2Uz + 6U + z^4 + 2z^2 - 3.$$

For fixed U, replace the odd powers of z by $x = |z|$. The resulting polynomial $P_2^+(U, x)$ satisfies

$$\partial_x P_2^+ = 2(3U^3 + 4U^2x + 6U^2 + 3Ux^2 + 6Ux + U + 2x^3 + 2x) > 0,$$

and therefore its supremum for $z^2 < \Omega$ occurs at $x = \sqrt{\Omega}$. There

$$P_2^+(U, \sqrt{\Omega}) = 4U(U + \sqrt{\Omega} - 3) < 0.$$

As $1 + U + z = 2a > 0$, this gives $A_0 < 0$. The positive junction value and negative endpoint value force the first root strictly between them, proving the L_+ half of (124); reflection proves the other half.

For the coordinate bounds, the exact identities

$$D^2 - (3U - z)^2 = 6U(1 + z - U), \qquad D^2 - (3U + z)^2 = 6U(1 - z - U)$$

give $J_u, J_v < 1/2$. It remains to control the coordinate sum along L_+. For $E = L_+(h^{-1})$, the sign of $3 - 2a - E_u - E_v$ is the sign of

$$D(6U + 2z + 6) + B(U, z),$$

where

$$B(U, z) = -9U^3 - 9U^2z - 9U^2 - 3Uz^2 - 18Uz + 3U \\ - 3z^3 + 3z^2 - 3z + 3.$$

Here

$$B_z = -3(3z^2 + 2Uz + 6U - 2z + 3U^2 + 1) < 0,$$

because the quadratic in parentheses has discriminant $-32U^2 - 80U - 8 < 0$. Hence its minimum on $z^2 < \Omega$ is at $z = \sqrt{\Omega}$, where $B(U, \sqrt{\Omega}) = 6(1 - 3U - \sqrt{\Omega}) > 0$ because $(1 - 3U)^2 - \Omega = 12U^2$. Thus $E_u + E_v < 3 - 2a$. Also $J_u + J_v < 1 < 3 - 2a$, and the coordinate sum is affine on the line segment, so the same strict bound holds at the first root. Now

$$\frac{\partial F}{\partial t}(P, t) = 1 + 2t - P_u - P_v.$$

Since $F(L_+(\mu_+), 1-a) = 0$, the coordinate-sum bound makes its t-derivative strictly positive for $t \geq 1 - a$. Together with the no-opposite-line result, this proves (125). Reflection gives both the bound on L_- and the complete $C_3(y)$ statement. □

Define the local and global witnesses by

$$Q_-^{\mathrm{loc}} = L_-(\lambda_-), \qquad Q_0^{\mathrm{loc}} = J, \qquad Q_+^{\mathrm{loc}} = L_+(\mu_+), \tag{126}$$

and by the displayed conversion from (u, v) to the plane.

Lemma 6.6 (Direct asymmetric forcing)**.** *Assume that $U_C, U_0, \ldots, U_5$ cover H. Then*

$$Q_-, Q_0, Q_+ \in \mathrm{int}(H) \setminus \bigcup_{i=0}^{5} U_i,$$

and consequently all three witnesses belong to U_C.

Proof. The root order and positivity of the line coordinates give $0 < u, v < 1$ for all three points. Each belongs to $\mathcal{R}(b,a)$ and hence lies in a closed unit triangle containing V_4, so $\mathsf{q}(u,v) \le 1$; hence $|u-v| < 1$ and all three points lie in $\mathrm{int}(H)$.

The closure of U_4 is one of the triangles represented by $\mathcal{R}(b,a)$, since it contains V_4, X_3, X_4. All three witnesses lie on its exact non-axis frontier. If one belonged to the open triangle U_4, a sufficiently small plane ball about it would lie both in U_4 and in the interior of the corner cone, making the witness an interior point of $\mathcal{R}(b,a)$. This contradiction excludes U_4.

For the triangles U_0, U_1, U_2, their contained vertices have local coordinates

i	0	1	2
V_i	$(2,1)$	$(2,2)$	$(1,2)$.

For $P = (u,v)$,

$$\|P-(2,1)\|^2 - 1 = \mathsf{q}(u,v) + 2 - 3u, \quad \|P-(1,2)\|^2 - 1 = \mathsf{q}(u,v) + 2 - 3v.$$

The inequalities $J_u, J_v < 1/2$ and the line order give distance greater than one from V_0 for Q_-, Q_0 and from V_2 for Q_0, Q_+. For the remaining pair, the first-root circle and the sum bound in Lemma 6.5 give

$$\|Q_+ - V_0\|^2 - 1 = a(3 - a - (Q_+)_u - (Q_+)_v) > a^2,$$

and, by reflection, $\|Q_- - V_2\|^2 - 1 > b^2$. All three points are farther than one from $V_1 = (2,2)$, because writing $u = 1-\xi$, $v = 1-\eta$ gives

$$\|P - V_1\|^2 = 1 + \xi + \eta + \xi^2 + \eta^2 - \xi\eta > 1.$$

Since two points in one open unit equilateral triangle have distance strictly less than one, these inequalities exclude U_0, U_1, U_2.

The actual handoff $X_5 = C_5(\xi_5)$ belongs to U_5, and $1-a < \xi_5 < b$ by (6.3). The three signs in (125) therefore put every witness at distance at least one from X_5, excluding U_5. Similarly, if $y = 1 - \xi_2$, then $1-b < y < a$, $X_2 = C_3(y) \in U_3$, and the reflected signs exclude U_3. Thus no vertex triangle contains any witness. The assumed cover now forces all three interior witnesses into U_C. □

The nine directly forced points are $D_0, \dots, D_5, Q_-, Q_0, Q_+$. Under a putative cover, Lemmas 6.4 and 6.6 therefore force the compact set

$$K_{\mathrm{wit}}(a,b) = \mathcal{D}_\eta \cup \{Q_-, Q_0, Q_+\} \subset U_C. \tag{127}$$

6.1.5. *Tangent reduction and exact mixed overlaps.* For the nontrivial regime $c_* > 2/3$, we first remove the square radicals in the first-root witnesses. Normalize $\bar{\alpha} = \alpha/h$, and similarly for the other three coefficients, and put

$$\xi = \lambda/h, \quad \zeta = \mu/h, \quad \xi_* = \zeta_* = \frac{8\rho}{3(D+3)}.$$

Let g_-, g_+ be the rescaled forms of (123), and take one Newton step at the junction:

$$\widehat{\xi}_- = \xi_* - \frac{g_-(\xi_*)}{g_-'(\xi_*)}, \qquad \widehat{\zeta}_+ = \zeta_* - \frac{g_+(\zeta_*)}{g_+'(\zeta_*)}.$$

Define, in local coordinates,

$$\begin{aligned} A &= \left(b - \frac{3}{4}\bar{\beta}\widehat{\xi}_-, \frac{3}{4}\bar{\alpha}\widehat{\xi}_-\right), \\ B &= J, \\ C &= \left(\frac{3}{4}\bar{\delta}\widehat{\zeta}_+, a - \frac{3}{4}\bar{\gamma}\widehat{\zeta}_+\right), \end{aligned}$$

and convert them to global vectors.

Lemma 6.7 (Newton inner reduction)**.** *The points above have coordinates rational in a, b, D and satisfy*

$$A \in (Q_0, Q_-), \qquad C \in (Q_0, Q_+).$$

Consequently

$$\widehat{K} = \operatorname{conv}(\mathcal{D}_\eta \cup \{A, B, C\}),$$

obeys $\widehat{K} \subseteq \operatorname{conv}(K_{\text{wit}})$.

Proof. Each quadratic g_-, g_+ is strictly convex, positive, and decreasing at the junction. Its tangent zero therefore lies strictly between the junction and its first root. This gives the two open-segment inclusions. All displayed coefficients and Newton quotients are rational in a, b, D, and convexity gives the asserted containment. $\square$

Put $W = C + RA$. For a disk $\mathbb{D}(X, r)$ let

$$I(X, r) = \{n : \|n\| = 1,\ \langle X, n\rangle + r \geq h\}.$$

6.1.6. *Ray order and four overlaps.*

Proposition 6.8 (Technical four-overlap verification)**.** *Assume* (117) *and* $c_* > 2/3$. *Then*

$$\|A\|, \|C\| > \eta,$$

the rays A, B, C, W, RA occur in that cyclic order, and the four consecutive cap pairs

$$I(A, 2\eta) \leftrightarrow I(B, 2\eta) \leftrightarrow I(C, 2\eta) \leftrightarrow I(W, \eta) \leftrightarrow I(RA, 2\eta)$$

overlap across their intervening short sectors.

Proof. Put $\tau = h - 2\eta = h(2c_* - 1)$. The rays of

$$A,\ B,\ C,\ W,\ RA \tag{128}$$

occur in this cyclic order from A to RA. Indeed the two line distances give

$$\frac{\operatorname{cross}(A, B)}{\|B - A\|} = \alpha(1 - b) + \beta > 0, \quad \frac{\operatorname{cross}(B, C)}{\|C - B\|} = \gamma + \delta(1 - a) > 0.$$

If $A = (-X, -Y)$ and $C = (-P, -Q)$ in the centered cone basis, then $0 < X, Y, P, Q < 1$, $X, Q > 1/2$, and

$$\frac{\operatorname{cross}(C, RA)}{h} = PX + (Q - P)Y > 0.$$

The last sign is immediate unless $Q < P$ and $X < Y$, in which case it is larger than $Q - P/2 > 0$. The ray of $W = C + RA$ lies between those of C and RA. The same coordinates give

$$\|A\|, \|C\| > h/2 > \eta. \tag{129}$$

For the two equal-radius overlaps it is enough to prove

$$\frac{\operatorname{cross}(A,B)}{\|B-A\|} \geq \tau, \qquad \frac{\operatorname{cross}(B,C)}{\|C-B\|} \geq \tau. \tag{130}$$

We now give the exact analytic verification. By reflection take $a \leq b$ and set $m = 1-b$, $M = 1-a$, $U = a+b-1$, $s = m+M$, and $w = M-m$. The smaller of the two normalized left sides in (130) is

$$d = \frac{\mathcal{A} + U(2m-1) + D(\mathcal{A}+U)}{2\rho}, \qquad \mathcal{A} = 1 - m + m^2. \tag{131}$$

On the C_L sheet, write $F_L(z) = z^4 - z^2 + mz - m^2$. This defining polynomial evaluated at $r = 1 - m/2 + m^2/2$ is

$$F_L(r) = \frac{m^2(m-1)}{16}(m^5 - 3m^4 + 11m^3 - 17m^2 + 28m - 12) > 0.$$

Indeed the polynomial in parentheses is increasing on $(0,1/2)$ and has value $-33/32$ at $1/2$; its derivative is $28 - 34m + m^2(5m^2 - 12m + 33) > 0$. Since $F_L(h) = -(m-h/2)^2 \leq 0$ and the selected root is unique, $2c_* - 1 \leq \mathcal{A}$. Put

$$X_0 = \mathcal{A} + U, \qquad Y_0 = 2\rho\mathcal{A} - \mathcal{A} - U(2m-1).$$

Then

$$d - \mathcal{A} = \frac{D(\mathcal{A}+U) - Y_0}{2\rho},$$

and exact expansion gives

$$\begin{aligned} Y_0 = {} & 2(m^2 - m + 1)U^2 + (2m^3 - 2m + 3)U \\ & + (m^2 - m + 1)(2m^2 - 2m + 1) > 0. \end{aligned}$$

Furthermore

$$D^2 X_0^2 - Y_0^2 = 4m\rho\Phi(U,m),$$

where

$$\begin{aligned} \Phi(U,m) = {} & (1-m)(m^2 - m + 2)U^2 \\ & + (2 - m^2)(m^2 - m + 1)U \\ & - m(1-m)^2(m^2 - m + 1). \end{aligned}$$

In these variables

$$\chi = U^4 - 4U^3 + 5U^2 - (m+2)U + m(1-m).$$

Since $U^2(U^2 - 4U + 5) > 0$, the selector $\chi \leq 0$ implies $U > m(1-m)/(m+2)$; Φ is increasing in U and at that lower endpoint is

$$\frac{m^2(1-m)(m^4 - 2m^3 + 6m^2 - 6m + 4)}{(m+2)^2} > 0.$$

The last quartic equals $(m^2 - m)^2 + 5m^2 - 6m + 4 > 0$. Thus $d > \mathcal{A} \geq 2c_* - 1$.

On the other radial sheet let $E = \sqrt{4s^2 - 3}$. Then $0 \leq w < sE$ and $c_* = (s+w)/(1+E)$. At fixed U, differentiation of (131) can be checked without an implicit estimate. Namely,

$$d = \frac{1+D}{2} - UB(D,w), \qquad B(D,w) = \frac{(1+U)(3+D) + w(1-D)}{D^2 + 3}.$$

Here $D^2 = w^2 + 6U + 3U^2$, so $0 \le D' = w/D \le 1$, and

$$B_w = \frac{1-D}{D^2+3} \ge 0.$$

Writing $N = (1+U)(3+D) + w(1-D)$, direct differentiation gives

$$B_D = \frac{(1+U-w)(D^2+3) - 2DN}{(D^2+3)^2} > -\frac{34}{27}.$$

Indeed the first term in the numerator is positive, while $1 + U - w = 2a > 0$, $N < (7/6)4 + 1 = 17/3$, $D < 1$, and $D^2 + 3 \ge 3$. If $B_D < 0$, then $B_D D' \ge B_D$; otherwise $B_D D' \ge 0$. Hence $B' = B_w + B_D D' > -34/27$, and $U < 1/6$ yields

$$d' < \frac{1}{2} + \frac{1}{6}\frac{34}{27} = \frac{115}{162} < 1,$$

whereas $(2c_* - 1)' = 2/(1+E) \ge 1$. Hence their difference decreases with w. At the boundary $w = sE$, where $\chi = 0$ and $c_* = s$, the preceding sheet applies: this is an admissible point because $h < s < 1$ and $D^2 = 4s^4 - 12s + 9 < 1$. Indeed $s^4 - 3s + 2 = (s-1)(s^3 + s^2 + s - 2) < 0$ on this interval; the second factor is increasing and already equals $(7h-5)/4 > 0$ at $s = h$. The preceding sheet gives $d \ge 2s - 1$. Therefore (130) holds on both sheets, and reflection supplies the other line.

It remains to check the two mixed overlaps. The exact rational envelopes, Gram reductions, and global positive-basis identities are recorded once in Appendix A. Theorem A.3 gives precisely the overlaps for $\mathbb{D}(C, 2\eta)$ with $\mathbb{D}(W, \eta)$ and for $\mathbb{D}(W, \eta)$ with $\mathbb{D}(RA, 2\eta)$.

Thus the four consecutive pairs of caps centered on the rays (128) overlap. $\square$

6.2. **Support arcs and the enclosure conclusion.** Let R denote counterclockwise rotation through $2\pi/3$. Retain the points A, B, C, the vector $W = C + \mathsf{R}A$, the forced disk $\mathcal{D}_\eta$, and the exact witness set K_{wit} constructed above. For uniform notation put

$$P_- = A, \qquad P_0 = B, \qquad P_+ = C, \qquad P_* = W, \qquad r_* = \eta,$$

and

$$\widehat{K} = \operatorname{conv}\big(\mathcal{D}_\eta \cup \{A, B, C\}\big).$$

For $X \in \mathbb{R}^2$ and $r \ge 0$, write

$$\mathbb{B}(X, r) = \{Y : \|Y - X\| \le r\},$$

and define the level-h, $h = \sqrt{3}/2$, support arc

$$\operatorname{Cap}(X, r) = \{n \in \mathbb{S}^1 : \langle X, n\rangle + r \ge h\}.$$

Lemma 6.9 (Cyclic support-arc covering)**.** *Let K be compact and put*

$$\Sigma_K = K + \mathsf{R}K + \mathsf{R}^2K.$$

Suppose Σ_K contains the full R-orbits of

$$\mathbb{B}(P_-, 2r_*), \quad \mathbb{B}(P_0, 2r_*), \quad \mathbb{B}(P_+, 2r_*), \quad \mathbb{B}(P_*, r_*).$$

Assume that the rays

$$P_-, P_0, P_+, P_*, \mathsf{R}P_-$$

occur in this cyclic order, that every corresponding support arc is connected with half-width less than $\pi/2$, and that each consecutive pair of arcs intersects across the intervening short angular sector. Then

$$\Lambda(K) \geq 1.$$

Proof. The four consecutive intersections and the cyclic order make the five arcs cover the sector from the ray of P_- to the ray of $\mathsf{R}P_-$. Their R-orbits cover $\mathbb{S}^1$. Hence for every unit vector n one of the displayed disks has support at least h in direction n, and therefore

$$h_{\Sigma_K}(n) \geq h.$$

Since

$$h_{\Sigma_K}(n) = \sum_{j=0}^{2} h_K(\mathsf{R}^j n),$$

Proposition 4.1 gives $\Lambda(K) \geq 1$. □

Proposition 6.10 (Four support-arc intersections)**.** *Assume the strict exact-one-supercritical domain and $c_* > 2/3$. The five rays in Lemma 6.9 occur in the required cyclic order, all four consecutive support-arc intersections are nonempty, and $\|P_-\|, \|P_+\| > r_*$.*

Proof. Proposition 6.8 proves the ray order, radius bounds, and the two equal-radius intersections. Theorem A.3 proves the two unequal-radius intersections by exact Gram reductions and Bernstein-positivity certificates. □

Theorem 6.11 (Witness enclosure)**.** *For every exact-one-supercritical parameter pair (a, b) constructed above,*

$$\Lambda(K_{\mathrm{wit}}(a, b)) \geq 1.$$

Proof. The centered disk alone gives

$$\Lambda(K_{\mathrm{wit}}) \geq 3(1 - c_*),$$

so the result follows when $c_* \leq 2/3$. Assume $c_* > 2/3$. Lemma 6.7 gives $\widehat{K} \subseteq \mathrm{conv}(K_{\mathrm{wit}})$. The Minkowski sum $\widehat{K} + \mathsf{R}\widehat{K} + \mathsf{R}^2\widehat{K}$ contains the disk orbits listed in Lemma 6.9. Proposition 6.10 supplies their cyclic order and all four intersections. Hence $\Lambda(\widehat{K}) \geq 1$, and monotonicity of the equilateral enclosure gauge yields $\Lambda(K_{\mathrm{wit}}) \geq 1$. □

Theorem 6.12 (Zero-gap obstruction)**.** *There is no cover of H by open unit equilateral triangles $U_C, U_0, \ldots, U_5$ such that every U_i is* Vd0*, the six vertex triangles cover ∂H, and exactly one vertex triangle is supercritical.*

Proof. The construction above forces the compact set K_{wit} into the open center triangle U_C, whereas Theorem 6.11 gives $\Lambda(K_{\mathrm{wit}}) \geq 1$. Every compact subset of an open unit equilateral triangle is contained in a closed equilateral triangle of side strictly below one, which would imply $\Lambda(K_{\mathrm{wit}}) < 1$. □

Proposition 6.13 (Cases excluded by the nine-point obstruction)**.** *Theorem 6.12 excludes*

(1) *the* CE0*, $N_+ = 1$, all-*Vd0 *case;*
(2) *the no-gap* CE1 *and* CE2*, $N_+ = 1$, all-*Vd0 *cases.*

Proof. In the CE0 case the six vertex triangles must cover the perimeter, since a missed boundary point would lie in the open center triangle and create a positive center trace. In the CE1 and CE2 cases, absence of a boundary gap means that the two incident vertex traces cover every edge. The unique-supercritical hypothesis is common to all three cases. □

7. Completion of the Proof

Proof of Theorem 1.1. Assume that seven open unit equilateral triangles cover H. By Proposition 2.13, the cover lies in exactly one row of Table 1. The rows assigned to trace length are excluded by Proposition 3.11; the rows assigned to area loss by Proposition 5.5; the boundary-reach propagation rows by Proposition 4.46, including the complete CE2 one-Vd1/Vd2 positional assembly in Proposition 4.49; and the no-gap, all-Vd0 rows by Proposition 6.13. Hence every row is impossible, contradicting the assumed cover. □

Corollary 1.2 follows from Proposition 2.1.

Appendix A. Exact Unequal-Radius Support-Arc Intersections

The two mixed cap overlaps in Method 4 are certified by exact symbolic arithmetic. This section states the complete mathematical reduction, explains the verification algorithms, and formally incorporates the code and sparse integer data listed below as an electronic proof supplement. No floating-point arithmetic, interval arithmetic, adaptive subdivision, branch-and-bound, or unrecorded numerical tolerance is used.

A.1. **Certificate manifest.** All paths below are relative to the Method 4 computation directory. The machine-readable provenance manifest stores the full Git blob identities; the paper prints compact identifiers and twelve-character prefixes.

certificate object	blob prefix
`core_data_00`	`359b219f9ded`
`core_data_01`	`bfc807f90635`
`core_data_02`	`92887696ef94`
`core_data_03`	`c2791a8a6add`
`core_data_04`	`7eeabecbbd17`
`core_data_05`	`d676114292e7`
`core_polynomials`	`07f57b178cf7`
`derivation verifier`	`0a55211ae37f`
`positivity verifier`	`821f9a6dc750`

The six data shards decode to one canonical sparse-polynomial transcript with SHA–256 digest

`dc46aaf263655d5159ecd3a81db72ee82477951d06172f4743b248df37209485`

The first verifier independently derives the transcript from the witness formulas and compares every coefficient with the authenticated object. The second verifier derives all Bernstein coefficients from that same object and checks their signs exactly.

A.2. **Rational radial upper envelopes.** Retain the notation of Section 6.1. Thus

$$p = 1 - b, \qquad q = 1 - a, \qquad s = p + q, \qquad m = \min\{p, q\}, \qquad M = \max\{p, q\},$$

$$\chi = s^4 - s^2 + mM, \qquad h = \frac{\sqrt{3}}{2}, \qquad c_* = c_{\max}(p, q).$$

The strict witness domain gives

$$mM > (1 - s)(2 - s), \qquad m > 1 - s, \qquad s > \frac{2}{3}. \tag{132}$$

Put

$$F_m(x) = x^4 - x^2 + mx - m^2, \qquad \kappa = F_m'(s) = 4s^3 - 2s + m.$$

Then

$$\kappa > 4s^3 - 3s + 1 = (s + 1)(2s - 1)^2 > 0.$$

Define

$$\bar{c}_L = s - \frac{\chi}{\kappa} \quad (\chi \le 0), \qquad \bar{c}_T = s - \frac{\chi}{1 - m} \quad (\chi \ge 0). \tag{133}$$

Lemma A.1 (Branchwise radial upper bounds)**.** *On the corresponding radial cells,*

$$c_* \le \bar{c}_L < 1, \qquad c_* \le \bar{c}_T < 1.$$

At $\chi = 0$ both upper bounds equal $s = c_*$.

Proof. On the L cell, $F_m(s) = \chi \le 0$ and $F_m(c_*) = 0$. For $x \ge s > 2/3$, $F_m''(x) > 0$, so convexity gives
$$0 \ge F_m(s) + F_m'(s)(c_* - s),$$
and hence $c_* \le \bar{c}_L$. If $U = 1 - s$, then
$$\chi + U\kappa > U\{m + 1 - 4U + 8U^2 - 3U^3\} > 0$$
by (132); therefore $\bar{c}_L < 1$.

On the T cell put
$$E = \sqrt{4s^2 - 3}, \qquad m_0 = \frac{s(1-E)}{2}, \qquad M_0 = \frac{s(1+E)}{2}.$$
Then
$$\chi = (m - m_0)(M_0 - m), \qquad s - c_* = \frac{2(m - m_0)}{1 + E}.$$
Consequently
$$\frac{\chi}{s - c_*} = \frac{1+E}{2}(M_0 - m) \le 1 - m,$$
which gives $c_* \le \bar{c}_T$. The strict inequality $\chi > 0$ gives $\bar{c}_T < s < 1$. □

Reflect so that $z = a - b \ge 0$ and put $U = a + b - 1$. Then
$$s = 1 - U, \qquad m = \frac{1 - U - z}{2}, \qquad D^2 = z^2 + 6U + 3U^2 =: K(U, z),$$
and
$$\Xi(U, z) = 4\chi = 1 - 10U + 21U^2 - 16U^3 + 4U^4 - z^2,$$
$$G(U, z) = 2\kappa = 5 - 21U + 24U^2 - 8U^3 - z.$$
Thus
$$\bar{c}_L = 1 - U - \frac{\Xi(U, z)}{2G(U, z)}, \qquad \bar{c}_T = 1 - U - \frac{\Xi(U, z)}{2(1 + U + z)}. \tag{134}$$

A.3. **Gram reduction and the geometric implication.** On the active cell let $\bar{c}$ denote the corresponding expression in (134), and put
$$\bar{\eta} = h(1 - \bar{c}), \qquad e = 1 - \bar{c}, \qquad t = 2\bar{c} - 1.$$
Lemma A.1 gives $0 < \bar{\eta} \le \eta = h(1 - c_*)$. It therefore suffices to prove the mixed intersections for the smaller disks with radii $2\bar{\eta}, \bar{\eta}, 2\bar{\eta}$.

Represent a global vector as (x, hy) and put $C' = \mathsf{R}^{-1}C$. Define
$$N_A = \|A\|^2, \qquad N_C = \|C\|^2, \qquad \nu = \langle A, C' \rangle,$$
$$\Delta_0 = x_A y_{C'} - y_A x_{C'} > 0.$$
Thus the Euclidean cross product is $h\Delta_0$, and the Gram identity is
$$h^2\Delta_0^2 = N_A N_C - \nu^2. \tag{135}$$
Define
$$Q_A = \Delta_0^2 - \|tA - eC'\|^2, \qquad Q_C = \Delta_0^2 - \|tC' - eA\|^2. \tag{136}$$

Lemma A.2 (Residual signs imply the mixed cap intersections). *If $Q_A, Q_C \geq 0$, then*

$$I(C, 2\bar{\eta}) \cap I(C + \mathsf{R}A, \bar{\eta}) \neq \varnothing,$$

and

$$I(C + \mathsf{R}A, \bar{\eta}) \cap I(\mathsf{R}A, 2\bar{\eta}) \neq \varnothing.$$

Consequently the same intersections hold with η in place of $\bar{\eta}$.

Proof. Rotate the first pair by R^{-1}. Its centers become C' and $A + C'$, so their difference is A. An outer common tangent normal n_A is chosen with

$$\langle A, n_A \rangle = \bar{\eta}.$$

Since $\|A\| > \bar{\eta}$ and $\Delta_0 > 0$, choose the tangent side for which

$$\langle C', n_A \rangle = \frac{\bar{\eta}\nu + \sqrt{N_A - \bar{\eta}^2}\, h\Delta_0}{N_A}.$$

The common support of the two disks is therefore

$$2\bar{\eta} + \frac{\bar{\eta}\nu + \sqrt{N_A - \bar{\eta}^2}\, h\Delta_0}{N_A}.$$

Let $\bar{\tau} = h - 2\bar{\eta} = ht$. Expanding (136) and using (135) gives

$$\bar{P}_A := (N_A - \bar{\eta}^2)h^2\Delta_0^2 - (\bar{\tau}N_A - \bar{\eta}\nu)^2 = h^2 N_A Q_A. \tag{137}$$

Thus $Q_A \geq 0$ implies

$$\sqrt{N_A - \bar{\eta}^2}\, h\Delta_0 \geq \bar{\tau}N_A - \bar{\eta}\nu,$$

and the common support is at least $2\bar{\eta}+\bar{\tau} = h$. Hence the two level-$h$ caps intersect. The same calculation with A and C' exchanged gives

$$\bar{P}_C := (N_C - \bar{\eta}^2)h^2\Delta_0^2 - (\bar{\tau}N_C - \bar{\eta}\nu)^2 = h^2 N_C Q_C, \tag{138}$$

and proves the second intersection. Enlarging the disk radii from $\bar{\eta}$ to η only enlarges the caps. ☐

A.4. **Eight exact integer-polynomial signs.** The Newton witnesses A, C are rational functions of U, z, D. Substituting (134) into (136), clearing denominators, and reducing by

$$D^2 - K(U, z) = 0$$

gives, for $B \in \{L, T\}$ and $X \in \{A, C\}$,

$$\mathrm{num}(Q_{B,X}) = 32(K + 3)^2\{A_{B,X}(U, z) + DB_{B,X}(U, z)\}, \tag{139}$$

where all eight core polynomials lie in $\mathbb{Z}[U, z]$. Every omitted denominator is strictly positive on the strict domain: its factors are positive constants, $(D+3)^4$, $G(U, z)^2$ or $(1+U+z)^2$, and squares of the two Newton-derivative numerators. Here $G = 2\kappa > 0$, and the fixed-line sign theorem proves both Newton derivatives strictly negative.

The derivation verifier constructs A, C, both rational envelopes, and all four residuals directly in the exact field $\mathbb{Q}(U, z, D)$. It performs the reduction modulo $D^2 - K$, removes the common positive factor, primitive-normalizes the eight integer polynomials, and compares their sparse term lists coefficient by coefficient with the authenticated transcript. Thus the data are checked against the geometric formulas rather than trusted as an independent precomputation.

A.5. **Three global positive-basis certificates.** Set

$$\mathcal{G}(U)=1-6U-3U^2,\qquad H(U)=1-10U+21U^2-16U^3+4U^4,$$

$$U_H=1-\frac{\sqrt{3}}{2},\qquad U_{\max}=\frac{2}{\sqrt{3}}-1.$$

The complete reflected domain is the union of the three closed cells

$$\begin{array}{ll} T: & 0\le U\le U_H,\quad 0\le z^2\le H(U),\\ L_0: & 0\le U\le U_H,\quad H(U)\le z^2\le \mathcal{G}(U),\\ L_1: & U_H\le U\le U_{\max},\quad 0\le z^2\le \mathcal{G}(U). \end{array}$$

For the T cell use

$$U=\frac{(x-1)(3-x)}{4x},\qquad z=y\frac{9-x^4}{8x^2},\qquad 1\le x\le\sqrt{3},\quad 0\le y\le 1,$$

and then $x=1+(\sqrt{3}-1)r$. This maps $[0,1]^2$ onto the complete cell. After multiplication by positive powers of 2 and x, the four transformed core polynomials have global tensor Bernstein expansions with data

	bidegree	coefficient count	zero count
$A_{T,A}$	$(88,22)$	2047	2
$B_{T,A}$	$(80,20)$	1701	0
$A_{T,C}$	$(88,22)$	2047	2
$B_{T,C}$	$(80,20)$	1701	0.

Every nonzero coefficient is strictly positive.

For an L-cell polynomial write

$$F(U,z)=E_F(U,w)+zO_F(U,w),\qquad w=z^2,$$

and

$$R_F(U,w)=E_F(U,w)^2-wO_F(U,w)^2.$$

The implications $E_F\ge 0$ and $R_F\ge 0$ give $E_F\ge\sqrt{w}\,|O_F|$, hence $F\ge 0$ for $z\ge 0$. Use the exact charts

$$U=U_Hr,\qquad w=H(U)+s\{\mathcal{G}(U)-H(U)\}$$

for L_0, and

$$U=U_H+(U_{\max}-U_H)r,\qquad w=s\mathcal{G}(U)$$

for L_1. The positivity verifier forms one global Bernstein expansion for each of E_F, R_F on each square. The resulting sixteen expansions have the bidegrees and coefficient counts recorded in the certificate tables below; every coefficient is strictly positive.

The verifier implements polynomial addition, multiplication, substitution, and power-to-Bernstein conversion over exact integers. Coefficients in $\mathbb{Q}(\sqrt{3})$ are represented as rational pairs and signed by exact rational comparisons. If $P(r,s)=\sum a_{ij}r^is^j$ has bidegree (m,n), its exact tensor Bernstein coefficients are

$$b_{k\ell}=\sum_{i\le k}\sum_{j\le\ell}a_{ij}\frac{\binom{k}{i}}{\binom{m}{i}}\frac{\binom{\ell}{j}}{\binom{n}{j}}.$$

This is precisely the conversion implemented by the verifier.

Theorem A.3 (Exact mixed-overlap certificate)**.** *On the complete strict witness domain with* $c_* > 2/3$*, all eight polynomials in* (139) *are nonnegative. Hence* $Q_A, Q_C \geq 0$*, and the two mixed cap intersections required by Proposition 6.8 hold.*

Proof. The T chart proves the four T-cell core signs by nonnegative Bernstein coefficients. On each L chart the positive Bernstein certificates for E_F and R_F prove the four L-cell signs. Since $D \geq 0$, (139) gives $Q_A, Q_C \geq 0$ for $z \geq 0$; reflection supplies $z \leq 0$. Lemma A.2 then gives the two cap intersections. □

Remark A.4 (Reproduction status)*.* The files above constitute the exact certificate used by the proof. A reproduction run should execute

```
verify_mixed_overlap_core_derivation.py
verify_global_core_positivity.py
```

from their package directory. This manuscript revision does not claim that those commands were rerun during the present source-only edit; the certificate is incorporated by its exact repository objects and authenticated transcript.